PHASE A

Team card

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Course (CdL)

Digital Humanities and Digital Knowledge

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Course (CdL)

Digital Humanities and Digital Knowledge

Client card

Name of the client:

Istituto Poligrafico e Zecca dello Stato

Description of the client:

The Istituto Poligrafico e Zecca dello Stato is an entity founded in 1928, wholly owned by the Ministero dell'Economia e delle Finanze. It represents a European leader for security printing solutions, digital identity and coin minting.

Name of the tool:

App CieID

How to access the tool:

In order to access the tool it is required to download the application available on App Store for iOS and on Google Play Store for Android. On the application it is possible to register an ID card by inserting the 8 digit PIN handed over at the time the card is issued and holding the ID card on the back of the mobile phone for the NFC recognition.
There are different levels of security. Level 1 and 2 require access through username, password, QR code and push notification while level 3, which represents the maximum level of security, must be obtained by the NFC reading of the card. The level 3 of security is usually only reachable through the mobile App registration unless the user does own a smart card contactless/NFC connected via USB to the computer.

Why should the client want a new tool?

Many SPID (Sistema Pubblico di Identità Digitale) have introduced an annual fee to access their services, ranging from 5,98 to 9,99 euros, including the SPID provided by Poste Italiane, which is currently the first provider in Italy, with 28,7 million digital identities activated at the end of

2024. Italian citizens may prefer to switch to a free service, which is now provided by the App CieID that carries out the same task of accessing the public administration applications.
Since, in order to register the ID card it is necessary to insert the PIN or alternatively the PUK provided at the moment of card handling on a paper sheet, people may need to retrieve those codes if lost. Now the process requires much time, effort and the assistance of an intermediary. A rethinking of the user experience that guides the user through the process and speeds it up could be a good reason to encourage users to trust the new application.

Product Objectives for the client

The main objective is to facilitate PIN recovery and NFC scanning for the CIE, dramatically speeding up a process that previously generated hesitation and required external assistance. In addition, the goal is to empower citizens, particularly senior and digitally anxious users, to become autonomous in registering their card, reducing initial distrust through a unified **single-scan flow** and step-by-step visual guidance.

Streamline first-time onboarding and code management: eliminate initial cognitive barriers by providing transparent upfront guidance on credential locations and replacing technical jargon (e.g., “Livello 1 e 2”, “Livello 3”, “NFC scanning”) with plain-language and contextual explanations.

Direct and context-aware support: eliminate indirect, multi-nested menu sequences during authentication or PIN errors by providing direct action shortcuts. Replace hidden hamburger submenus with a persistently visible navigation bar, ensuring onboarding guides and support tools remain instantly accessible

Integrated in-app guidance and support: embed contextual visual tutorials, such as NFC alignment illustrations and dynamic progress feedback, directly within the core application flow. Ensure onboarding guides and help resources remain permanently accessible on-demand

during subsequent app sessions, rather than redirecting users to external web portals.

Business Goals

Reduce time and effort related to the tasks:
By designing a system focused on saving time, the application will present itself as the best option among the SPID services by accordingly diminishing the mental load and effort required by the user.

Capturing the migration from SPID to CieID:
Redesigning the user experience of the application in order to transform it into an appealing free substitute of more diffused SPID providers, leveraging on no fee policy and a fast recovery of lost codes.

Overcome the initial effort of PIN recovery:
Insist on the simplification of this first step in order to allow the user to be able to use all the other functionality of CieID autonomously, installing trust in the overall system.

Transform intermediaries from executors into digital mentors: shift the operational dynamic at public assistance centers by equipping direct users with self-service tools, enabling intermediaries to guide citizens toward long-term digital autonomy rather than performing tasks on their behalf.

Reduce support operations and administrative costs: decrease the volume of in-person visits, municipal support requests, and help-desk calls stemming from lost PIN/PUK codes, failed card registrations, or confusing menu navigation.

Brand identity

Trustable and accessible system: become an accessible and free system, trusted by citizens and usable according to the efficiency, effectiveness and emotion metrics.

Transparency and security: leverage on ensuring a secure system in which the user trusts and relies on.

Positioning as the sovereign digital identity: affirming the application as the ideal application in the field of digital identities, keeping the service free and open to everyone.

Empowering and non-patronizing tone: ensure all interface messaging, error handling, and instructional dialogues speak in a simple, empathetic, and constructive register, building user confidence without making low-tech citizens feel incompetent.

Inclusive and accessible visual standards: adhere strictly to public design and accessibility guidelines by ensuring high-contrast UI layouts (e.g., readable text over colored backgrounds) and offering multi-language support for non-native citizens.

Success metrics

Active adoption and usage rate: a percentage increase of citizens completing the on-boarding process and using CieID as the primary service to access public administration applications.

Task completion time: reducing execution time while carrying out the recovery of the PIN, transforming the procedure from a multiple day activity to a rapid task.

First-time unassisted completion rate: the percentage of first-time users who successfully complete card registration or PIN recovery autonomously without requiring intermediary intervention.

Reduction in intermediary support volume: a measurable decrease in repeat visits to assistance centers and municipal offices for routine code retrieval and onboarding support.

System usability scale (SUS) benchmark: achieving a System Usability Scale score above 68/100 evaluated across first-time users, low-digital-literacy demographics and assisting intermediaries.

User Segmentation card

Segmentation dimensions chosen (demographic & psychological)

- **Dimension #1 (demographic)**: age
- **Dimension #2 (demographic)**: schooling
- **Dimension #3 (psychological)**: attitude
- **Dimension #4 (contextual)**: device
- **Dimension #5 (contextual)**: physical constraints

Reason for choice, relationship with client's tool

Age:
A fundamental parameter to provide an overview of the user's technological skills and familiarity with technological devices, the level of confidence while approaching new applications and the ability to understand technical terminology. The age segment provides essential information about the alleged level of confidence shown by the user during the autonomous completion of task.

Schooling:
Indicates the knowledge acquired through the user's educational background, influencing their ability to understand administrative vocabulary and complex concepts at the core of the application. Based on the user's level of education, the application must simplify complex sequences of tasks and guide the user by leveraging on learnability.

Attitude:

Dimension reflecting the user's mindset and emotional disposition when interacting with the application. Attitude can either act as a positive driver when the user feels confident performing tasks or represent a barrier to overcome by minimizing friction and complexity, thereby instilling trust and shifting their perception towards the app.

Device:
A contextual parameter identifying the device's technical characteristics and intrinsic limitations.The type of device available significantly influences the user experience while interacting with the app. The mobile application differs drastically from the browser interface and the absence of NFC technology on older smartphones or desktop PCs restricts the registrations process required to achieve level 3 of security. Ultimately, device modernity introduces specific hardware advantages and constraints that directly affect user autonomy.

Physical constraints:
Represents the user's temporary or permanent physical limitations that impede fluid interaction with the application. Strongly influenced by the age, these constraints include minor visual or motor impairments such as shortsightedness or reduced reaction times, as well as situational limitations derived from the context of use like forgetting reading glasses or trying to interact with the device in suboptimal physical conditions.

Segments within dimension #1 (Age)

1. **Segment 1A:** 35-40 years old
2. **Segment 1B:** 50-62 years old
3. **Segment 1C:** over 70
4. **Segment 1D:** 18-32 years old

Segments within dimension #2 (Schooling)

1. **Segment 2A:** high school graduates
2. **Segment 2B:** lower secondary education
3. **Segment 2C:** master's degree holder

Segments within dimension #3 (Attitude)

1. **Segment 3A:** an ambitious individual, confident in using technology and motivated to discover new tools.
2. **Segment 3B:** a passive user hesitant to leave their routine, exhibiting low motivation to adopt digital tools.
3. **Segment 3C:** an active person, always open to learning but skeptical about new technology.
4. **Segment 3D:** an autonomous person, reticent to ask for help, motivated to reach their goals.

Segments within dimension #4 (Device)

1. **Segment 4A:** a mobile device lacking NFC hardware.
2. **Segment 4B:** a modern smartphone equipped with NFC sensor and biometric authentication.
3. **Segment 4C:** a personal computer used via web browser.

Segments within dimension #5 (Physical constraints)

1. **Segment 5A:** an individual with age-related visual decline, difficulties in reading small text.
2. **Segment 5B:** reduced reaction time individual, struggling with fine gestures.
3. **Segment 5C:** a fully abled individual experiencing no physical constraints.

Total number of segment combinations: 4 age x 3 schooling x 4 attitude x 3 device x 3 physical constraints = **432 combinations**

Chosen segmentation combinations

- **The Autonomy-Seeking Senior** - Direct user (1C + 2B + 3C +4B + 5A): an elderly citizen, motivated to maintain their autonomy, seeking guidance rather than complete delegation.
- **The Trusted Digital Mentor** - Intermediary (1D + 2C + 3A + 4B + 5C): digital native, confident while using technology and capable of stepping into the role of a mentor for less experienced individuals when needed.

Why the chosen combinations are more representative, challenging, expressive than the others

The chosen combinations highlight two complementary profiles that reflect the real world dynamics of public identity tools like CieID. The inevitable shift toward digital self-service forces the **Autonomy-Seeking senior**, despite being open minded and eager to learn into a vulnerable position. They must now learn autonomously how to perform tasks traditionally assisted by an operator at a physical front-desk. While this category often faces digital anxiety, they remain motivated to learn by relying heavily on “knowledge in the world” rather than “knowledge in the head”.
To support this transition, the intermediary is embodied by the **Trusted Digital Mentor**, the younger, educated user whom the Autonomy-Seeking senior turns to for guidance.The gap in education and technical abilities represents a key obstacle to overcome. Redesigning the user experience equips the intermediary with plain language guidance and intuitive mentoring flows to help the elderly user achieve digital autonomy, ultimately reducing their skepticism and fear of new technologies.

Expected impact on design

Facilitated terminology and contextual guidance:
Relying on a non technical vocabulary, accessible across all education levels, transforming complex instruction into clear, step-by-step guidance. By integrating contextual tutorials directly into the task workflow, the app offloads complex explanations onto the UX itself, helping the intermediary instruct the senior citizen more efficiently.

Intuitive User Interaction and simplified flows:
Designing an intuitive system that directly supports vulnerable users who are the most impacted by the digital transition of public services. The interface reduces digital anxiety, encouraging senior citizens to build technological skills while alleviating pressure on physical front-desk operators.

Information design improvement:

Streamlining information design to clearly map the user journey for critical operations, reducing navigation time and cognitive effort while enhancing learnability for both direct user and intermediaries.

Market research card

Segment of choice

The Autonomy-Seeking Senior: an elderly citizen, motivated to maintain their autonomy, seeking guidance rather than complete delegation.

Reason for choice

Since the digital transition of public administration services requires the use of web technologies, elderly citizens risk being left behind by the progress. New devices, such as smartphones, are usually a source of anxiety and many avoid using them autonomously by delegating tasks to an intermediary, usually a front-desk operator. The redesign must take in consideration this segment of population to simplify the design and target vulnerable users, guiding them through the app with the goal of making them autonomous.

User research method employed

- Available statistics
  Sources:
  - **Cittadini e ITC - Anno 2025:** official statistics provided by Istat on the use of technology across all age groups, comparing the situation of precedent years with 2025. The results are overall positive since the percentage of citizens with digital skills between **16-74 years** has increased by 8,4 percentage points and the access to the internet in families composed only of seniors is increasing too. The age group from **65-74 years** shows an improvement in their basic digital

skills since 2023 (from 19,4 to 27,4%) but it is still the less competent one with a gap of 41,7%.
- **Research report by BVA DOXA:** from a questionnaire performed in 2021 on the elderly population of Bologna, citizens over 75 declare to rely on external support while using digital devices. 27,4% of the sample declares to ask for help from younger family members, friends or experts while 33,1% does not use digital services at all. More broadly, citizens above 65 feel the need for easier digital tools, specifically designed to accommodate their needs.

- **Field observation #1**

Location of observation

Home

Main outcomes

Inexperience with NFC technology:
Questioning of the system and its functionality. The user needs clarifications when NFC is mentioned otherwise the subject's doubts and anxiety persists, creating cognitive overload.

Reliance on mental model versus interface suggestions:
Digital signifiers about the correct sequence of interaction are often ignored. The user mainly relies on mental models acquired during precedent experiences with other apps.

Persistent questioning one's actions and general doubting:
The user demonstrated to be partially autonomous but still very skeptical about his performance, often requiring external confirmations by the observer.

Overall results of the analysis

The research, integrating both statistics about digital skills across the elderly citizens and direct observation on the field, highlights a gap between younger samples of population and the target of the analysis.

Digital competences are increasingly improving even among seniors but there is still a segment of the population which is left behind during the digital transition.
The observations on two different direct users allowed us to recognize persistent points of friction: inaccessible, technical language, general misunderstanding of complex concepts, a persistent need for guidance and confirmations. These aspects represent the reasons that lead seniors to seek external help, limiting their autonomy. …

Impact on design

Reorganization of information design:
Adapting the interface in order to immediately show the most important feature of the app and facilitate primary tasks such as the registration of the card or the recovery of the PIN. Simplifying the main navigation making it explicit by substituting the hamburger menu, which collects the links to other sections of the app with clear, unconcealed links inside a navigation bar.

Adoption of an accessible vocabulary:
Vulnerable users need to be encouraged in the use of the app by avoiding technical language and ensuring the comprehension of the content relying on a simple vocabulary. Replace complex terms and concepts with simplified, yet equally explicative versions.

Clear instruction and tutorials:
Each main task is going to be guided through its execution in order to avoid ambiguity and limit user's doubting. Existent tutorial, currently hidden inside the menu will be integrated during the process of carrying out an action presenting the user to spend excessive time navigating the system.

22 APRILE 2026

statistiche report

Istat

CITTADINI E ICT | ANNO 2025

Aumentano le competenze digitali

Nel 2025 è pari al 54,3% la quota di persone di 16-74 anni con competenze digitali almeno di base, 8,4 punti percentuali in più rispetto all'anno precedente. Resta però lontano l'obiettivo europeo che punta, entro il 2030, a una quota dell'80%.

In Italia, come in altri Paesi europei, le competenze digitali sono caratterizzate da forti divari associati a specifiche caratteristiche socio-culturali della popolazione.

Nel 2025 il 71,7% dei ragazzi di 20-24 anni che hanno usato Internet negli ultimi 3 mesi ha competenze digitali almeno di base. Una quota che scende rapidamente con l'età, per arrivare al 49,1% tra i 55-59enni e attestarsi al 27,4% tra le persone di 65-74 anni.

47%

Gli individui di 15 anni e più che hanno usato l'identità digitale SPID o CIE negli ultimi 12 mesi

+5,7 punti percentuali sul 2023

17,8%

Gli individui di 14 anni e più che hanno usato strumenti di intelligenza artificiale negli ultimi 3 mesi

49%

Gli individui di 14 anni e più che hanno fatto acquisti *online* negli ultimi 12 mesi

+2,2 punti percentuali sul 2024

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Fig, 1 - Istat Research (Cittadini e ITC - Anno 2025)

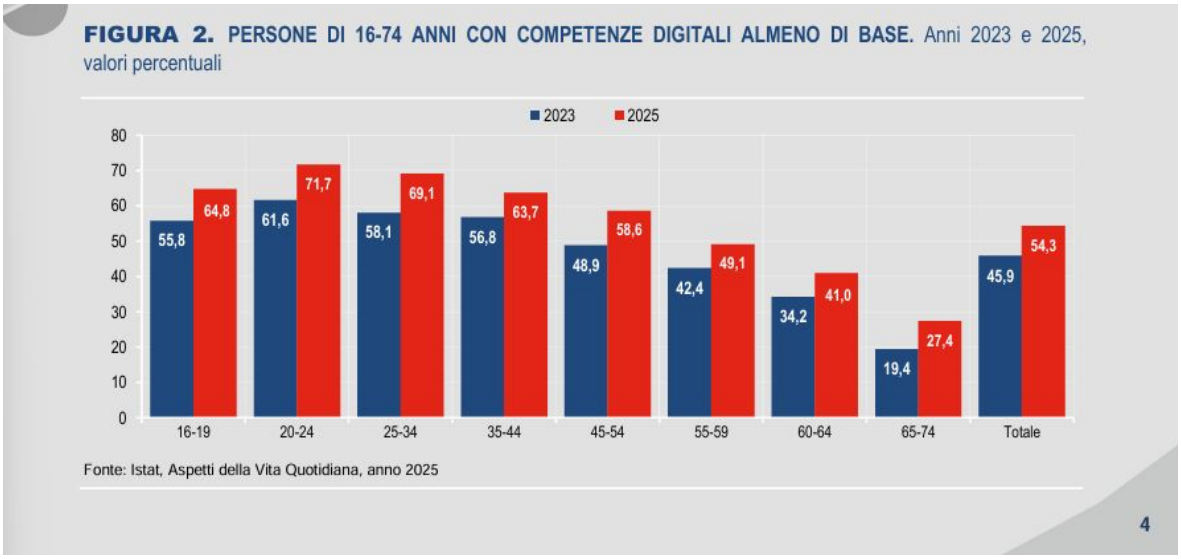


Fig 2 - Bar chart of population and their digital skills (Cittadini e ITC - Anno 2025)

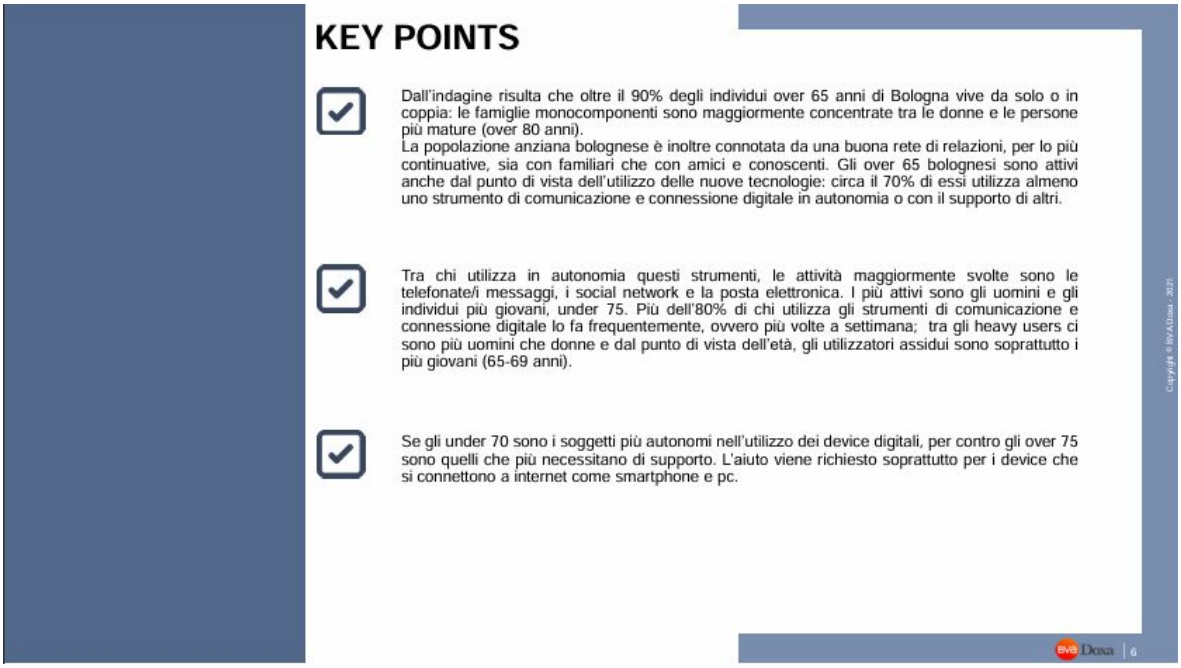


Fig 3. Bva Doxa research report Key Points p.6 (Indagine sulla capacità della popolazione over-65 del Comune di Bologna di utilizzo delle tecnologie digitali - 2021)

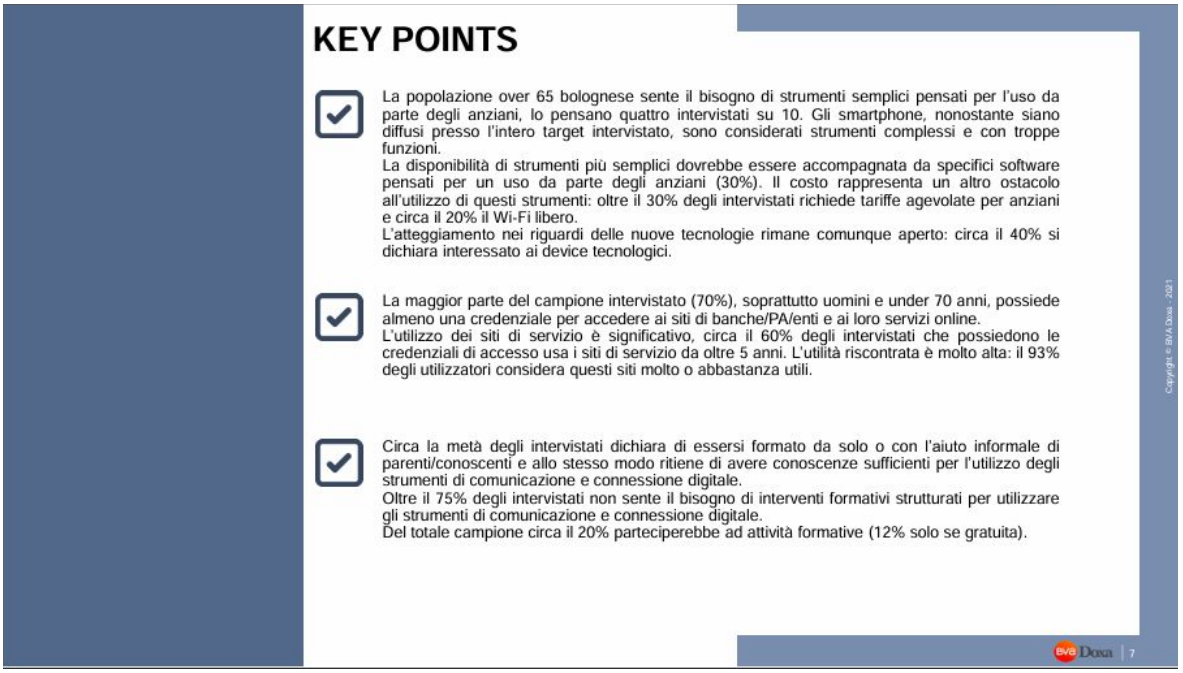


Fig 4. Bva Doxa research report Key Points p.7 (Indagine sulla capacità della popolazione over-65 del Comune di Bologna di utilizzo delle tecnologie digitali - 2021)

FIELD OBSERVATION #1

(Transcription of the user commentary while using the CieID application)
User: Giovanni
Observer: Alice
Task: Registration of the CIE

Observer: Commenta quello che fai.

User: Sto andando avanti... poi non so che cos'è questa tecnologia NFC.

Observer: Ok, ora dimmi cosa faresti.

User: [legge dall'app] *Allora, configura per accedere in modo semplificato*, quindi faccio *accesso semplificato*. Vabbè, metto il codice fiscale, giusto?

Observer: Sì.

User: Password... che password? Non l'avevo inventata?

Observer: No, in questo caso no. Se non hai le credenziali dovresti attivarle.

User: Attivarle ora? Ok.

Observer: Perché se non ti sei mai registrato…

User: Ah no, questo è solo in caso di registrazione. Qui, *dati di contatto*. Ah, codice fiscale.

Observer: [osservazione] Qua siamo stati rimandati sulla pagina web per la registrazione.

User: Ah, *numero di serie della carta d'identità*, giusto?

Observer: Sì.

User: Che dovrebbe essere questo qua? No, questo. Io non ci vedo... [tenta la lettura del CAPTCHA] *7061? 70514... 70514... LM... Livorno, Milano?* Il codice qua. Beh, *1, 8. 1, 8, i* minuscolo. *A* maiuscolo? *4? 4, I* minuscolo, *E* minuscolo. Mi sembra una *Q*. *Q*? E questa è una *U*, giusto?

Observer: Sì.

User: [schermata di richiesta della 1ª e 3ª cifra del PUK] *La prima e la terza cifra. La prima e la terza cifra.* Quindi la prima è *, la terza è una *. In teoria. Ecco qua. Quindi, * e *. Ok. Ok, mi dice... *Ti arriverà il messaggio col codice.* **** giusto?

Observer: Sì.

User: Un numero, metto il **...

Observer: Quello che vuoi, minuscola va bene, sì sì. Allora qua stiamo creando il codice.

User: [legge dall'app] *Crea il nome del dispositivo, ti aiuterai a identificare...* Qua io direi annulla perché... [pop-up] *Attenzione, continuando annullerai il processo di certificazione*

Observer: Allora no, fai la certificazione.

User: Ora che ci metto, "Giovanni"? Allora, *certifica dispositivo*, metto il numero di telefono?

Observer: No.

User: Metto "Giovanni", o se no, ecco, "smartphone di Giovanni". No, metto "Giuseppe".

Observer: Va bene, sì, procedi, ok. Ora prova a chiudere, ok. Facciamo questo: *registra carta*. E per registrarla io sono andata qua, quindi *gestisci carta*.

User: Ok, che devo fare? Registra carta.

Observer: Esatto, *registra carta*.

User: [legge] *Inserisci PIN a 8 cifre*. Quindi il PIN è formato da questa cifra, che è ****. **** più ****, ah proprio è ********. Allora, [legge] *Leggi attentamente le prossime pagine, ti spiegheremo come fare*, ok.

Observer: [l'utente tenta di scorrere in basso] L'utente sta cercando di scorrere in giù, ma in realtà si scorre laterale. Ok, vai, continua, vai... però dovresti leggere così

capisci [riferendosi al tutorial].

User: *...quando mi chiede di appoggiare la carta d'identità elettronica sul retro dello smartphone...*

User: [legge la guida] *La carta è in contatto con il cellulare per tutta l'operazione? Per tutto il tempo? ... La carta è utilizzata per proteggere lo smartphone...* Io provo a rimuovere [la cover del cellulare] *...potrebbero impedire la corretta interazione con la carta... Puoi mettere la carta su una superficie piana*. Ok, ora sono finite le istruzioni, appoggio qua?

Observer: Sì, ok, inizia. Così, sì.

User: O devo proprio appoggiare?

Observer: Appoggiala anche, ora.

User: Appoggiare così... *inizia*.

Observer: [appare il messaggio di errore finale] No vabbè, ok... c'è un problema: non c'è l'NFC, non possiamo registrare la carta.

Market research card

Segment of choice

The Trusted Digital Mentor: digital native, confident while using technology and capable of stepping into the role of a mentor for less experienced individuals when needed.

Reason for choice

This segment represents the typical digital mentor whose help is necessary for most senior citizens approaching a public administration

application. The role of intermediary often requires empathizing with people with different educational backgrounds and digital skills which can be very complex at times.
Intermediary may find themselves completing the task on behalf of the delegator without involving the subject in the process. The reasons are often the learned helplessness and the lack of confidence many seniors experience. This could result in the frustration of the intermediary that often fails to transform the subject into an autonomous user.

User research method employed

- Available statistics
  Sources:

  - **Report Commissione Consiliare Customer Service (2024):** a survey conducted in the Municipality of Milan on the assistance and support system for citizens. The numbers show that the CIE is among the first three reasons leading the citizen to rely on the infoline. In 2023 the total help requests including infoline, “scrivi” and chatbot amounted to one million (although decreasing compared to previous years). Acting on the services related to the CIE such as the CieID would mean decreasing the assistance request, weighing less on infolines.

- One-on-one interview
  Number of interviews

  1

  Topics discussed

  Autonomy of the user and assistance:
  The interview reveals the complete reliance on front-desk assistance and the resignation of older adults who, once they have completed the registration of the card, do not return to the app and completely forget functions and features.

  Waiting times regarding the recovery of the PUK:

We discussed the recovery of the PUK and the 48 hours waiting time and the necessity to schedule a second appointment to the front-desk to complete the task, which both weighs on the operators and on the senior that, not yet independent, has to return to the office.

Complexity of tasks:
During the execution of some task, the interviewer admitted giving up on explaining the process to the user, due to the gap in language and skills, completing the action without involving the senior under a tacit agreement.

Potential improvements of the CieID app:
The interviewer suggested several improvements that could increase the quality of the User Experience such as clearer tutorials, constant guidance, and support numbers at your fingertips to reassure the user.

Interface legibility:
The low contrast between the electric blue of the background and the white of the text and tool's icon makes interaction more complex in the presence of physical issues such as poor eyesight.

Main outcomes

Complexity of the intermediary role:
The intermediary finds themselves in a complex position since they acknowledge the importance of making the user autonomous but at the same time recognize that explaining the task could result in a difficult discussion complicated by the gap in digital skills.

Resignation of the elderly direct user:
The user, however eager to learn, tends to completely delegate the task to a trusted operator, often giving up on the possibility of learning while actively understanding what the intermediary is doing.

Lack of accessible tutorials:
Despite the use of tutorials inside the app, they are not easily accessible. The navigation involuntarily hides them from less experienced navigators such as over 70 and inexperienced citizens.

- **Field observation #2**

  Location of observation

  Home

  Main outcomes

  Technical vocabulary as an obstacle:
  From direct observation, one of the obstacles observed is the technicality of the language. During the authentication the user had to choose between level 1 and 2 or level 3 of security, without having familiarity with these technicalities.
  Excerpt from the field observation: *Ciao Mauro, la tua app è configurata solo per l'accesso semplificato*, livello 1 e 2, non so cosa voglia dire.

  Perceived ambiguity on how to use NFC recognition:
  During the scanning of the card through NFC recognition, the intermediary underlines how the NFC concept can affect users unfamiliar with the technology and the scanning modality, misunderstanding the timing for a correct recognition of the card.

  The hurdle of the 48 hours waiting period:
  The user highlights how the alert of the 48 hours waiting period required for PUK recovery can discourage and annoy the user which will likely wonder how to finish the task in less time and avoid relying for a second time on an intermediary.

  Lack of straight to the point explanation:
  Partially explained information deceives the direct user. This is observable while the app asks the user to insert the PIN of the

CIE. The user will likely misinterpret the instruction searching for the PIN (actually foundable in the sheet of paper handled together with the ID card from the front-desk operator) on the CIE itself before understanding the requirement.

Overall results of the analysis

The statistics and the discussion with the intermediaries provide an overview on the role of external support, which is still essential. Most Italian citizens rely on infolines, front-desk offices, chatbots seeking for help. The result of a research conducted in the city of Milan in 2024 views the CIE and presumably the services related to that as the primary reason for requesting assistance. This is confirmed by the interviewed affirming that her work strongly consists in taking over the task to carry out it on behalf of the senior citizen asking for help. During her working hours she did not manage to transform a citizen facing difficulties into an autonomous user. This is attributed to the learned helplessness of the senior and the desire for fast execution of the intermediary. Most direct users give up on learning even before approaching the app, considering a success the possibility to avoid it completely. The intermediaries manage to help them but rarely to teach them.

Impact on design

Enhancement of User experience:
Simplification and redesign of the application interface, leveraging on learnability, memorability and external consistency in order to encourage the use of the app. Tasks like the registration and the PIN recovery are going to be guided step-by-step to offer guidance to the user and User Interface conventions are going to be followed to ensure the user that the system is friendly and accessible, relying on previously developed mental models.

Redesign to specifically accommodate vulnerable user:
Replace hamburger menu, difficult to intercept in presence of shortsightedness with clear wording of other sections of the application. Express level of security concepts (level 1, level 2 and level 3) rephrasing them to seek transparency.

Include contextual tutorial: integrate in each screen involved in a task a question a direct link to reach the contextual tutorial, offering the user an easy way to immediately access important information fundamental to correctly carry out a task. This will lighten the burden of questions directed at intermediaries transforming helpless users in autonomous subjects.

Report Commissione Consiliare Customer Service (2024): https://www.palazzorealemilano.it/documents/20126/464296132/04062416+Aggiornamento+Customer+Service.pdf/25dc73eb-1c18-28e1-efb4-36831912855e?t=1725279222364

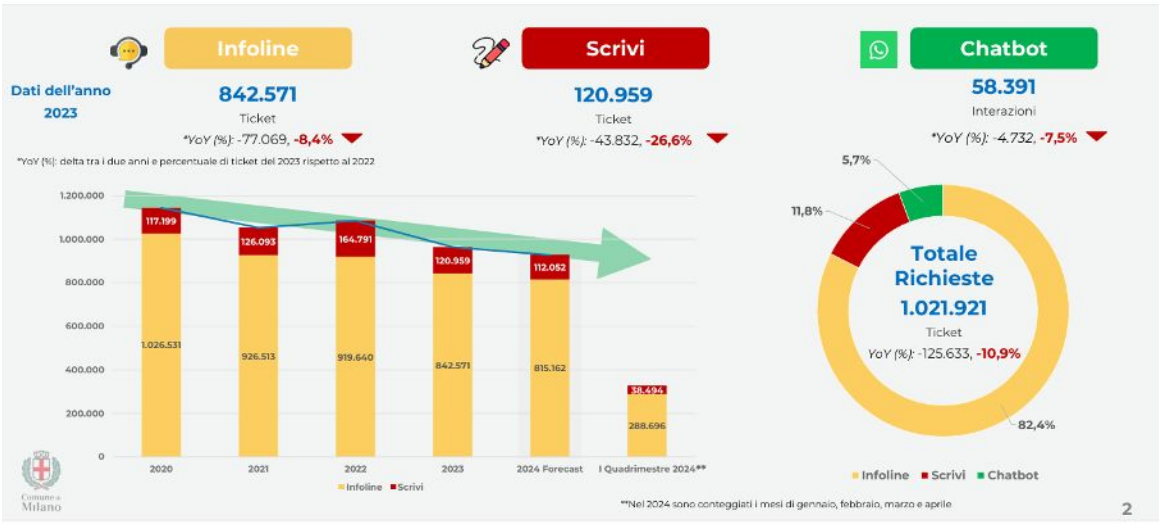


Fig. 5 - Report Commissione Consiliare Customer Service (2024) p.5

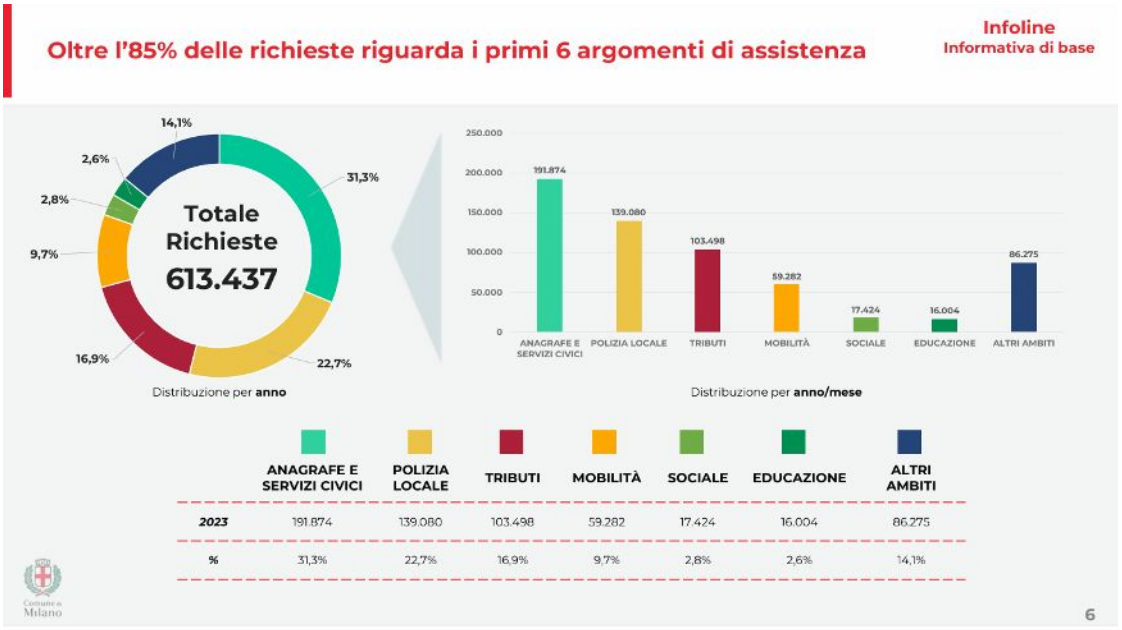


Fig. 6 - Report Commissione Consiliare Customer Service (2024) p.6



Fig. 7 - Report Commissione Consiliare Customer Service (2024) p.7

One-on-one interview

(Transcription of the one-to-one interview with a intermediary)

Intermediary: Camilla

Interviewer 1: Alice

Interviewer 2: Maryam

Interviewer 1: Quando offrivi supporto durante il servizio civile digitale ti è capitato di aiutare qualcuno con l'app CieID?

Speaker: Sì, la proponevo come una soluzione alternativa durante tutto l'anno, perché la maggior parte degli utenti avevano lepida che aveva svariate cose che non funzionavano e molti di questi anziani, mentre io ero lì, gli veniva rinnovata la carta d'identità e gli veniva data la lettera in cui gli si spiegava che potevano fare la CieID e quindi con quell'occasione lì, mi sembrava una buona scusa, molte persone le avevo fatte fare in quell'occasione lì perché le circostanze facevano sì che avessero finalmente una carta d'identità elettronica, avevano lo smartphone perché venivano da noi e avevano appena ricevuto tutti i foglietti che poi sarebbero finiti e quindi sembrava una buona occasione.

Interviewer 1: Cosa si potrebbe implementare per aiutare l'utente a portare a termine i task autonomamente?

Interviewer 2: I task come la registrazione della carta d'identità o il recupero del PIN o del PUK.

Speaker: Ok, quindi la prima parte, l'enroll. Con noi erano completamente seguiti chiaramente perché ricevevano una prima indicazione al comune quando ricevevano la carta d'identità. Poi però già banalmente una delle modalità, di nuovo se non erro perché non è che mi ricordo esattamente tutto, era di dover appoggiare la carta d'identità sul sensore dell'NFC, quindi ovviamente spiegare che cos'è l'NFC, senza tecnicismi però comunque cercavo di solito di spiegare le logiche di quello che facevamo.
Comunque glielo appoggiavamo noi, dicevamo guarda puoi solo appoggiarlo, poi a volte erano in coppia quindi magari lo fa prima uno, poi l'altro lo fa già un po' più autonomamente perché hanno visto sulla prima persona. Quindi sì, quello perché noi facevamo quella modalità lì. Poi noi lavoravamo dal PC perché a mio avviso la parte dell'enroll sul dispositivo mobile è un po' complessa.
Quindi di base io avevo il mio PC con cui facevo tutta la parte di registrazione iniziale con la loro mail, il loro numero di telefono, gli dicevo adesso ti arriverà un messaggio per confermarlo eccetera. E poi finito quello gli spiegavo come utilizzare l'app ma solo una volta attivato tramite il computer. E la cosa che facevamo noi è perché CieID a differenza degli speed, della maggior parte degli speed poi non so tutti, non devi per forza usare l'app, puoi semplicemente ricevere un messaggio e questo per loro era più facile perché comunque in realtà molti utenti avevano un doppio dispositivo, cioè fanno le cose serie a casa con computer e poi venivano a negoziare.

E nel momento dell'enrollment tu puoi scegliere vuoi ricevere la notifica, vuoi ricevere il messaggio e io per tutti loro attivavo direttamente il messaggino che era una cosa un po' più facile anche perché come abbiamo visto anche quando abbiamo fatto il nostro progetto CieID è ben fatta nel senso che quando arriva il messaggino si autocompila praticamente quando schiacci quindi questa cosa era molto comoda. Spiegavo a loro che non era una scelta irrevocabile e lo spiega anche il sito, cioè ti dice poi volendo se tu vuoi avere l'app però è comunque binaria, cioè quindi o chiedi solo l'attivazione tramite l'app, cioè la conferma la seconda...
L'autenticazione. Esatto, il secondo fattore con l'app oppure il secondo fattore tramite messaggio, quindi questa cosa era obbligatoria, bisognava scegliere fra questi due, io ovviamente a loro le attivavo così. Quindi poi di base l'app alcuni di loro finivano per non usarla mai perché entravano nelle varie cose che gli serviva da internet, schiacciavano in automatico, io gli avevo salvato la password su Google che non cambia mai, quindi nome utente password sempre quello, poi ci sono le tre nome utenti diverse che è molto comodo e poi gli arrivava il messaggino, quindi a volte l'app veniva completamente evitata per un'utenza anziana, ovviamente per altri no.

Interviewer 1: Ok, quindi esistono secondo te competitor meno problematici, se sì quali?

Speaker: No, e lo dico per il lavoro che abbiamo fatto e a parte per quello che abbiamo rilevato in quell'anno, cioè io gli facevo quello perché io sapevo che era il migliore, poi sì, ne ho potuti vedere un paio, Amiral, Lepida, Poste, di base erano quelli.

Interviewer 2: Però se SPID non era a pagamento comunque CieID era meglio.

Speaker: Sì, secondo me sì.

Interviewer 1: Domanda extra - Per il recupero di PUK, spesso capita che ti consegnano il PIN e il PIK quando fai la carta d'identità, molti le hanno perse col tempo, è un compito che tu hai svolto per qualcun altro? Sono venuti spesso per recuperare il PUK?

Speaker: Allora, no per quella cosa che avevi detto prima, cioè che spesso era la prima volta che loro ricevono la carta d'identità elettronica, era la prima volta che scoprivano questi codici, quindi venivano a noi dicendo, mi hanno dato questa cosa, cosa posso farne? E quindi ce l'avevano già. Mi pare che forse a qualcuno ho dovuto fare un recupero PIN e PUK, forse addirittura tipo l'ho dovuto fare Marianna per se stessa, cioè una roba tipo tra di noi e no, o forse qualcuno mi ha detto, bisognava prendere un appuntamento?

Non ricordo, se si trattava di prendere un appuntamento senz'altro poi lo prendevamo noi. Cioè nel senso, adesso non ricordo se c'era una piattaforma, una cosa doveva riguardare, però glielo potevamo prendere noi semplicemente, però sì, cioè il fatto che comunque ti richiede di dover tornare di nuovo da qualche parte, riottenere i tuoi codici, o forse no, come funzionava? Ricevi una prima parte via mail, una seconda parte, cioè non lo ricordo.

Interviewer 1: Di base quando l'ho fatto io recentemente facevi tutto online, è un po' complicato secondo me arrivare alla gestione della carta, perché ti dice semplicemente che se non hai il PUK vai in questa parte del sito, che non è linkata da nessuna parte e quindi te la devi cercare. Una volta là ti dice inserisci il PIN, e anche il PIN tu non ce l'hai perché è su carta, e quindi tu devi fare una serie di passaggi per poi mandare una richiesta e ricevere dopo due giorni una mail con il PUK.

Speaker: Allora forse mi ricordo questa cosa perché mi ricordo che c'erano state occasioni in cui gli prendevamo l'appuntamento due giorni dopo, perché poi ha una scadenza quella mail che viene inviata, giusto?

Interviewer 1: 48 ore.

Speaker: E io mi ricordo che gli dovevamo prendere l'appuntamento dopo 48 ore, cioè dovevano tornare. Sarebbe utile, non lo so, esiste l'NFC, sfruttiamolo, cioè se io ti appoggio la mia carta d'identità lì dietro, forse non è abbastanza come sistema di controllo, però potrebbe già essere qualcosa, capito? Perché comunque in fase di registrazione della carta d'identità elettronica, quando c'è proprio la carta d'identità, tu dai la tua mail e il tuo numero di telefono, quindi quei due sono già certificati come tuoi.
Quindi è possibile che comunque i due che sono certificati tuoi, più l'NFC, più la carta d'identità, non si riesce a trovare una soluzione alternativa a questo task che non debba aspettare di ricevere una mail ulteriore tra due giorni che scade entro 24 ore, quindi lo devi fare automaticamente. Non è facilissimo, ecco.

Interviewer 1: Eh no, per quello, soprattutto, a parte la registrazione della carta, che vabbè non è forse automatico il fatto di appoggiare la carta dietro al telefono, però il ritiro del PIN/PUK è una cosa che richiede sia tempo, richiede il fatto che tu vai ad esplorare un po' il sito e trovi, diciamo, questo task in modo autonomo, se non hai gli intermediari, quindi per quello era un'altra…

Speaker: Infatti non ricordo, c'era un video spiegazioni? Cioè c'è un video che spiega come recuperare pin e book sul sito? E come è fatto?
È tipo un video? È facile?

Interviewer 1
Sì, sul sito. Non c'è però nell'applicazione e non appare mai quando la usi. Allora, invece, la quarta domanda è quante volte avete reso un utente autonomo nella gestione dei task rispetto alle volte in cui avete eseguito il task senza la collaborazione dell'utente? In questo caso sia il recupero del PUK che la registrazione della carta, se è mai successo.

Speaker: Ovviamente, cioè, è raro che poi... Allora, nel senso che secondo me con CieID era il migliore, nel senso che era quello per cui una volta fatto la registrazione carta, almeno che era la parte complessa, e una volta capito come funzionava, la password non sarebbe più cambiata, quindi anche se scrivevano il loro biglietto, era sempre lo stesso, ricevevano semplicemente un messaggino, quindi l'app veniva proprio tagliata fuori, era abbastanza facile.
Il problema è che molti di questi utenti, fino a che non avevano CieID, utilizzavano gli speed e l'app del fascicolo sanitario dell'Emilia Romagna non accetta come accesso dall'app, dal sito CieID. Quindi per loro era un po' complesso, cioè ci sono state persone che secondo me, tipo quella che avevamo intervistato noi, io mi ricordo che a Maria lei l'aveva fatto, lei è tornata a lepida per il semplice fatto che era più facile utilizzare l'app del servizio sanitario nel caso del fascicolo sanitario, perché non ha l'accesso, quindi questa è anche una cosa un po' problematica, però sono due cose che non c'è, nel senso finché è lepida che è fatto da lepida e il fascicolo sanitario è fatto da lepida, ovviamente incentivano l'utilizzo di speed, non hanno messo l'opzione con CieID.

Interviewer 1: *Quando svolgete l'operazione al posto loro, qual è il momento esatto, il passaggio specifico nel flusso di CieID in cui pensate che questo è troppo difficile da spiegare e forse prima farlo io?*

Speaker: Sicuramente appoggiare la carta all'NFC, cioè è un acronimo di una cosa che non si conosce, sì, mi sa che c'era una piccola iconcina che faceva vedere, però se tu non hai idea che il cellulare è in grado di leggere una carta, è un po' complesso, quindi senz'altro quello secondo me. Poi la creazione postulare è abbastanza facile, anche il nome utente è facile, perché a parte la tua mail, se non ce l'hai non importa, il tuo codice fiscale è sempre quello e il tuo numero di carte d'identità è sempre quello, quindi a parte il nome utente è meno complesso, però senz'altro l'NFC è proprio un concetto che gli sfugge giustamente, sfugge a me.

Interviewer 1: *Qual è il task per cui viene richiesta la maggiore assistenza?*

Speaker: La registrazione, senz'altro.

Interviewer 1: *Cosa cambieresti in generale dell'applicazione CieID?*

Speaker: Se effettivamente delle cose video sono già state fatte, delle guide, perché non metterle dentro l'app? Oppure dentro l'app mettere almeno un icona a punto di domanda che non rimandi a qualcos'altro, ma che lì dentro ci sia qualcosa, potrebbe essere interessante.

Interviewer 1: *Quali sono le caratteristiche di un'applicazione ideale in questo ambito, quindi cosa manca e quali sarebbero le aspettative?*

Speaker: Che dia la possibilità di parlare con un essere umano e essere assistito da una persona, tra l'altro il telefono, quindi magari rimandare un numero di telefono o qualcosa. Non mi ricordo com'era a livello di colori, perché è sia il blu e il bianco, ma il problema è che il bianco sul blu è troppo sottile ed è bianco sul blu, il contrasto non è adeguato. La cosa che fa più ridere in assoluto è che in una delle guidelines c'è proprio una sezione in cui l'unica cosa che parla di colori è, non ci deve essere blu elettrico con cose bianche e c'è in tutte le cose del governo, perché questa è la guidelines dei colori della UI governativa, quindi secondo me è già questo il complesso.

Interviewer 1: E' più l'aspetto forse di colori, grandezza dei testi.

Speaker: Sì, secondo me dovrebbe essere più grande.

Interviewer 1: E forse posso permettermi di mettere sia numero assistenza che tutorial come prima cosa che vedi.

Speaker: Sì, potrebbe essere benissimo qua in fondo, non capisco perché deve stare qua dentro e qua dentro, c'è nel senso potresti anche metterlo qua il tutorial e l'assistenza.

Interviewer 1: Secondo me la problematica di quest'app è il fatto che le cose sono per una persona non nativa digitale molto nascoste.

Speaker: Sì, sì, sì. Cioè ci devi mettere un po' a trovarle. Ma anche c'è capito che il primo messaggio sia la tua app è configurata per accedere con tutti i livelli di sicurezza, livello 1, 2, 3.
Cioè nel senso, loro non hanno idea di cosa significa livello 1, livello 2, livello 3.

Interviewer 1: Anzi, quando non hai registrato la carta c'è scelto. E là è una cosa che tu dici, ma cosa vuol dire? Cioè anch'io, nativa digitale, mi sono chiesta, ma come dovrei scegliere? Forse è bene che ci sia qualche informazione in più.

Speaker: Sì, almeno non lo so, tipo un guiding alla prima volta che apri l'app.

Interviewer 1: Allora, le ultime domande. *Potreste descrivere, potresti descrivere il profilo delle persone che vi chiedono maggiore assistenza, quali sono le loro caratteristiche principali?*

Speaker: Vabbè, per quello che faccio io, erano anziani. Però, in realtà, per questa cosa qua, di CieID, arrivavano anche dei diciottenni, perché anche loro si stavano rifacendo la carta, perché si rifà a diciottenni, e ricevevano per la prima volta...
E ricevevano per la prima volta la carta d'identità elettronica e anche loro scoprirono di poter attivare questa cosa. Però con loro era più facile. Cioè, mi sa che sono stati solo due casi, mi ricordo i due ragazzi che sono venuti, che conoscevo perché magari i genitori avevano figli più piccoli che andavano a scuola, quindi venivano da noi per altre cose, e visto che sapevano questa cosa, mi dicevano, ah, mio figlio è...
possiamo attivargliela? Sì. Però venivano raramente, perché secondo me...
e venivano persone non italofone, quindi immagino che poi ci fosse un limite dato dalla lingua, perché qua non mi pare che nell'app si possa cambiare la lingua.

Interviewer 1: *Oltre alla scarsa abilità tecnologica, sono emerse altre difficoltà frequentemente? Ad esempio difficoltà linguistiche o problemi visivi?*

Speaker: Secondo me, problemi visivi è quello che abbiamo detto, e linguistiche è quello che abbiamo appena detto, perché mi era capitato dei giovani, ma soltanto giovani non italofoni, oppure di seconda generazione italiana, quindi magari che non hanno un livello italiano altissimo, e loro avevano bisogno di una mano, altri no. Quindi, quello, senz'altro.

Interviewer 2: Ok, va bene.

Interviewer 1: Perfetto, è tutto. Grazie. Tu hai qualcosa da dire?

Interviewer 2
No, no. Ok, se mi dovesse venire in mente qualcosa, adesso lo sapevo.
Grazie, ciao.

FIELD OBSERVATION #2

(Transcription of the user commentary while using the CieID application)
User: Mauro
Observer: Maryam

Task: Recovery of PUK

Sono sull'app

Ciao Mauro, la tua app è configurata solo per l'accesso semplificato, livello 1 e 2, un utente anziano non sa cosa voglia dire. [legge il messaggio]

Accesso semplificato, se accedi da pc inquadra il QR code, non ho un pc, *accedi con lettura a carta, registra la carta per accedere con il massimo livello di sicurezza, livello 3. Inserisci il PIN a 8 cifre, il codice PIN è composto dalle 4 cifre ricevute al momento della presentazione della domanda.*

Quindi per questo mi serve sapere il PIN della carta d'identità che non c'è scritto mi sembra sulla carta d'identità.

Fai vedere se c'è sulla carta d'identità, non penso, non avrebbe senso poiché scrivessero il PIN direttamente sulla carta, *quindi hai smarrito il codice PIN della carta, vai in gestione carta e usa la funzionalità sblocco carta per impostare un nuovo PIN.*

Ok, fammi rileggere, *gestione carta, sblocco carta per impostare un nuovo PIN*, facciamolo, *gestione carta* dall'hamburger menu in alto a sinistra, *nessuna carta d'identità elettronica registrata, sblocca carta*, bottone terziario.

Inserisci il codice PUK, anche questo non ce l'ho, *il codice PUK è composto dalle prime 4 cifre fornite al momento della richiesta della carta. Hai dimenticato il PUK, recuperalo e vai*, bottone text, *ti serviranno carta d'identità elettronica, CIE, numero di cellulare o email comunicati in fase richiesta della CIE.*

Ricorda, puoi effettuare il recupero PUK per una sola carta d'identità elettronica dopo 48 ore dalla richiesta.

Ok, quindi avrò bisogno di 48 ore, ma facciamolo.

[Scansione NFC] *Pronto a scansionare, appoggio la carta sul retro nella parte superiore dello smartphone per effettuare la lettura della carta.*

Errore, hai rimosso la carta troppo presto, va bene rifacciamolo.

Lettura carta terminata, perfetto, *inserisci il numero di serie della CIE* che è questo qui in alto. [inserisce il codice] Ok, C-A-N-O, ok, bom bom, bom bom, bom bom, bom.

Fatto, prosegui, *inserisci il tuo email*, lo mettiamo. [inserisce l'email]

Facciamo next, prosegui, *richiesta inviata*, e mi arriverà per mail. *Hai completato la prima parte della procedura, un avviso su email, dopo 48 ore potrai completare il recupero PUK.*

Ok, faccio *verifica stato richiesta, richiesta inviata, richiesta in lavorazione.* Ok,

vediamo se mi è arrivata per mail. [controlla la mail] No, ancora no. Ancora no.

Assessment of existing resource card

This is the client's tool

Name of tool

CieID

Platforms for tool (devices, operating systems, etc.)

Mobile and web application

Main purpose of tool, main operations allowed, perceived level of adequacy to professional standards in its design and implementation

The tool allows users to register CIEs on the app as a digital identity in order to access public administrations services. Operations allowed consist in the authentication (level 1 and 2 of security) and registration of the card through NFC scanning (level 3 of security). The app conveys professionality through the interface's mere aesthetic but does not seem adequate for novice and vulnerable users lacking digital skills. Both the language and terminology used do not foster accessibility. The app resembles a domain specific system that targets experts rather than an accessible application open to all user segmentations.

Main strengths of tool, reason for choice

- **Free digital identity** connected to the CIE (Carta d'Identità Elettronica) which is the current standard for citizens' identification in Italy.

- Allows access to public administration services.
- **High level of security** through three-factor authentication

Main weaknesses of tool

- Use of **domain specific vocabulary**, preventing accessibility to vulnerable and elderly users.
- Partial or **hidden information** on how to execute tasks
- Recovery of PUK is a **time consuming task**, forcing the user in need of assistance to rely on the intermediary's help at two different stages, 48 hours apart.
- **Unclear information design**, unintuitive to novice users.

First Impressions

The first access on the CieID app features a tutorial in five steps that explains the main functionality of the system. Although necessary, especially to people unfamiliar with the concept of digital identity or authentication levels, the size of the text's chunks and the low contrast between the white background and light-grey color of the texts prevent people suffering from poor eyesight from correctly interiorizing the explanations. Since vision problems are especially diffused through the elderly portion of the population, which is the same segment that would most benefit from a clear tutorial, the app immediately fails to reduce the anxiety and properly welcome novices on a new application.
The problem of the texts being unreadable persists, offering an inadequate pairing of colors (electric blue background and thin white texts) making the reading process complex for elderly people.
The home page shows the two main operations the user can choose but the wording choices (using exclusively terms such as “Athentication”, “level 1 and 2 configuration”) results are very specific and domain related. The initial choice can create confusion in the novice user leading him to seek support from an intermediate able to teach them. The hamburger menu, although its familiar look to most digital native users, inevitably hides all the other app functions, such as “Gestione carta”, “Recupero PUK”, “Tutorial” and “Assistenza”. While approaching the NFC recognition of the card, another brief tutorial is provided but the small dimension of the text can lead the user to ignore it due to

illegibility. While performing a task there are no instructions available. Assistance can only be provided through the abandonment of the task, by navigating to either the “Tutorial” or “Assistenza” sections.
The overall information design can be improved: a contextual navigation can be added in order to access task related content in one click rather than abandoning the current task in execution to further navigate an already improvable system.

Direct Analysis

List of outstanding issues

- **Crucial options hidden:** Fundamental features such as "Recupero PUK" (PUK recovery), "Gestione carta" (Card management), tutorials, and assistance contacts are not immediately visible on the Home Page, but are instead hidden inside a hamburger menu.
- **Lack of introductory explanations:** The Home Page immediately presents the user with complex authentication options without providing any upfront guidance or context, generating confusion for first-time users.
- **Missing navigation feedback:** The system does not provide a site map or visual feedback to show the user where they currently are within the task flow, limiting their sense of control.
- **Unexplained jargon:** Complex concepts such as "authentication levels" or "NFC reading" are not translated into simple language that is accessible to novices.
- **Absence of contextual help:** The app does not adapt its level of explanation based on the user's experience. During critical operations (such as adding the ID card or recovering the PIN/PUK), there are no contextual on-screen instructions available. The user is forced to abandon the task and return to the Home page to seek help in the general menus.
- **No progression feedback:** In multi-step tasks, the progress is not shown to the user, which increases uncertainty.
- **Visual accessibility issues:** Overall legibility is severely compromised by the use of texts with a very thin weight

overlapping a saturated electric blue background, creating insufficient contrast for elderly users or those with poor eyesight.

- **Command ambiguity (Affordance):** Non-clickable visual elements present characteristics that suggest interaction. For example, the text "Gestione carta" inside the "Inserisci il PIN" section looks like a link that could redirect the user, but it is not interactive.
- **Unclear iconography:** The hamburger menu icon is drawn with lines that are too thin, making it easily misunderstood or effectively invisible to less experienced users.
- **Misleading format indications:** The data entry fields do not proactively suggest the correct format. For example, in the "Crea il tuo codice app id" section, the visual fields suggest the insertion of a purely numeric code, while the system actually requires at least one letter as well.
- **Poorly legible labels:** The text labels that explain what data needs to be entered in the forms are displayed in a font size that is too small.

Direct Analysis (CielD)

Summary of results

	Raw score	# Questions	# Answers	Score
Home Page	1	20	8	56%
Task Orientation	3	44	16	59%
Navigation & IA	5	29	12	71%
Forms & Data Entry	7	23	9	89%
Trust & Credibility	7	13	8	94%
Writing & Content Quality	9	23	19	74%
Page Layout & Visual Design	10	38	24	71%
Search	-1	20	2	25%
Help, Feedback & Error Tolerance	-3	37	3	0%
Overall score		247	101	60%

Summary of result

Checkpoint		Comments
The items on the home page are clearly focused on users' key tasks ("featuritis" has been avoided)	1	
The home page contains a search input box	-1	There in no search box input in the app
Product categories are provided and clearly visible on the homepage		
Useful content is presented on the home page or within one click of the home page		
The home page shows good examples of real site content		
Links on the home page begin with the most important keyword (e.g. "Sun holidays" not "Holidays in the sun")		
There is a short list of items recently featured on the homepage, supplemented with a link to archival content		
Navigation areas on the home page are not over-formatted and users will not mistake them for adverts		
The value proposition is clearly stated on the home page (e.g. with a tagline or welcome blurb)		
The home page contains meaningful graphics, not clip art or pictures of models	1	
Navigation choices are ordered in the most logical or task-oriented manner (with the less important corporate information at the bottom)	0	Many important information re hidden in the hamburger menu and are not available immediately in the homepage
The title of the home page will provide good visibility in search engines like Google		
All corporate information is grouped in one distinct area (e.g. "About Us")	1	
Users will understand the value proposition		
By just looking at the home page, the first time user will understand where to start	-1	Homepage shows two options of authentication without further explanation
The home page shows all the major options	0	It does not show the "Recupero PUK" and "Gestione carta"
The home page of the site has a memorable URL		
The home page is professionally designed and will create a positive first impression		
The design of the home page will encourage people to explore the site	0	
The home page looks like a home page; pages lower in the site will not be confused with it		

Fig 9 - Homepage

Checkpoint		Comments
The site is free from irrelevant, unnecessary and distracting information	1	
Excessive use of scripts, applets, movies, audio files, graphics and images has been avoided	1	
The site avoids unnecessary registration	1	
The critical path (e.g. purchase, subscription) is clear, with no distractions on route		
Information is presented in a simple, natural and logical order	-1	Concepts such as level of authentication and NFC reading should be disambiguated through a simpler explanation
The number of screens required per task has been minimised		
The site requires minimal scrolling and clicking		
The site correctly anticipates and prompts for the user's probable next activity	0	
When graphs are shown, users have access to the actual data (e.g. numeric annotation on bar charts)		
Activities allocated to the user or the computer take full advantage of the strengths of each (look for actions that can be done automatically by the site, e.g. postcode lookup)		
Users can complete common tasks quickly	-1	During the authentication through NFC reading two bugs occured.
Items can be compared easily when this is necessary for the task (e.g. product comparisons)		
The task sequence parallels the user's work processes		
The site makes the user's work easier and quicker than without the system	0	Not the PUK recovery.
The most important and frequently used topics, features and functions are close to the centre of the page, not in the far left or right margins		
The user does not need to enter the same information more than once		
Important, frequently needed topics and tasks are close to the 'surface' of the web site		
Typing (e.g. during purchase) is kept to an absolute minimum, with accelerators ("one-click") for return users		
The path for any given task is a reasonable length (2-5 clicks)	1	
When there are multiple steps in a task, the site displays all the steps that need to be completed and provides feedback on the user's current position in the workflow	0	It does not provide feedback on the position for any task.
Price is always clearly displayed next to any product		

Task Orientation pt.1

The site's privacy policy is easy to find, especially on pages that ask for personal information, and the policy is simple and clear		
Users of the site do not need to remember information from place to place		
The use of metaphors is easily understandable by the typical user		
Data formats follow appropriate cultural conventions (e.g. miles for UK)		
Details of the software's internal workings are not exposed to the user	1	
The site caters for users with little prior experience of the web	-1	The app looks confusing for someone who's got no experience
The site makes it easy for users to explore the site and try out different options before committing themselves		
A typical first-time visitor can do the most common tasks without assistance	-1	Could be hard to perform a task independently for an elder user
When they return to the site, users will remember how to carry out the key tasks		
The functionality of novel device controls is obvious		
On the basket page, there is a highly visible 'Proceed to checkout' button at the top and bottom of the page		
Important calls to action, like 'Add to basket', are highly visible		
Action buttons (such as "Submit") are always invoked by the user, not automatically invoked by the system when the last field is completed	1	
Command and action items are presented as buttons (not, for example, as hypertext links)		
If the user is half-way through a transaction and quits, the user can later return to the site and continue from where he left off		
When a page presents a lot of information, the user can sort and filter the information		
If there is an image on a button or icon, it is relevant to the task	1	
The site prompts the user before automatically logging off the user, and the time out is appropriate		
Unwanted features (e.g. Flash animations) can be stopped or skipped	1	
The site is robust and all the key features work (i.e. there are no javascript exceptions, CGI errors or broken links)		
The site supports novice and expert users by providing different levels of explanation (e.g. in help and error messages)	-1	The app does not adapt to user's experience level
The site allows users to rename objects and actions in the interface (e.g. naming delivery addresses or accounts)		
The site allows the user to customise operational time parameters (e.g. time until automatic logout)		

Task Orientation pt.2

Checkpoint		Comments
There is a convenient and obvious way to move between related pages and sections and it is easy to return to the home page	0	
The information that users are most likely to need is easy to navigate to from most pages	-1	During PIN and PUK recovery, you have to got to the menu on the home page, so you have to go back on the home page.
Navigation choices are ordered in the most logical or task-oriented manner	-1	The navigation is not task oriented, because in the process of adding card, you may have to use "Gestione carta" back in the menu but the app does not offer a shortcut.
The navigation system is broad and shallow (many items on a menu) rather than deep (many menu levels)	1	
The site structure is simple, with a clear conceptual model and no unnecessary levels	0	The definitions of level 1, 2 and 3 of authentication aren't comprehensible enough for elderly users
The major sections of the site are available from every page (persistent navigation) and there are no dead ends	-1	
Navigation tabs are located at the top of the page, and look like clickable versions of real-world tabs		
There is a site map that provides an overview of the site's content		
The site map is linked to from every page		
The site map provides a concise overview of the site, not a rehash of the main navigation or a list of every single topic		
Good navigational feedback is provided (e.g. showing where you are in the site)		
Category labels accurately describe the information in the category	0	
Links and navigation labels contain the "trigger words" that users will look for to achieve their goal	1	
Terminology and conventions (such as link colours) are (approximately) consistent with general web usage		
Links look the same in the different sections of the site		
Product pages contain links to similar and complementary products to support cross-selling		
The terms used for navigation items and hypertext links are unambiguous and jargon-free		
Users can sort and filter catalogue pages (e.g. by listing in price order, or showing 'most popular')		
There is a visible change when the mouse points at something clickable (excluding cursor changes)		
Important content can be accessed from more than one link (different users may require different link labels)		
Navigation-only pages (such as the home page) can be viewed without scrolling	1	

Navigation and AI pt.1

Hypertext links that invoke actions (e.g downloads, new windows) are clearly distinguished from hypertext links that load another page		
The site allows the user to control the pace and sequence of the interaction		
There are clearly marked exits on every page allowing the user to bale out of the current task without having to go through an extended dialog	0	Yes but they are not consistent
The site does not disable the browser's "Back" button and the "Back" button appears on the browser toolbar on every page		
Clicking the back button always takes the user back to the page the user came from	1	
A link to both the basket and checkout is clearly visible on every page		
If the site spawns new windows, these will not confuse the user (e.g. they are dialog-box sized and can be easily closed)		
Menu instructions, prompts and messages appear on the same place on each screen		

Navigation and AI pt.2

Checkpoint		Comments
Fields in data entry screens contain default values when appropriate and show the structure of the data and the field length	1	
When a task involves source documents (such as a paper form), the interface is compatible with the characteristics of the source document		
The site automatically enters field formatting data (e.g. currency symbols, commas for 1000s, trailing or leading spaces). Users do not need to enter characters like £ or %.		
Field labels on forms clearly explain what entries are desired	0	Text that explain that are too small
Text boxes on forms are the right length for the expected answer	1	
There is a clear distinction between "required" and "optional" fields on forms		
The same form is used for both logging in and registering (i.e. it's like Amazon)	1	
Forms pre-warn the user if external information is needed for completion (e.g. a passport number)	1	
Questions on forms are grouped logically, and each group has a heading		
Fields on forms contain hints, examples or model answers to demonstrate the expected input		
When field labels on forms take the form of questions, the questions are stated in clear, simple language		
Pull-down menus, radio buttons and check boxes are used in preference to text entry fields on forms (i.e. text entry fields are not overused)	1	
With data entry screens, the cursor is placed where the input is needed		
Data formats are clearly indicated for input (e.g. dates) and output (e.g. units of values).		
Users can complete simple tasks by entering just essential information (with the system supplying the non-essential information by default)	1	
Forms allow users to stay with a single interaction method for as long as possible (i.e. users do not need to make numerous shifts from keyboard to mouse to keyboard).		
The user can change default values in form fields		
Text entry fields indicate the amount and the format of data that needs to be entered	0	In "Crea il tuo codice app id" the entries look all the same, suggesting the inserting of a numeric only code while at least a letter is also required.
Forms are validated before the form is submitted		
With data entry screens, the site carries out field-level checking and form-level checking at the appropriate time		
The site makes it easy to correct errors (e.g. when a form is incomplete, positioning the cursor at the location where correction is required)		
There is consistency between data entry and data display	1	
Labels are close to the data entry fields (e.g. labels are right justified)		

Forms and data entry

Checkpoint		Comments
The content is up-to-date, authoritative and trustworthy	1	
The site contains third-party support (e.g. citations, testimonials) to verify the accuracy of information.		
It is clear that there is a real organisation behind the site (e.g. there is a physical address or a photo of the office)	1	
The company comprises acknowledged experts (look for credentials)	1	
The site avoids advertisements, especially pop-ups.	1	
Delivery costs are highlighted at the very beginning of checkout		
The site avoids marketing waffle		
Each page is clearly branded so that the user knows he is still in the same site	1	
It is easy to contact someone for assistance and a reply is received quickly	0	Contacts are hidden under the hamburger menu
The content is fresh: it is updated frequently and the site includes recent content		
The site is free of typographic errors and spelling mistakes	1	
The visual design complements the brand and any offline marketing messages	1	
There are real people behind the organisation and they are honest and trustworthy (look for bios)		

Trust and credibility

Checkpoint		Comments
The site has compelling and unique content	1	
Text is concise, with no needless instructions or welcome notes	1	
Each content page begins with conclsuions or implications and the text is written with an inverted pyramid style	1	
Pages use bulleted and numbered lists in preference to narrative text	1	
Lists are prefaced with a concise introduction (e.g. a word or phrase), helping users appreciate how the items are related to one another		
The most important items in a list are placed at the top		
Information is organised hierarchically, from the general to the specific, and the organisation is clear and logical	0	
Content has been specifically created for the web (web pages do not comprise repurposed material from print publications such as brochures)	1	
Product pages contain the detail necessary to make a purchase, and users can zoom in on product images		
Hypertext has been appropriately used to structure content	0	
Sentences are written in the active voice	1	
Pages are quick to scan, with ample headings and sub-headings and short paragraphs	-1	The legibility is compromised by thin texts and low constrast with the bacground
The site uses maps, diagrams, graphs, flow charts and other visuals in preference to wordy blocks of text	1	
Each page is clearly labelled with a descriptive and useful title that makes sense as a bookmark	1	
Links and link titles are descriptive and predictive, and there are no "Click here!" links		
The site avoids cute, clever, or cryptic headings	0	It uses domain specific vocabulary in some occasions
Link names match the title of destination pages, so users will know when they have reached the intended page	1	
Button labels and link labels start with action words	1	
Headings and sub-headings are short, straightforward and descriptive	0	
The words, phrases and concepts used will be familiar to the typical user	-1	They can be complex for novices
Numbered lists start at "1" not at "0"	1	
Acronyms and abbreviations are defined when first used	-1	
Text links are long enough to be understood, but short enough to minimise wrapping (especially when used as a navigation list)	1	

Writing and content quality

Checkpoint		Comments
The screen density is appropriate for the target users and their tasks	0	
The layout helps focus attention on what to do next	1	
On all pages, the most important information (such as frequently used topics, features and functions) is presented on the first screenful of information ("above the fold")		
The site can be used without scrolling horizontally	1	
Things that are clickable (like buttons) are obviously pressable	1	
Items that aren't clickable do not have characteristics that suggest that they are	0	"Gestione carta" inside "Inserisci il PIN" section looks like it could be clicked to redirect the user to the correct page.
The functionality of buttons and controls is obvious from their labels or from their design	0	
Clickable images include redundant text labels (i.e. there is no 'mystery meat' navigation)		
Hypertext links are easy to identify without needing to 'minesweep' (e.g. underlined)	-1	Links are all underlined
Fonts are used consistently	1	
The relationship between controls and their actions is obvious	0	
Icons and graphics are standard and/or intuitive (concrete and familiar)	0	The hamburger menu icon is very thin, could be ambiguous to non experts
There is a clear visual "starting point" to every page		
Each page on the site shares a consistent layout	1	
Pages on the site are formatted for printing, or there is a printer-friendly version		
Buttons and links show that they have been clicked	-1	
GUI components (like radio buttons and check boxes) are used appropriately	0	

Page layout and visual design pt. 1

Fonts are readable	0	Fonts are, the dimension is the problem
The site avoids italicised text and uses underlining only for hypertext links	1	
There is a good balance between information density and use of white space	0	
The site is pleasant to look at	1	
Pages are free of "scroll stoppers" (headings or page elements that create the illusion that users have reached the top or bottom of a page when they have not)		
The site avoids extensive use of upper case text	1	
The site has a consistent, clearly recognisable look and feel that will engage users	1	
Saturated blue is avoided for fine detail (e.g. text, thin lines and symbols)	-1	It is used in many pages
Colour is used to structure and group items on the page	1	
Graphics will not be confused with banner ads		
Emboldening is used to emphasise important topic categories	1	
On content pages, line lengths are neither too short (<50 characters per line) nor too long (>100 characters per line) when viewed in a standard browser width window	1	
Pages have been designed to an underlying grid, with items and widgets aligned both horizontally and vertically	1	
Meaningful labels, effective background colours and appropriate use of borders and white space help users identify a set of items as a discrete functional block		
The colours work well together and complicated backgrounds are avoided		
Individual pages are free of clutter and irrelevant information		
Standard elements (such as page titles, site navigation, page navigation, privacy policy etc.) are easy to locate		
The organisation's logo is placed in the same location on every page, and clicking the logo returns the user to the most logical page (e.g. the home page)		
Attention-attracting features (such as animation, bold colours and size differentials) are used sparingly and only where relevant		
Icons are visually and conceptually distinct yet still harmonious (clearly part of the same family)		
Related information and functions are clustered together, and each group can be scanned in a single fixation (5-deg. about 4.4cm diam circle on screen)		

Page layout and visual design pt.2

Checkpoint		Comments
The default search is intuitive to configure (no Boolean operators)		
The search results page shows the user what was searched for and it is easy to edit and resubmit the search		
Search results are clear, useful and ranked by relevance		
The search results page makes it clear how many results were retrieved, and the number of results per page can be configured by the user		
If no results are returned, the system offers ideas or options for improving the query based on identifiable problems with the user's input		
The search engine handles empty queries gracefully		
The most common queries (as reflected in the site log) produce useful results		
The search engine includes templates, examples or hints on how to use it effectively		
The site includes a more powerful search interface available to help users refine their searches (preferably named "revise search" or "refine search", not "advanced search)	-1	There are no search input in the app
The search results page does not show duplicate results (either perceived duplicates or actual duplicates)		
The search box is long enough to handle common query lengths		
Searches cover the entire web site, not a portion of it		
If the site allows users to set up a complex search, these searches can be saved and executed on a regular basis (so users can keep up-to-date with dynamic content)		
The search interface is located where users will expect to find it (top right of page)		
The search box and its controls are clearly labelled (multiple search boxes can be confusing)		
The site supports people who want to browse and people who want to search	0	The app allowes only to browse the system, not to search
The scope of the search is made explicit on the search results page and users can restrict the scope (if relevant to the task)		
The search results page displays useful meta-information, such as the size of the document, the date that the document was created and the file type (Word, pdf etc.)		
The search engine provides automatic spell checking and looks for plurals and synonyms		
The search engine provides an option for similarity search ("more like this")		

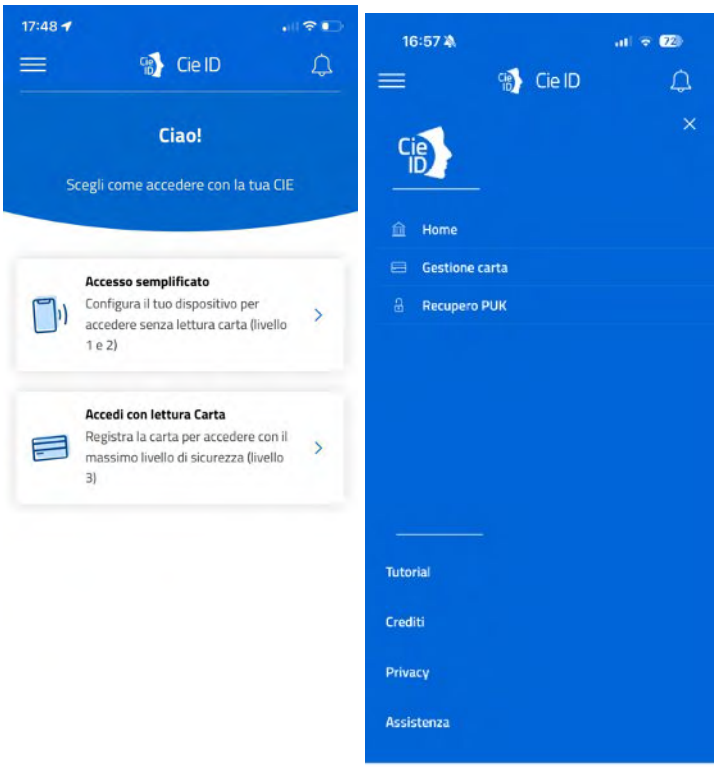
Search

Checkpoint		Comments
The FAQ or on-line help provides step-by-step instructions to help users carry out the most important tasks		
It is easy to get help in the right form and at the right time	-1	When adding ID card or in "recupero PUK and PIN", there aren't any help options. You have to go back to the home page, but even there it's not enough.
Prompts are brief and unambiguous	-1	prompts regarding recupero, are ambiguous.
The user does not need to consult user manuals or other external information to use the site		
The site uses a customised 404 page, which includes tips on how to find the missing page and links to "Home" and Search		
The site provides good feedback (e.g. progress indicators or messages) when needed (e.g. during checkout)		
Users are given help in choosing products		
User confirmation is required before carrying out potentially "dangerous" actions (e.g. deleting something)		
Confirmation pages are clear		
Error messages contain clear instructions on what to do next		
Immediately prior to commiting to the purchase, the site shows the user a clear summary page and this will not be confused with a purchase confirmation page		
When the user needs to choose between different options (such as in a dialog box), the options are obvious		
The site keeps users informed about unavoidable delays in the site's response time (e.g. when authorising a credit card transaction)		
Error messages are written in a non-derisory tone and do not blame the user for the error		
Pages load quickly (5 seconds or less)		
The site provides immediate feedback on user input or actions		
The user is warned about large, slow-loading pages (e.g. "Please wait…"), and the most important information appears first		
Where tooltips are used, they provide useful additional help and do not simply duplicate text in the icon, link or field label		
When giving instructions, pages tell users what to do rather than what to avoid doing		
The site shows users how to do common tasks where appropriate (e.g. with demonstrations of the site's functionality)	-1	The explanations and help sections aren't easily available, they are in the menu part rather than in the task section when it's needed.
The site provides feedback (e.g. "Did you know?") that helps the user learn how to use the site		

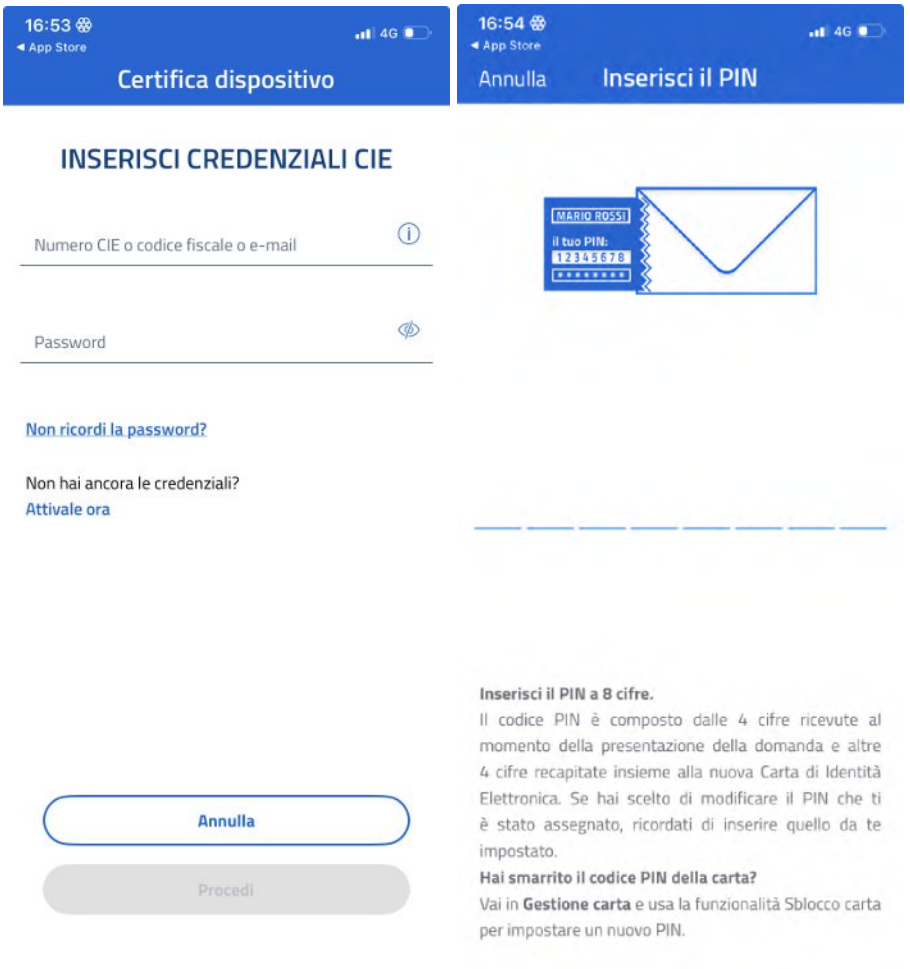
Fig 20 - Help, feedback, error tolerance pt.1

Help is clear and direct and simply expressed in plain English, free from jargon and buzzwords		
The site provides clear feedback when a task has been completed successfully		
Important instructions remain on the screen while needed, and there are no hasty time outs requiring the user to write down information		
Fitts' Law is followed (the distance between controls and the size of the controls is appropriate, with size proportional to distance)		
There is sufficient space between targets to prevent the user from hitting multiple or incorrect targets		
There is a line space of at least 2 pixels between clickable items		
The site makes it obvious when and where an error has occurred (e.g. when a form is incomplete, highlighting the missing fields)		
The site uses appropriate selection methods (e.g. pull-down menus) as an alternative to typing		
The site does a good job of preventing the user from making errors		
The site prompts the user before correcting erroneous input (e.g. Google's "Did you mean…")		
The site ensures that work is not lost (either by the user or site error)		
Error messages are written in plain language with sufficient explanation of the problem		
When relevant, the user can defer fixing errors until later in the task		
The site can provide more detail about error messages if required		
It is easy to "undo" (or "cancel") and "redo" actions		

Help, feedback, error tolerance pt.2



The home page and the hamburger menu hiding all the essential features inside the application



The confusing pin and password part

Reverse Analysis

List of outstanding issues

- **Assistance contacts** are only provided in the required section despite the frequent need for help while performing tasks.
- The home page does not provide a clear overview of all the possible action and task performable through the app.
- The home page presents the non expert user with a crossroads: choose between **two authentication modes** the elderly user is probably not familiar with, increasing insecurity and digital anxiety.
- **NFC scanning**, also known as the most complicated task for elderly users, is required three times during the PIN recovery.
- The app does not facilitate the novice user's experience and interaction.

- The app does not offer the possibility of **adapting it to the user's level of experience**

Reverse Analysis (CielD)

Summary of results

	Raw score	# Questions	# Answers	Score
Home Page	2	20	16	56%
Task Orientation	5	44	22	61%
Navigation & IA	3	29	13	62%
Forms & Data Entry	7	23	21	67%
Trust & Credibility	6	13	9	83%
Writing & Content Quality	-1	23	15	47%
Page Layout & Visual Design	14	38	26	77%
Search	-1	20	2	25%
Help, Feedback & Error Tolerance	13	37	30	72%
Overall score		247	154	61%

Summary of result

Checkpoint		Comments
The items on the home page are clearly focused on users' key tasks ("featuritis" has been avoided)	0	Some important tasks such as "Recupero PIN/PUK" are not featured in the home page.
The home page contains a search input box	-1	There are no search input box in the application
Product categories are provided and clearly visible on the homepage		
Useful content is presented on the home page or within one click of the home page	1	
The home page shows good examples of real site content	0	Many of the options and functions are hidden in the menu
Links on the home page begin with the most important keyword (e.g. "Sun holidays" not "Holidays in the sun")	1	
There is a short list of items recently featured on the homepage, supplemented with a link to archival content	-1	
Navigation areas on the home page are not over-formatted and users will not mistake them for adverts	1	
The value proposition is clearly stated on the home page (e.g. with a tagline or welcome blurb)	0	
The home page contains meaningful graphics, not clip art or pictures of models	1	
Navigation choices are ordered in the most logical or task-oriented manner (with the less important corporate information at the bottom)	0	
The title of the home page will provide good visibility in search engines like Google	1	
All corporate information is grouped in one distinct area (e.g. "About Us")	1	
Users will understand the value proposition		
By just looking at the home page, the first time user will understand where to start	-1	
The home page shows all the major options	0	
The home page of the site has a memorable URL		
The home page is professionally designed and will create a positive first impression	0	
The design of the home page will encourage people to explore the site	-1	
The home page looks like a home page; pages lower in the site will not be confused with it		

Homepage

Checkpoint		Comments
The site is free from irrelevant, unnecessary and distracting information	1	
Excessive use of scripts, applets, movies, audio files, graphics and images has been avoided	1	
The site avoids unnecessary registration	1	
The critical path (e.g. purchase, subscription) is clear, with no distractions on route		
Information is presented in a simple, natural and logical order	0	
The number of screens required per task has been minimised		
The site requires minimal scrolling and clicking	0	
The site correctly anticipates and prompts for the user's probable next activity	0	
When graphs are shown, users have access to the actual data (e.g. numeric annotation on bar charts)		
Activities allocated to the user or the computer take full advantage of the strengths of each (look for actions that can be done automatically by the site, e.g. postcode lookup)		
Users can complete common tasks quickly	0	The app does not simplify the tasks for vulnerable users, therefore the completion of the task is time-consuming
Items can be compared easily when this is necessary for the task (e.g. product comparisons)		
The task sequence parallels the user's work processes		
The site makes the user's work easier and quicker than without the system		
The most important and frequently used topics, features and functions are close to the centre of the page, not in the far left or right margins	-1	They are hidden in the hamburger menu on the left.
The user does not need to enter the same information more than once	-1	During the PUK recovery the user must scan the CIE through NFC technology three times
Important, frequently needed topics and tasks are close to the 'surface' of the web site	0	
Typing (e.g. during purchase) is kept to an absolute minimum, with accelerators ("one-click") for return users		
The path for any given task is a reasonable length (2-5 clicks)	1	
When there are multiple steps in a task, the site displays all the steps that need to be completed and provides feedback on the user's current position in the workflow	0	Not for every multi-action task
Price is always clearly displayed next to any product		

Task Orientation pt.1

The site's privacy policy is easy to find, especially on pages that ask for personal information, and the policy is simple and clear	0	Hidden inside the menu
Users of the site do not need to remember information from place to place	1	
The use of metaphors is easily understandable by the typical user		
Data formats follow appropriate cultural conventions (e.g. miles for UK)		
Details of the software's internal workings are not exposed to the user		
The site caters for users with little prior experience of the web	-1	Display of complex concept on the home page
The site makes it easy for users to explore the site and try out different options before committing themselves	0	The app is not easy to navigate without prior experience with digital tools
A typical first-time visitor can do the most common tasks without assistance	0	Strongly depends on digital skill levels
When they return to the site, users will remember how to carry out the key tasks		
The functionality of novel device controls is obvious		
On the basket page, there is a highly visible 'Proceed to checkout' button at the top and bottom of the page		
Important calls to action, like 'Add to basket', are highly visible		
Action buttons (such as "Submit") are always invoked by the user, not automatically invoked by the system when the last field is completed	1	
Command and action items are presented as buttons (not, for example, as hypertext links)	1	
If the user is half-way through a transaction and quits, the user can later return to the site and continue from where he left off		
When a page presents a lot of information, the user can sort and filter the information		
If there is an image on a button or icon, it is relevant to the task	1	
The site prompts the user before automatically logging off the user, and the time out is appropriate		
Unwanted features (e.g. Flash animations) can be stopped or skipped		
The site is robust and all the key features work (i.e. there are no javascript exceptions, CGI errors or broken links)	1	
The site supports novice and expert users by providing different levels of explanation (e.g. in help and error messages)	-1	The site does not adapt to novice and vulnerable users
The site allows users to rename objects and actions in the interface (e.g. naming delivery addresses or accounts)		
The site allows the user to customise operational time parameters (e.g. time until automatic logout)		

Task Orientation pt.2

Checkpoint		Comments
There is a convenient and obvious way to move between related pages and sections and it is easy to return to the home page	0	
The information that users are most likely to need is easy to navigate to from most pages	-1	During PIN and PUK recovery, you have to got to the menu on the home page, so you have to go back on the home page.
Navigation choices are ordered in the most logical or task-oriented manner	-1	The navigation is not task oriented, because in the process of adding card, you may have to use "Gestione carta" back in the menu but the app does not offer a shortcut.
The navigation system is broad and shallow (many items on a menu) rather than deep (many menu levels)	1	
The site structure is simple, with a clear conceptual model and no unnecessary levels	0	The definitions of level 1, 2 and 3 of authentication aren't comprehensible enough for elderly users
The major sections of the site are available from every page (persistent navigation) and there are no dead ends	-1	
Navigation tabs are located at the top of the page, and look like clickable versions of real-world tabs		
There is a site map that provides an overview of the site's content		
The site map is linked to from every page		
The site map provides a concise overview of the site, not a rehash of the main navigation or a list of every single topic		
Good navigational feedback is provided (e.g. showing where you are in the site)		
Category labels accurately describe the information in the category	0	
Links and navigation labels contain the "trigger words" that users will look for to achieve their goal	1	
Terminology and conventions (such as link colours) are (approximately) consistent with general web usage		
Links look the same in the different sections of the site		
Product pages contain links to similar and complementary products to support cross-selling		
The terms used for navigation items and hypertext links are unambiguous and jargon-free		
Users can sort and filter catalogue pages (e.g. by listing in price order, or showing 'most popular')		
There is a visible change when the mouse points at something clickable (excluding cursor changes)		
Important content can be accessed from more than one link (different users may require different link labels)		
Navigation-only pages (such as the home page) can be viewed without scrolling	1	

Navigation and AI pt.1

Navigation-only pages (such as the home page) can be viewed without scrolling	1	
Hypertext links that invoke actions (e.g downloads, new windows) are clearly distinguished from hypertext links that load another page	1	
The site allows the user to control the pace and sequence of the interaction	1	
There are clearly marked exits on every page allowing the user to bale out of the current task without having to go through an extended dialog	0	
The site does not disable the browser's "Back" button and the "Back" button appears on the browser toolbar on every page		
Clicking the back button always takes the user back to the page the user came from		
A link to both the basket and checkout is clearly visible on every page		
If the site spawns new windows, these will not confuse the user (e.g. they are dialog-box sized and can be easily closed)		
Menu instructions, prompts and messages appear on the same place on each screen	1	

Navigation and AI pt.2

Checkpoint		Comments
Fields in data entry screens contain default values when appropriate and show the structure of the data and the field length	-1	it's field with instructions, not default values
When a task involves source documents (such as a paper form), the interface is compatible with the characteristics of the source document	1	
The site automatically enters field formatting data (e.g. currency symbols, commas for 1000s, trailing or leading spaces). Users do not need to enter characters like £ or %	-1	there are no restrictions on the field formatting data.
Field labels on forms clearly explain what entries are desired	1	
Text boxes on forms are the right length for the expected answer	1	
There is a clear distinction between "required" and "optional" fields on forms	0	
The same form is used for both logging in and registering (i.e. it's like Amazon)	1	
Forms pre-warn the user if external information is needed for completion (e.g. a passport number)	1	
Questions on forms are grouped logically, and each group has a heading	1	
Fields on forms contain hints, examples or model answers to demonstrate the expected input	-1	the differemce between letters and numbers in the boxes aren't clear. you can't undrestand if you have to put a number or letter
When field labels on forms take the form of questions, the questions are stated in clear, simple language	1	
Pull-down menus, radio buttons and check boxes are used in preference to text entry fields on forms (i.e. text entry fields are not overused)	0	
With data entry screens, the cursor is placed where the input is needed		
Data formats are clearly indicated for input (e.g. dates) and output (e.g. units of values).	-1	it doesn't distinguish if a number is required or a letter.
Users can complete simple tasks by entering just essential information (with the system supplying the non-essential information by default)	0	
Forms allow users to stay with a single interaction method for as long as possible (i.e. users do not need to make numerous shifts from keyboard to mouse to keyboard)		
The user can change default values in form fields	1	
Text entry fields indicate the amount and the format of data that needs to be entered	-1	it doesn't distinguish if a number is required or a letter.
Forms are validated before the form is submitted	1	
With data entry screens, the site carries out field-level checking and form-level checking at the appropriate time	1	
The site makes it easy to correct errors (e.g. when a form is incomplete, positioning the cursor at the location where correction is required)	0	
There is consistency between data entry and data display	1	
Labels are close to the data entry fields (e.g. labels are right justified)	1	

Forms and data entry

Checkpoint		Comments
The content is up-to-date, authoritative and trustworthy	1	
The site contains third-party support (e.g. citations, testimonials) to verify the accuracy of information.	1	
It is clear that there is a real organisation behind the site (e.g. there is a physical address or a photo of the office)	1	
The company comprises acknowledged experts (look for credentials)		
The site avoids advertisements, especially pop-ups.	1	
Delivery costs are highlighted at the very beginning of checkout		
The site avoids marketing waffle	1	
Each page is clearly branded so that the user knows he is still in the same site	1	
It is easy to contact someone for assistance and a reply is received quickly	-1	There are no direct links or numbers on the pages where a need for help might be necessary.
The content is fresh: it is updated frequently and the site includes recent content	0	
The site is free of typographic errors and spelling mistakes	1	
The visual design complements the brand and any offline marketing messages		
There are real people behind the organisation and they are honest and trustworthy (look for bios)		

Trust and credibility

Checkpoint		Comments
The site has compelling and unique content		
Text is concise, with no needless instructions or welcome notes	1	
Each content page begins with conclsuions or implications and the text is written with an inverted pyramid style	0	
Pages use bulleted and numbered lists in preference to narrative text	0	
Lists are prefaced with a concise introduction (e.g. a word or phrase), helping users appreciate how the items are related to one another		
The most important items in a list are placed at the top	0	
Information is organised hierarchically, from the general to the specific, and the organisation is clear and logical	0	
Content has been specifically created for the web (web pages do not comprise repurposed material from print publications such as brochures)		
Product pages contain the detail necessary to make a purchase, and users can zoom in on product images		
Hypertext has been appropriately used to structure content	0	
Sentences are written in the active voice	1	
Pages are quick to scan, with ample headings and sub-headings and short paragraphs		
The site uses maps, diagrams, graphs, flow charts and other visuals in preference to wordy blocks of text	0	
Each page is clearly labelled with a descriptive and useful title that makes sense as a bookmark	0	
Links and link titles are descriptive and predictive, and there are no "Click here!" links	-1	Some texts like "gestione carta" seem clickable, but they aren't.
The site avoids cute, clever, or cryptic headings		
Link names match the title of destination pages, so users will know when they have reached the intended page		
Button labels and link labels start with action words	-1	Some texts like "gestione carta" seem clickable, but they aren't and it's confusing.
Headings and sub-headings are short, straightforward and descriptive	0	
The words, phrases and concepts used will be familiar to the typical user	-1	The explanations regarding the "livelli di sicurezza", PUK, PIN, recupero PUK and PIN and etc., are all confusing.
Numbered lists start at "1" not at "0"		
Acronyms and abbreviations are defined when first used	-1	PUK, PIN, livelli di sicurezza and etc., aren't properly explained.
Text links are long enough to be understood, but short enough to minimise wrapping (especially when used as a navigation list)	1	

Writing and content quality

Checkpoint		Comments
The screen density is appropriate for the target users and their tasks		
The layout helps focus attention on what to do next	1	
On all pages, the most important information (such as frequently used topics, features and functions) is presented on the first screenful of information ("above the fold")		
The site can be used without scrolling horizontally	1	
Things that are clickable (like buttons) are obviously pressable	1	
Items that aren't clickable do not have characteristics that suggest that they are	-1	"Gestione carta" in the Leve 3 authentication page looks clickable even though it is just in bold.
The functionality of buttons and controls is obvious from their labels or from their design	1	
Clickable images include redundant text labels (i.e. there is no 'mystery meat' navigation)		
Hypertext links are easy to identify without needing to 'minesweep' (e.g. underlined)	0	
Fonts are used consistently	1	
The relationship between controls and their actions is obvious		
Icons and graphics are standard and/or intuitive (concrete and familiar)	0	The hamburger menu does not immediately stand out
There is a clear visual "starting point" to every page		
Each page on the site shares a consistent layout	1	
Pages on the site are formatted for printing, or there is a printer-friendly version		
Buttons and links show that they have been clicked		
GUI components (like radio buttons and check boxes) are used appropriately		

Page layout and visual design pt. 1

Fonts are readable	0	Low contrast and small text limit vulnerable users' possibility to properly read instructions
The site avoids italicised text and uses underlining only for hypertext links		
There is a good balance between information density and use of white space	0	Unbalanced use of white space in the home page
The site is pleasant to look at	1	
Pages are free of "scroll stoppers" (headings or page elements that create the illusion that users have reached the top or bottom of a page when they have not)		
The site avoids extensive use of upper case text	1	
The site has a consistent, clearly recognisable look and feel that will engage users	1	
Saturated blue is avoided for fine detail (e.g. text, thin lines and symbols)	-1	Saturated blue is massively used also for fine details
Colour is used to structure and group items on the page	1	
Graphics will not be confused with banner ads	1	
Emboldening is used to emphasise important topic categories	1	
On content pages, line lengths are neither too short (<50 characters per line) nor too long (>100 characters per line) when viewed in a standard browser width window	1	
Pages have been designed to an underlying grid, with items and widgets aligned both horizontally and vertically	1	
Meaningful labels, effective background colours and appropriate use of borders and white space help users identify a set of items as a discrete functional block	1	
The colours work well together and complicated backgrounds are avoided	0	Saturated blue and fine white texts are hardly readable by people suffering of poor sight
Individual pages are free of clutter and irrelevant information	1	
Standard elements (such as page titles, site navigation, page navigation, privacy policy etc.) are easy to locate	0	These elements are hard to find for novice and elderly users
The organisation's logo is placed in the same location on every page, and clicking the logo returns the user to the most logical page (e.g. the home page)	-1	
Attention-attracting features (such as animation, bold colours and size differentials) are used sparingly and only where relevant		
Icons are visually and conceptually distinct yet still harmonious (clearly part of the same family)	1	

Page layout and visual design pt.2

Checkpoint		Comments
The default search is intuitive to configure (no Boolean operators)		
The search results page shows the user what was searched for and it is easy to edit and resubmit the search		
Search results are clear, useful and ranked by relevance		
The search results page makes it clear how many results were retrieved, and the number of results per page can be configured by the user		
If no results are returned, the system offers ideas or options for improving the query based on identifiable problems with the user's input		
The search engine handles empty queries gracefully		
The most common queries (as reflected in the site log) produce useful results		
The search engine includes templates, examples or hints on how to use it effectively		
The site includes a more powerful search interface available to help users refine their searches (preferably named "revise search" or "refine search", not "advanced search)	-1	There are no search input in the app
The search results page does not show duplicate results (either perceived duplicates or actual duplicates)		
The search box is long enough to handle common query lengths		
Searches cover the entire web site, not a portion of it		
If the site allows users to set up a complex search, these searches can be saved and executed on a regular basis (so users can keep up-to-date with dynamic content)		
The search interface is located where users will expect to find it (top right of page)		
The search box and its controls are clearly labelled (multiple search boxes can be confusing)		
The site supports people who want to browse and people who want to search	0	The app allowes only to browse the system, not to search
The scope of the search is made explicit on the search results page and users can restrict the scope (if relevant to the task)		
The search results page displays useful meta-information, such as the size of the document, the date that the document was created and the file type (Word, pdf etc.)		
The search engine provides automatic spell checking and looks for plurals and synonyms		
The search engine provides an option for similarity search (“more like this”)		

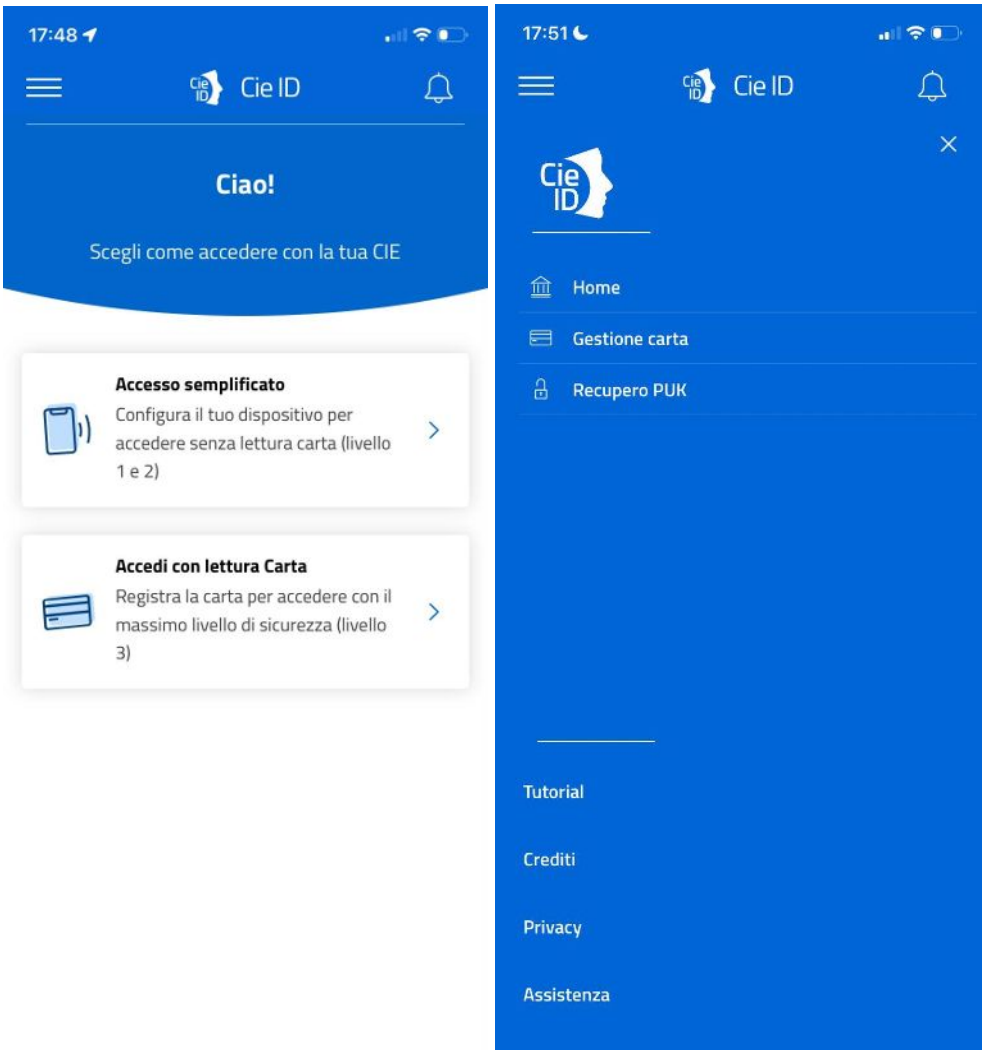
Search

Checkpoint		Comments
The FAQ or on-line help provides step-by-step instructions to help users carry out the most important tasks	1	
It is easy to get help in the right form and at the right time	-1	When adding ID card or in "recupero PUK and PIN", there aren't any help options. You have to go back to the home page, but even there it's not enough.
Prompts are brief and unambiguous	-1	prompts regarding recupero, are ambiguous.
The user does not need to consult user manuals or other external information to use the site	0	For high level tutorials, the user may consult external resources on the web.
The site uses a customised 404 page, which includes tips on how to find the missing page and links to "Home" and Search		
The site provides good feedback (e.g. progress indicators or messages) when needed (e.g. during checkout)	0	
Users are given help in choosing products		
User confirmation is required before carrying out potentially "dangerous" actions (e.g. deleting something)	1	
Confirmation pages are clear		
Error messages contain clear instructions on what to do next	0	
Immediately prior to commiting to the purchase, the site shows the user a clear summary page and this will not be confused with a purchase confirmation page		
When the user needs to choose between different options (such as in a dialog box), the options are obvious		
The site keeps users informed about unavoidable delays in the site's response time (e.g. when authorising a credit card transaction)		
Error messages are written in a non-derisory tone and do not blame the user for the error	1	
Pages load quickly (5 seconds or less)	1	
The site provides immediate feedback on user input or actions	1	
The user is warned about large, slow-loading pages (e.g. "Please wait..."), and the most important information appears first	1	
Where tooltips are used, they provide useful additional help and do not simply duplicate text in the icon, link or field label	0	
When giving instructions, pages tell users what to do rather than what to avoid doing	1	
The site shows users how to do common tasks where appropriate (e.g. with demonstrations of the site's functionality)	-1	The explanations and help sections aren't easily available, they are in the menu part rather than in the task section when it's needed.
The site provides feedback (e.g. "Did you know?") that helps the user learn how to use the site	0	

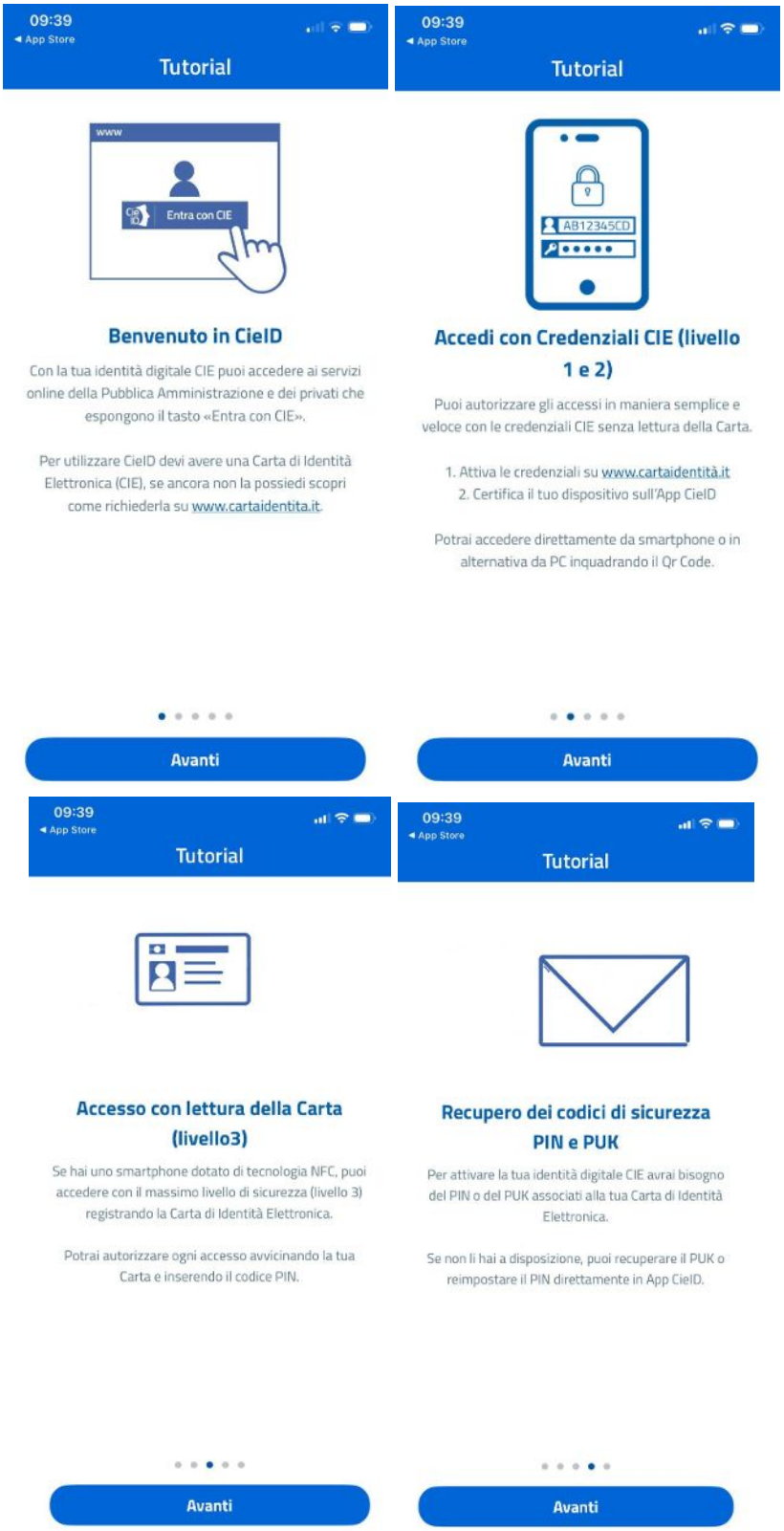
Help, feedback, error tolerance pt.1

The site provides context sensitive help	-1	The explanations and help sections aren't easily available, they are in the menu part rather than in the task section when it's needed.
Help is clear and direct and simply expressed in plain English, free from jargon and buzzwords	0	
The site provides clear feedback when a task has been completed successfully	1	
Important instructions remain on the screen while needed, and there are no hasty time outs requiring the user to write down information	1	
Fitts' Law is followed (the distance between controls and the size of the controls is appropriate, with size proportional to distance)	1	
There is sufficient space between targets to prevent the user from hitting multiple or incorrect targets	1	
There is a line space of at least 2 pixels between clickable items	1	
The site makes it obvious when and where an error has occurred (e.g. when a form is incomplete, highlighting the missing fields)	1	
The site uses appropriate selection methods (e.g. pull-down menus) as an alternative to typing	1	
The site does a good job of preventing the user from making errors	0	
The site prompts the user before correcting erroneous input (e.g. Google's "Did you mean...")		
The site ensures that work is not lost (either by the user or site error)	0	
Error messages are written in plain language with sufficient explanation of the problem	1	
When relevant, the user can defer fixing errors until later in the task	1	
The site can provide more detail about error messages if required	0	
It is easy to "undo" (or "cancel") and "redo" actions	1	

Help, feedback, error tolerance pt.2



The home page only shows the two types of authentication and it lacks a search input box. The majority of important information is hidden inside the hamburger menu which is very thin and not very visible at first sight. The initial information is specifically mentioning level 1, 2 and 3 of authentication omitting further clearer explanation. Texts are too thin in order to be understandable by all segments of populations.



Initial tutorial explaining both levels of authentication with thin texts and low contrast. This results in illegibility to people suffering from poor sighting.


15:52
App Store
Prima di iniziare
Cie ID
Leggi attentamente le prossime pagine, ti spiegheremo come far interagire correttamente il tuo iPhone con la Carta di Identità Elettronica
AVANTI

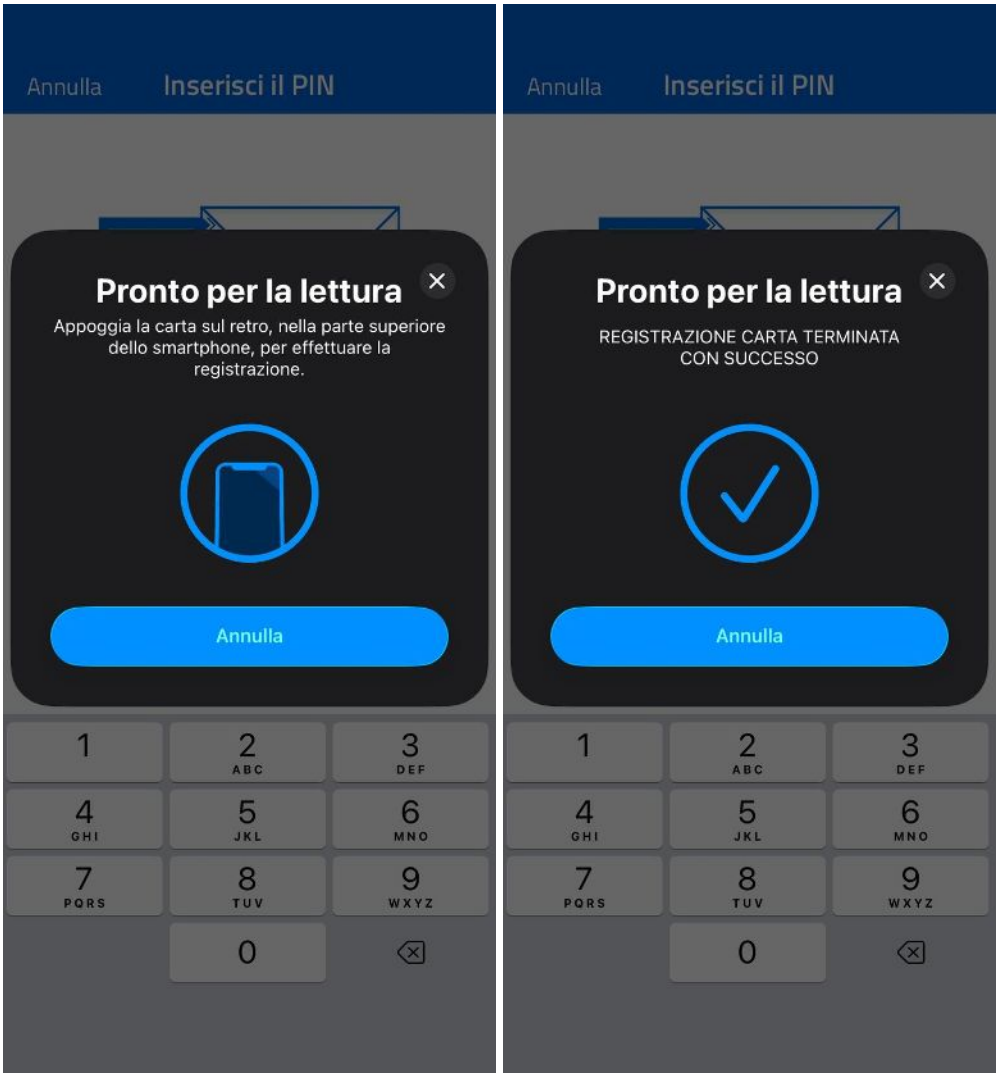

15:53
App Store
Suggerimento 2
Mantieni la carta in contatto per tutto il tempo della lettura. L'operazione di lettura richiede solitamente meno di 10 secondi
AVANTI


15:53
App Store
Suggerimento 1
Quando richiesto appoggia la Carta di Identità Elettronica sul retro dello smartphone, nella parte alta del tuo smartphone come mostrato nell'immagine
AVANTI

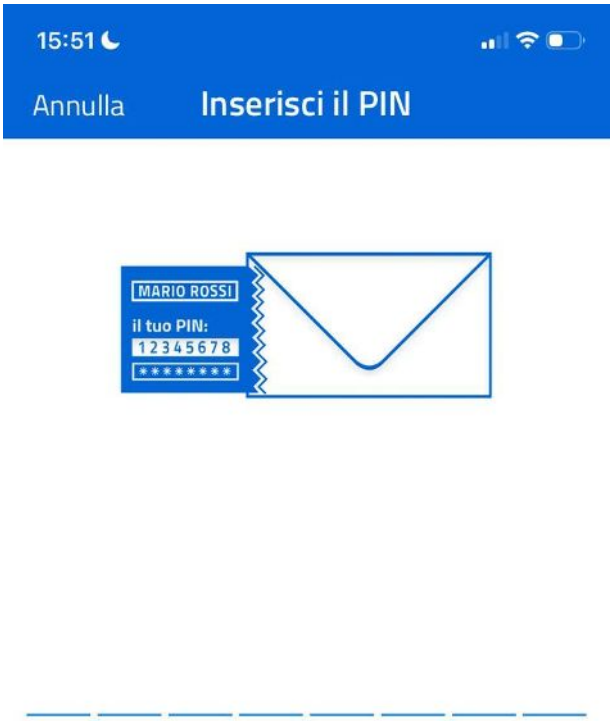

15:53
App Store
Suggerimento 3
Se hai difficoltà a leggere la carta ed utilizzi una cover per proteggere lo smartphone dagli urti, prova a rimuoverla, potrebbe impedire la corretta interazione con la carta
AVANTI



When trying to register the card for the first time this tutorial is shown. The white texts on the electric blue background lead poor sighted people to ignore the instructions. This tutorial is only shown the first time. If the user abandons the action and tries later the instructions are not repeated.



No visual cues indicating how to execute NFC reading. **Lack of dynamic feedback** to signal how much time is required to properly read the card. The only instructions are provided through thin and low contrast texts that complicate the users' experience



The “Inserisci il PIN” page provides an explanation on how to act if the user loses the card’s PIN. The indication “Vai in **Gestione carta**” which could have been a direct link to the recovery page, is only a bold text that can mislead people instead of providing a direct link to redirect the user.

Assessment of existing resource card

This is a competitor's tool

Name of tool

PosteID

Platforms for tool (devices, operating systems, etc.)

Mobile and web application

Main purpose of tool, main operations allowed, perceived level of adequacy to professional standards in its design and implementation

App PosteID allows the user to access Public Administration services online after the registration. From the application interface the user can authorize access to online services, manage the PosteID profile and recover user name and password. Through the app the user can view the access request history. The user can complete their registration by using the CIE. The user must own the PIN in order to register the card without any fee. To recover the pin, a payment of 10€ is needed. The activation of PosteID's SPID used to be free; currently it is free the first year and from the second year the cost is 6€ per year.

Main strengths of tool, reason for choice

- In Italy, this has become the most used SPID system. Estimates indicate that it accounts for 60% of digital identities.
- Simple interface, fostering recognizability and memorability. Familiar to most Italian citizens.
- Use clear terminology and avoids ambiguity
- Provides most of the important information and functions in the homepage.

- Accessing Public Administration services is possible even without entering the app through the PIN.

Main weaknesses of tool

- Has become a paid service starting from the second year.
- Lacks a search box input in the home page
- Does not provide assistance in the home page. In order to access the FAQ the user must first click on “Entra nell’app”.

First Impressions

PosteID represents the main SPID system in Italy. In use for many years, it represents a familiar, simple, and clear choice for online access. The first approach to the app is featured with a simple skippable tutorial/overview of the app’s function. Texts are short and avoid technicalities although the dimensions of text chunks can complicate the reading for people suffering from poor eyesight. The app allows the registration with the credential of the first activation of the SPID. As mentioned before the service is free the first year; then it costs 6€. The user can avoid paying by using the CIE and the PIN. If the PIN is lost the user must pay 10€ to recover it. Generally the application presents most of the essential functions on the home page avoiding technical and domain specific vocabulary. Important texts in the home page are highlighted in bold and the dimension is adequate for all types of users. Assistance is structured in clear categories providing simple and straightforward answers but its location (after clicking “Entra nell’app”) obliges the user to look for it instead of having it close at hand. It generally proves to be a less confusing and more straightforward option for user groups that struggle most with technology.

Direct Analysis

List of outstanding issues

- Missing search box input and, causing the impossibility to articulate a question without browsing the site.
- Lack of contacts and assistance information directly in the home page, obliging the user to conduct research throughout the site.

- Contact information hidden in some of the FAQ answers.
- Wordy FAQ answers, causing the possible distraction of the user and abandonment of the task.
- The app automatically logs the user out after less than an hour of inactivity.

Direct Analysis (PosteID)

Summary of results

	Raw score	# Questions	# Answers	Score
Home Page	6	20	13	73%
Task Orientation	20	44	29	84%
Navigation & IA	15	29	18	92%
Forms & Data Entry	7	23	7	100%
Trust & Credibility	5	13	11	73%
Writing & Content Quality	10	23	17	79%
Page Layout & Visual Design	23	38	36	82%
Search	-4	20	5	10%
Help, Feedback & Error Tolerance	17	37	23	87%
Overall score		247	159	76%

Summary of results

Checkpoint		Comments
The items on the home page are clearly focused on users' key tasks ("featuritis" has been avoided)	1	
The home page contains a search input box	-1	
Product categories are provided and clearly visible on the homepage		
Useful content is presented on the home page or within one click of the home page	1	
The home page shows good examples of real site content		
Links on the home page begin with the most important keyword (e.g. "Sun holidays" not "Holidays in the sun")	1	
There is a short list of items recently featured on the homepage, supplemented with a link to archival content		
Navigation areas on the home page are not over-formatted and users will not mistake them for adverts	1	
The value proposition is clearly stated on the home page (e.g. with a tagline or welcome blurb)		
The home page contains meaningful graphics, not clip art or pictures of models	1	
Navigation choices are ordered in the most logical or task-oriented manner (with the less important corporate information at the bottom)	0	No assistance is provided in the "home page" before accessing the app
The title of the home page will provide good visibility in search engines like Google		
All corporate information is grouped in one distinct area (e.g. "About Us")	-1	Lack of About us section
Users will understand the value proposition	1	
By just looking at the home page, the first time user will understand where to start	0	It is quite user friendly also for novices
The home page shows all the major options	1	
The home page of the site has a memorable URL		
The home page is professionally designed and will create a positive first impression	1	
The design of the home page will encourage people to explore the site	0	
The home page looks like a home page; pages lower in the site will not be confused with it		

Home Page

Checkpoint		Comments
The site is free from irrelevant, unnecessary and distracting information	1	
Excessive use of scripts, applets, movies, audio files, graphics and images has been avoided	1	
The site avoids unnecessary registration	1	
The critical path (e.g. purchase, subscription) is clear, with no distractions on route		
Information is presented in a simple, natural and logical order	1	
The number of screens required per task has been minimised		
The site requires minimal scrolling and clicking	1	
The site correctly anticipates and prompts for the user's probable next activity	1	
When graphs are shown, users have access to the actual data (e.g. numeric annotation on bar charts)		
Activities allocated to the user or the computer take full advantage of the strengths of each (look for actions that can be done automatically by the site, e.g. postcode lookup)	1	
Users can complete common tasks quickly	1	
Items can be compared easily when this is necessary for the task (e.g. product comparisons)		
The task sequence parallels the user's work processes	1	
The site makes the user's work easier and quicker than without the system		
The most important and frequently used topics, features and functions are close to the centre of the page, not in the far left or right margins	0	Yes, except for the assistance in the first Home page. When accessed the app you find it in menu unter "Aiuto"
The user does not need to enter the same information more than once	1	
Important, frequently needed topics and tasks are close to the 'surface' of the web site	1	
Typing (e.g. during purchase) is kept to an absolute minimum, with accelerators ("one-click") for return users	1	
The path for any given task is a reasonable length (2-5 clicks)	1	
When there are multiple steps in a task, the site displays all the steps that need to be completed and provides feedback on the user's current position in the workflow	0	It tells you what is the task but not the position in the sequence
Price is always clearly displayed next to any product	1	
The site's privacy policy is easy to find, especially on pages that ask for personal information, and the policy is simple and clear	0	It is accessible but consists of information-dense documents

Task Orientation pt. 1

Users of the site do not need to remember information from place to place	1	
The use of metaphors is easily understandable by the typical user	1	
Data formats follow appropriate cultural conventions (e.g. miles for UK)		
Details of the software's internal workings are not exposed to the user	1	
The site caters for users with little prior experience of the web	1	
The site makes it easy for users to explore the site and try out different options before committing themselves		
A typical first-time visitor can do the most common tasks without assistance	1	
When they return to the site, users will remember how to carry out the key tasks	1	
The functionality of novel device controls is obvious		
On the basket page, there is a highly visible 'Proceed to checkout' button at the top and bottom of the page		
Important calls to action, like 'Add to basket', are highly visible		
Action buttons (such as "Submit") are always invoked by the user, not automatically invoked by the system when the last field is completed	-1	The system automatically proceeds to the next page when the last field is completed
Command and action items are presented as buttons (not, for example, as hypertext links)	1	
If the user is half-way through a transaction and quits, the user can later return to the site and continue from where he left off		
When a page presents a lot of information, the user can sort and filter the information		
If there is an image on a button or icon, it is relevant to the task	1	
The site prompts the user before automatically logging off the user, and the time out is appropriate	-1	The app logs out authomatically after less than an hour
Unwanted features (e.g. Flash animations) can be stopped or skipped		
The site is robust and all the key features work (i.e. there are no javascript exceptions, CGI errors or broken links)	1	
The site supports novice and expert users by providing different levels of explanation (e.g. in help and error messages)	-1	The site does not adapt to user's experience level but it is very simple to understant for any segment of the population
The site allows users to rename objects and actions in the interface (e.g. naming delivery addresses or accounts)		
The site allows the user to customise operational time parameters (e.g. time until automatic logout)		

Task orientation pt.2

Checkpoint		Comments
There is a convenient and obvious way to move between related pages and sections and it is easy to return to the home page	1	
The information that users are most likely to need is easy to navigate to from most pages	1	
Navigation choices are ordered in the most logical or task-oriented manner	1	
The navigation system is broad and shallow (many items on a menu) rather than deep (many menu levels)	1	
The site structure is simple, with a clear conceptual model and no unnecessary levels	1	
The major sections of the site are available from every page (persistent navigation) and there are no dead ends	1	
Navigation tabs are located at the top of the page, and look like clickable versions of real-world tabs	-1	They are on the bottom of the screen
There is a site map that provides an overview of the site's content		
The site map is linked to from every page		
The site map provides a concise overview of the site, not a rehash of the main navigation or a list of every single topic		
Good navigational feedback is provided (e.g. showing where you are in the site)	1	
Category labels accurately describe the information in the category	1	
Links and navigation labels contain the "trigger words" that users will look for to achieve their goal	1	
Terminology and conventions (such as link colours) are (approximately) consistent with general web usage	1	
Links look the same in the different sections of the site	1	
Product pages contain links to similar and complementary products to support cross-selling		
The terms used for navigation items and hypertext links are unambiguous and jargon-free	1	
Users can sort and filter catalogue pages (e.g. by listing in price order, or showing 'most popular')		
There is a visible change when the mouse points at something clickable (excluding cursor changes)		
Important content can be accessed from more than one link (different users may require different link labels)		
Navigation-only pages (such as the home page) can be viewed without scrolling		

Navigation and AI pt.1

Hypertext links that invoke actions (e.g downloads, new windows) are clearly distinguished from hypertext links that load another page	1
The site allows the user to control the pace and sequence of the interaction	0
There are clearly marked exits on every page allowing the user to bale out of the current task without having to go through an extended dialog	1
The site does not disable the browser's "Back" button and the "Back" button appears on the browser toolbar on every page	
Clicking the back button always takes the user back to the page the user came from	1
A link to both the basket and checkout is clearly visible on every page	
If the site spawns new windows, these will not confuse the user (e.g. they are dialog-box sized and can be easily closed)	
Menu instructions, prompts and messages appear on the same place on each screen	1

Navigation and AI pt.2

Fields in data entry screens contain default values when appropriate and show the structure of the data and the field length	1			
When a task involves source documents (such as a paper form), the interface is compatible with the characteristics of the source document				
The site automatically enters field formatting data (e.g. currency symbols, commas for 1000s, trailing or leading spaces). Users do not need to enter characters like £ or %.				
Field labels on forms clearly explain what entries are desired	1			
Text boxes on forms are the right length for the expected answer	1			
There is a clear distinction between "required" and "optional" fields on forms				
The same form is used for both logging in and registering (i.e. it's like Amazon)	1			
Forms pre-warn the user if external information is needed for completion (e.g. a passport number)				
Questions on forms are grouped logically, and each group has a heading				
Fields on forms contain hints, examples or model answers to demonstrate the expected input				
When field labels on forms take the form of questions, the questions are stated in clear, simple language				
Pull-down menus, radio buttons and check boxes are used in preference to text entry fields on forms (i.e. text entry fields are not overused)				
With data entry screens, the cursor is placed where the input is needed				
Data formats are clearly indicated for input (e.g. dates) and output (e.g. units of values).				
Users can complete simple tasks by entering just essential information (with the system supplying the non-essential information by default)				
Forms allow users to stay with a single interaction method for as long as possible (i.e. users do not need to make numerous shifts from keyboard to mouse to keyboard).				
The user can change default values in form fields				
Text entry fields indicate the amount and the format of data that needs to be entered				
Forms are validated before the form is submitted				
With data entry screens, the site carries out field-level checking and form-level checking at the appropriate time				
The site makes it easy to correct errors (e.g. when a form is incomplete, positioning the cursor at the location where correction is required)	1			
There is consistency between data entry and data display	1			
Labels are close to the data entry fields (e.g. labels are right justified)	1			

Forms and data entry

Checkpoint		Comments
The content is up-to-date, authoritative and trustworthy	1	
The site contains third-party support (e.g. citations, testimonials) to verify the accuracy of information.	-1	
It is clear that there is a real organisation behind the site (e.g. there is a physical address or a photo of the office)	0	This type of information id not available on the site
The company comprises acknowledged experts (look for credentials)	-1	No contacts are provided
The site avoids advertisements, especially pop-ups.	1	
Delivery costs are highlighted at the very beginning of checkout		
The site avoids marketing waffle	1	
Each page is clearly branded so that the user knows he is still in the same site	1	
It is easy to contact someone for assistance and a reply is received quickly	0	Contacts are provided on the app but are hidden in the FAQ section
The content is fresh: it is updated frequently and the site includes recent content	1	
The site is free of typographic errors and spelling mistakes	1	
The visual design complements the brand and any offline marketing messages	1	
There are real people behind the organisation and they are honest and trustworthy (look for bios)		

Trust and credibility

The site has compelling and unique content	1		
Text is concise, with no needless instructions or welcome notes	0		Could be shortened in some cases
Each content page begins with conclsuions or implications and the text is written with an inverted pyramid style			
Pages use bulleted and numbered lists in preference to narrative text	-1		Some FAQ answers are structured as big chuck of text
Lists are prefaced with a concise introduction (e.g. a word or phrase), helping users appreciate how the items are related to one another			
The most important items in a list are placed at the top			
Information is organised hierarchically, from the general to the specific, and the organisation is clear and logical			
Content has been specifically created for the web (web pages do not comprise repurposed material from print publications such as brochures)			
Product pages contain the detail necessary to make a purchase, and users can zoom in on product images			
Hypertext has been appropriately used to structure content	1		
Sentences are written in the active voice	1		
Pages are quick to scan, with ample headings and sub-headings and short paragraphs	1		
The site uses maps, diagrams, graphs, flow charts and other visuals in preference to wordy blocks of text	0		
Each page is clearly labelled with a descriptive and useful title that makes sense as a bookmark	0		
Links and link titles are descriptive and predictive, and there are no "Click here!" links	1		
The site avoids cute, clever, or cryptic headings	1		
Link names match the title of destination pages, so users will know when they have reached the intended page	1		
Button labels and link labels start with action words	1		
Headings and sub-headings are short, straightforward and descriptive	1		
The words, phrases and concepts used will be familiar to the typical user	0		
Numbered lists start at "1" not at "0"	1		
Acronyms and abbreviations are defined when first used	1		
Text links are long enough to be understood, but short enough to minimise wrapping (especially when used as a navigation list)	0		Generally yes but some answers in the FAQ section are too wordy

Writing and content quality

Checkpoint		Comments
The screen density is appropriate for the target users and their tasks	1	
The layout helps focus attention on what to do next	1	
On all pages, the most important information (such as frequently used topics, features and functions) is presented on the first screenful of information ("above the fold")	0	Missing assistance before accessing the app, in the first home page screen
The site can be used without scrolling horizontally	1	
Things that are clickable (like buttons) are obviously pressable	1	
Items that aren't clickable do not have characteristics that suggest that they are	1	
The functionality of buttons and controls is obvious from their labels or from their design	1	
Clickable images include redundant text labels (i.e. there is no 'mystery meat' navigation)	1	
Hypertext links are easy to identify without needing to 'minesweep' (e.g. underlined)	1	
Fonts are used consistently	1	
The relationship between controls and their actions is obvious	1	
Icons and graphics are standard and/or intuitive (concrete and familiar)	1	
There is a clear visual "starting point" to every page	0	
Each page on the site shares a consistent layout	0	"Profilo" and "Impostazioni" lack yellow banner typical of other sections
Pages on the site are formatted for printing, or there is a printer-friendly version		
Buttons and links show that they have been clicked	-1	
GUI components (like radio buttons and check boxes) are used appropriately	1	
Fonts are readable	0	Some secondary text chunck use thin texts
The site avoids italicised text and uses underlining only for hypertext links	1	
There is a good balance between information density and use of white space	1	
The site is pleasant to look at	1	

Page layout and visual design pt.1

Pages are free of "scroll stoppers" (headings or page elements that create the illusion that users have reached the top or bottom of a page when they have not)		
The site avoids extensive use of upper case text	1	
The site has a consistent, clearly recognisable look and feel that will engage users	0	
Saturated blue is avoided for fine detail (e.g. text, thin lines and symbols)	-1	Saturated blue is used for links and contacts information
Colour is used to structure and group items on the page	1	
Graphics will not be confused with banner ads	1	
Emboldening is used to emphasise important topic categories	1	
On content pages, line lengths are neither too short (<50 characters per line) nor too long (>100 characters per line) when viewed in a standard browser width window	-1	FAQ answers are very lverbose and do not make use of bullets or numbered sequences to simplify the reading
Pages have been designed to an underlying grid, with items and widgets aligned both horizontally and vertically	1	
Meaningful labels, effective background colours and appropriate use of borders and white space help users identify a set of items as a discrete functional block	1	
The colours work well together and complicated backgrounds are avoided	1	
Individual pages are free of clutter and irrelevant information	1	
Standard elements (such as page titles, site navigation, page navigation, privacy policy etc.) are easy to locate	1	
The organisation's logo is placed in the same location on every page, and clicking the logo returns the user to the most logical page (e.g. the home page)	-1	
Attention-attracting features (such as animation, bold colours and size differentials) are used sparingly and only where relevant	1	
Icons are visually and conceptually distinct yet still harmonious (clearly part of the same family)	1	
Related information and functions are clustered together, and each group can be scanned in a single fixation (5-deg, about 4.4cm diam circle on screen)	1	

Page layout and visual design pt.2

Checkpoint		Comments
The default search is intuitive to configure (no Boolean operators)		
The search results page shows the user what was searched for and it is easy to edit and resubmit the search		
Search results are clear, useful and ranked by relevance		
The search results page makes it clear how many results were retrieved, and the number of results per page can be configured by the user		
If no results are returned, the system offers ideas or options for improving the query based on identifiable problems with the user's input		
The search engine handles empty queries gracefully		
The most common queries (as reflected in the site log) produce useful results		
The search engine includes templates, examples or hints on how to use it effectively		
The site includes a more powerful search interface available to help users refine their searches (preferably named "revise search" or "refine search", not "advanced search)	-1	
The search results page does not show duplicate results (either perceived duplicates or actual duplicates)		
The search box is long enough to handle common query lengths	-1	The app totally lacks seach box input
Searches cover the entire web site, not a portion of it	-1	
If the site allows users to set up a complex search, these searches can be saved and executed on a regular basis (so users can keep up-to-date with dynamic content)		
The search interface is located where users will expect to find it (top right of page)	-1	
The search box and its controls are clearly labelled (multiple search boxes can be confusing)		
The site supports people who want to browse and people who want to search	0	It supports navigation but not searching
The scope of the search is made explicit on the search results page and users can restrict the scope (if relevant to the task)		
The search results page displays useful meta-information, such as the size of the document, the date that the document was created and the file type (Word, pdf etc.)		
The search engine provides automatic spell checking and looks for plurals and synonyms		
The search engine provides an option for similarity search ("more like this")		

Search

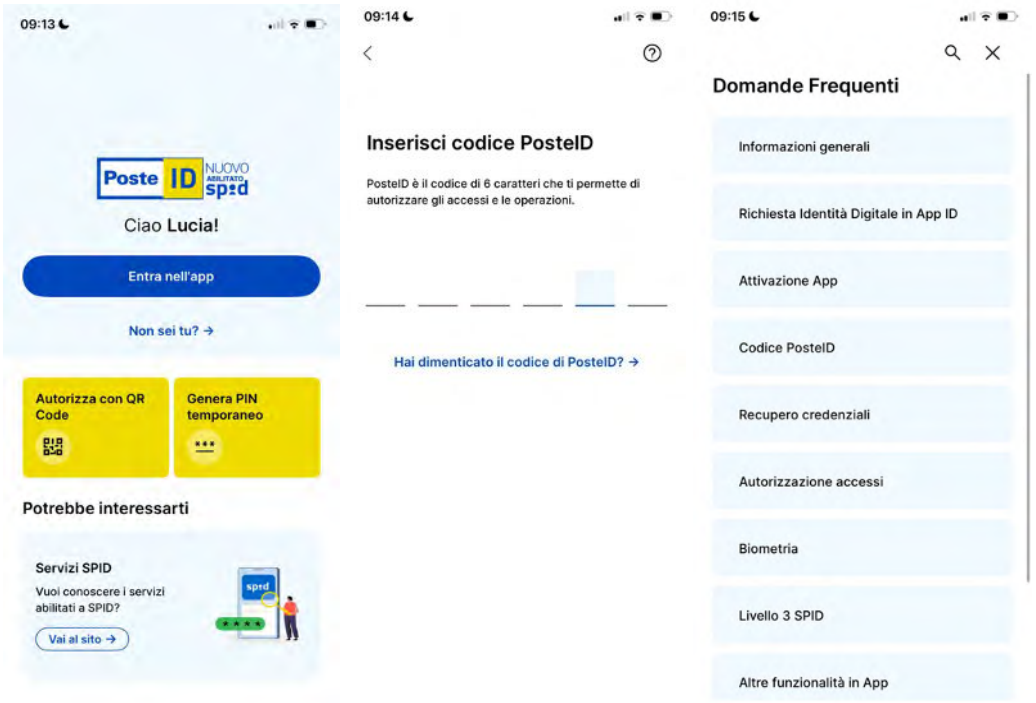
Checkpoint		Comments
The FAQ or on-line help provides step-by-step instructions to help users carry out the most important tasks	1	
It is easy to get help in the right form and at the right time	0	To get access to contacts the user can directly search on the web or explore the page. Contacts are not explicitely offered to the user in the home page
Prompts are brief and unambiguous		
The user does not need to consult user manuals or other external information to use the site	-1	
The site uses a customised 404 page, which includes tips on how to find the missing page and links to "Home" and Search		
The site provides good feedback (e.g. progress indicators or messages) when needed (e.g. during checkout)	1	
Users are given help in choosing products		
User confirmation is required before carrying out potentially "dangerous" actions (e.g. deleting something)	1	
Confirmation pages are clear	1	
Error messages contain clear instructions on what to do next	1	
Immediately prior to commiting to the purchase, the site shows the user a clear summary page and this will not be confused with a purchase confirmation page		
When the user needs to choose between different options (such as in a dialog box), the options are obvious	1	
The site keeps users informed about unavoidable delays in the site's response time (e.g. when authorising a credit card transaction)		
Error messages are written in a non-derisory tone and do not blame the user for the error	1	
Pages load quickly (5 seconds or less)	1	
The site provides immediate feedback on user input or actions	1	
The user is warned about large, slow-loading pages (e.g. "Please wait..."), and the most important information appears first	1	
Where tooltips are used, they provide useful additional help and do not simply duplicate text in the icon, link or field label		
When giving instructions, pages tell users what to do rather than what to avoid doing	1	
The site shows users how to do common tasks where appropriate (e.g. with demonstrations of the site's functionality)	0	
The site provides feedback (e.g. "Did you know?") that helps the user learn how to use the site		
The site provides context sensitive help	0	
Help is clear and direct and simply expressed in plain English, free from jargon and buzzwords	0	In some cases, explanations are too long

Help, Feedback and error tolerance pt.1

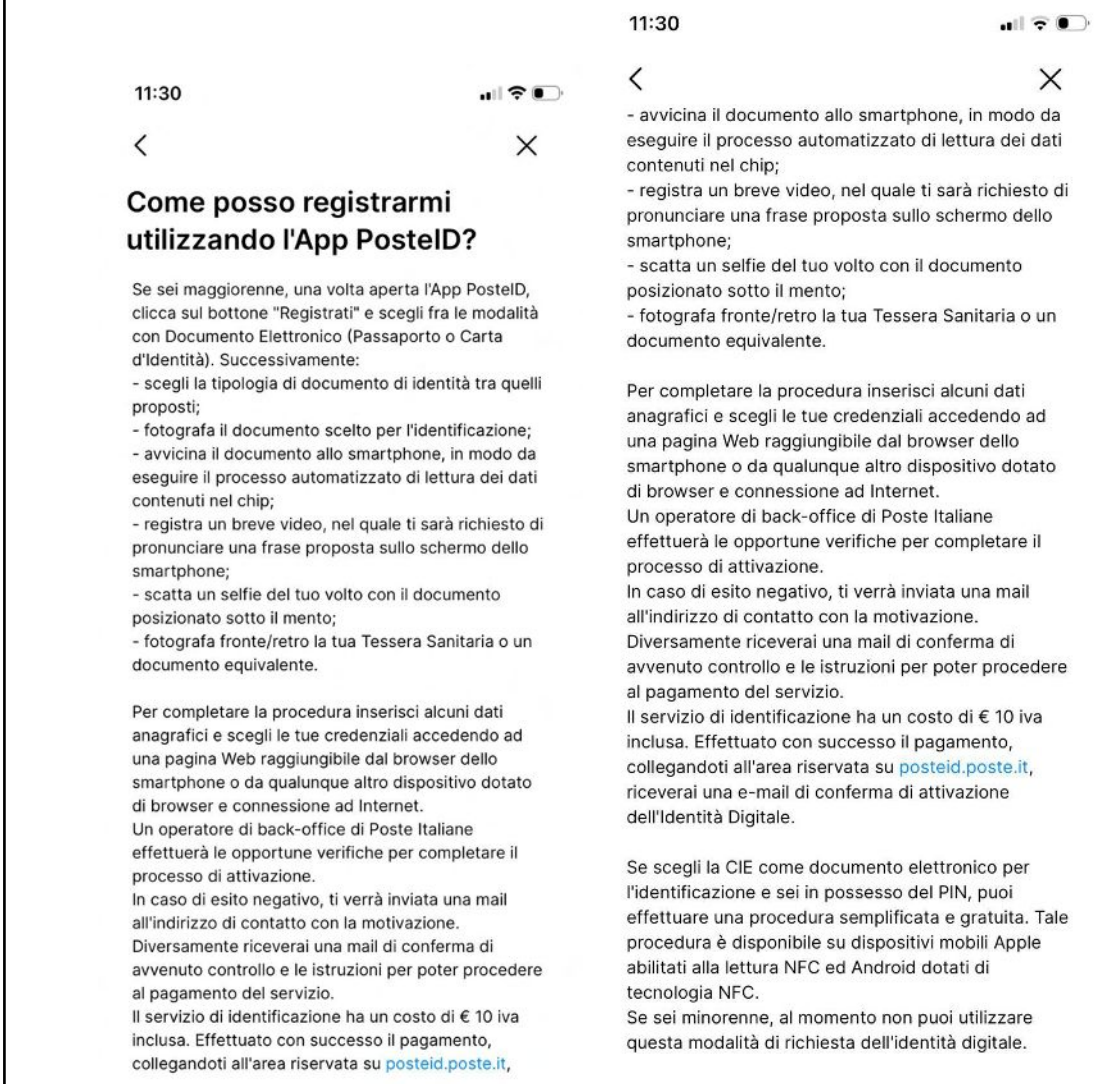
The site provides clear feedback when a task has been completed successfully	1			
Important instructions remain on the screen while needed, and there are no hasty time outs requiring the user to write down information				
Fitts' Law is followed (the distance between controls and the size of the controls is appropriate, with size proportional to distance)				
There is sufficient space between targets to prevent the user from hitting multiple or incorrect targets	1			
There is a line space of at least 2 pixels between clickable items				
The site makes it obvious when and where an error has occurred (e.g. when a form is incomplete, highlighting the missing fields)	1			
The site uses appropriate selection methods (e.g. pull-down menus) as an alternative to typing	1			
The site does a good job of preventing the user from making errors	1			
The site prompts the user before correcting erroneous input (e.g. Google's "Did you mean...")				
The site ensures that work is not lost (either by the user or site error)				
Error messages are written in plain language with sufficient explanation of the problem	1			
When relevant, the user can defer fixing errors until later in the task				
The site can provide more detail about error messages if required				
It is easy to "undo" (or "cancel") and "redo" actions	1			

Help, Feedback and error tolerance pt.2

PosteID screens



The page before clicking the “Entra nell’app” button lacks a search input box, contacts and assistance information. In order to be assisted the user must navigate the system. Only by clicking on “Entra nell’app”, the question icon appears allowing the user to access the “Domande frequenti” section.



Too verbose answers that lead the user to distract easily from its purpose.



Contacts are only provided in some of the FAQ answers leading the user to navigate the system before having the possibility to talk with an assistance operator.



This warning message occurs while inserting the wrong PIN. The five trials limit until the code is blocked generates anxiety and blocks the user to try to insert the PIN again.

Reverse Analysis

List of outstanding issues

- **Lack of search functionality:** The application completely lacks a search input box on the Home Page, making it impossible to search for specific features or help.
- **Cluttered Information Architecture:** The Home Page mixes notifications, last access logs ("ultimi accessi"), and a pop-up for scanning the QR code, which creates a confusing hierarchy for the user. Additionally, the list of "ultimi accessi" is confined to a small scrolling box at the bottom, which is difficult to browse.

- **Hidden Privacy Policy:** Essential corporate information, such as the privacy policy, is hidden deep within the menu rather than being easily accessible.
- **Long and rigid registration flow:** The registration process is excessively long. Furthermore, if a user quits halfway through the transaction, the progress is not saved, forcing them to restart the entire process from the beginning.
- **Ambiguous Field Labels:** Form labels lack clear examples. Specifically, the "VIA/ VIALE/ PIAZZA" address field is highly confusing, as it is unclear whether the user must type their full address or just select one of those three words.
- **Lack of input formatting:** The application does not automatically adapt the keyboard format to the required data type (e.g., the number keyboard does not automatically appear when typing the street number "civico").
- **Redundant text fields:** Asking users to re-enter their email multiple times introduces friction and increases the likelihood of user errors.
- **Confusing login/registration overlap:** The app uses the same initial page for logging in and registering, which frequently causes confusion for first-time users trying to sign up.
- **Poorly timed feedback:** The app does not validate forms field-by-field. Errors are only communicated *after* the user presses "Prosegui" (Continue), resulting in delayed feedback.
- **Unhelpful Error Messages:** Error messages lack clarity. For example, if a user makes a mistake in the address section, the error prompt does not provide actionable instructions on what exactly to fix.
- **System Freezes without feedback:** The application sometimes gets stuck on "Caricando..." (Loading) during the registration upload process, failing to communicate whether a timeout or a specific error occurred.
- **Lack of adaptive help:** There is only one static type of explanation for tasks. The app does not adapt its help or provide different levels of explanation to support users of varying ages, languages, or technical expertise.

- **Illogical control placement:** In navigation flows, the placement of buttons is counter-intuitive (e.g., "Prosegui" is placed on the left side, while "Indietro" is placed on the right side).
- **Inconsistent "Starting Points":** The visual starting point is not always obvious. For instance, in the registration flow, the starting point/button is placed at the very bottom of the page in a small font.
- **Lack of sorting/filtering:** Complex lists like notifications and "ultimi accessi" cannot be sorted or filtered by the user.

Reverse Analysis (PosteID)

Summary of results

	Raw score	# Questions	# Answers	Score
Home Page	4	20	14	64%
Task Orientation	1	44	33	52%
Navigation & IA	2	29	16	56%
Forms & Data Entry	-1	23	23	48%
Trust & Credibility	5	13	9	78%
Writing & Content Quality	4	23	19	61%
Page Layout & Visual Design	-4	38	37	45%
Search	-3	20	3	0%
Help, Feedback & Error Tolerance	-12	37	36	33%
Overall score		247	190	48%

Summary of results

Checkpoint		Comments
The items on the home page are clearly focused on users' key tasks ("featuritis" has been avoided)	0	
The home page contains a search input box	-1	there are no search boxes
Product categories are provided and clearly visible on the homepage		
Useful content is presented on the home page or within one click of the home page	1	
The home page shows good examples of real site content		
Links on the home page begin with the most important keyword (e.g. "Sun holidays" not "Holidays in the sun")	1	
There is a short list of items recently featured on the homepage, supplemented with a link to archival content	1	
Navigation areas on the home page are not over-formatted and users will not mistake them for adverts	1	
The value proposition is clearly stated on the home page (e.g. with a tagline or welcome blurb)		
The home page contains meaningful graphics, not clip art or pictures of models	-1	some graphics are confusing or repetitive
Navigation choices are ordered in the most logical or task-oriented manner (with the less important corporate information at the bottom)	0	
The title of the home page will provide good visibility in search engines like Google		
All corporate information is grouped in one distinct area (e.g. "About Us")	1	
Users will understand the value proposition		
By just looking at the home page, the first time user will understand where to start	1	
The home page shows all the major options	1	
The home page of the site has a memorable URL		
The home page is professionally designed and will create a positive first impression	-1	the order of items and notfications are a bit confusing
The design of the home page will encourage people to explore the site	0	
The home page looks like a home page; pages lower in the site will not be confused with it	0	

Homepage

Checkpoint		Comments
The site is free from irrelevant, unnecessary and distracting information	-1	notifications and last access are on the main page, while the "scan with QR code", is a pop up
Excessive use of scripts, applets, movies, audio files, graphics and images has been avoided	1	
The site avoids unnecessary registration	1	
The critical path (e.g. purchase, subscription) is clear, with no distractions on route	1	
Information is presented in a simple, natural and logical order	1	
The number of screens required per task has been minimised	-1	the registration process is so long and at some parts confusing
The site requires minimal scrolling and clicking	-1	the to part is bar as the notification section, and at the bottom, it's a scrolling box for "ultimi accessi", which is a bit confusing.
The site correctly anticipates and prompts for the user's probable next activity	0	
When graphs are shown, users have access to the actual data (e.g. numeric annotation on bar charts)		
Activities allocated to the user or the computer take full advantage of the strengths of each (look for actions that can be done automatically by the site, e.g. postcode lookup)		
Users can complete common tasks quickly	0	
Items can be compared easily when this is necessary for the task (e.g. product comparisons)		
The task sequence parallels the user's work processes	1	
The site makes the user's work easier and quicker than without the system	1	
The most important and frequently used topics, features and functions are close to the centre of the page, not in the far left or right margins	0	
The user does not need to enter the same information more than once	0	
Important, frequently needed topics and tasks are close to the 'surface' of the web site		
Typing (e.g. during purchase) is kept to an absolute minimum, with accelerators ("one-click") for return users	1	
The path for any given task is a reasonable length (2-5 clicks)	0	
When there are multiple steps in a task, the site displays all the steps that need to be completed and provides feedback on the user's current position in the workflow	0	
Price is always clearly displayed next to any product		

Task Orientation pt. 1

The site's privacy policy is easy to find, especially on pages that ask for personal information, and the policy is simple and clear	-1	the privacy policy is hidden in the menu
Users of the site do not need to remember information from place to place	0	
The use of metaphors is easily understandable by the typical user	0	
Data formats follow appropriate cultural conventions (e.g. miles for UK)	1	
Details of the software's internal workings are not exposed to the user	1	
The site caters for users with little prior experience of the web		
The site makes it easy for users to explore the site and try out different options before committing themselves		
A typical first-time visitor can do the most common tasks without assistance	0	
When they return to the site, users will remember how to carry out the key tasks	-1	
The functionality of novel device controls is obvious		
On the basket page, there is a highly visible 'Proceed to checkout' button at the top and bottom of the page		
Important calls to action, like 'Add to basket', are highly visible	-1	the "scan QR code", is a pop up
Action buttons (such as "Submit") are always invoked by the user, not automatically invoked by the system when the last field is completed	1	
Command and action items are presented as buttons (not, for example, as hypertext links)	-1	the "scan QR code", is a pop up
If the user is half-way through a transaction and quits, the user can later return to the site and continue from where he left off	-1	the registration process, is'nt saved, and once you quit, you have to start again.
When a page presents a lot of information, the user can sort and filter the information	-1	no sorting action on "ultimi accessi" or notifications.
If there is an image on a button or icon, it is relevant to the task	0	
The site prompts the user before automatically logging off the user, and the time out is appropriate	1	
Unwanted features (e.g. Flash animations) can be stopped or skipped	1	tutorials can be skipped
The site is robust and all the key features work (i.e. there are no javascript exceptions, CGI errors or broken links)	-1	it is sometimes blocked on uploading in the registration process.
The site supports novice and expert users by providing different levels of explanation (e.g. in help and error messages)	-1	there's only one type of explanation, although the app is meant to be used by different ages with different languages.
The site allows users to rename objects and actions in the interface (e.g. naming delivery addresses or accounts)		
The site allows the user to customise operational time parameters (e.g. time until automatic logout)		

Task Orientation pt. 2

Checkpoint		Comments
There is a convenient and obvious way to move between related pages and sections and it is easy to return to the home page	1	
The information that users are most likely to need is easy to navigate to from most pages	-1	the "scan qr code" is a pop up.
Navigation choices are ordered in the most logical or task-oriented manner	-1	on the left side, there is "prosegui", and next to it on the right side, there is indietro which isn't logical
The navigation system is broad and shallow (many items on a menu) rather than deep (many menu levels)	0	
The site structure is simple, with a clear conceptual model and no unnecessary levels	0	the notifications and ultimi accessi
The major sections of the site are available from every page (persistent navigation) and there are no dead ends	1	
Navigation tabs are located at the top of the page, and look like clickable versions of real-world tabs		
There is a site map that provides an overview of the site's content		
The site map is linked to from every page		
The site map provides a concise overview of the site, not a rehash of the main navigation or a list of every single topic		
Good navigational feedback is provided (e.g. showing where you are in the site)		
Category labels accurately describe the information in the category	1	
Links and navigation labels contain the "trigger words" that users will look for to achieve their goal	0	
Terminology and conventions (such as link colours) are (approximately) consistent with general web usage		
Links look the same in the different sections of the site		
Product pages contain links to similar and complementary products to support cross-selling		
The terms used for navigation items and hypertext links are unambiguous and jargon-free		
Users can sort and filter catalogue pages (e.g. by listing in price order, or showing 'most popular')	-1	notifications and ultimi accessi aren't sortable, and there are no applicable filter on them
There is a visible change when the mouse points at something clickable (excluding cursor changes)		
Important content can be accessed from more than one link (different users may require different link labels)	0	
Navigation-only pages (such as the home page) can be viewed without scrolling	1	

Navigation and AI pt. 1

Hypertext links that invoke actions (e.g downloads, new windows) are clearly distinguished from hypertext links that load another page	0
The site allows the user to control the pace and sequence of the interaction	0
There are clearly marked exits on every page allowing the user to bale out of the current task without having to go through an extended dialog	0
The site does not disable the browser's "Back" button and the "Back" button appears on the browser toolbar on every page	
Clicking the back button always takes the user back to the page the user came from	1
A link to both the basket and checkout is clearly visible on every page	
If the site spawns new windows, these will not confuse the user (e.g. they are dialog-box sized and can be easily closed)	
Menu instructions, prompts and messages appear on the same place on each screen	0

Navigation and AI pt. 2

Checkpoint		Comments
Fields in data entry screens contain default values when appropriate and show the structure of the data and the field length	-1	for the "VIA/ VIALE/ PIAZZA" part it's confusing if it is asking for the address or asking to enter one of these three words.
When a task involves source documents (such as a paper form), the interface is compatible with the characteristics of the source document	1	
The site automatically enters field formatting data (e.g. currency symbols, commas for 1000s, trailing or leading spaces). Users do not need to enter characters like £ or %.	0	for "civico" it doen't turn to number keyboard
Field labels on forms clearly explain what entries are desired	-1	for re-entering email or re-entring password, it's confusing and creates errors sometimes.
Text boxes on forms are the right length for the expected answer	0	
There is a clear distinction between "required" and "optional" fields on forms	1	
The same form is used for both logging in and registering (i.e. it's like Amazon)	-1	The first page is for logging in, and it causes confusion for registering.
Forms pre-warn the user if external information is needed for completion (e.g. a passport number)	0	
Questions on forms are grouped logically, and each group has a heading	0	
Fields on forms contain hints, examples or model answers to demonstrate the expected input	-1	for the "VIA/ VIALE/ PIAZZA" part it's confusing if it is asking for the address or asking to enter one of these three words. it only says "inserisci" with no example.
When field labels on forms take the form of questions, the questions are stated in clear, simple language	-1	for the "VIA/ VIALE/ PIAZZA" part it's confusing if it is asking for the address or asking to enter one of these three words. it only says "inserisci" with no example.
Pull-down menus, radio buttons and check boxes are used in preference to text entry fields on forms (i.e. text entry fields are not overused)	1	
With data entry screens, the cursor is placed where the input is needed	0	
Data formats are clearly indicated for input (e.g. dates) and output (e.g. units of values).	1	
Users can complete simple tasks by entering just essential information (with the system supplying the non-essential information by default)	-1	all information should be entered manually
Forms allow users to stay with a single interaction method for as long as possible (i.e. users do not need to make numerous shifts from keyboard to mouse to keyboard).	0	
The user can change default values in form fields	-1	all information should be entered manually
Text entry fields indicate the amount and the format of data that needs to be entered	-1	for the "VIA/ VIALE/ PIAZZA" part it's confusing if it is asking for the address or asking to enter one of these three words. it only says "inserisci" with no example.
Forms are validated before the form is submitted	1	
With data entry screens, the site carries out field-level checking and form-level checking at the appropriate time	0	
The site makes it easy to correct errors (e.g. when a form is incomplete, positioning the cursor at the location where correction is required)	0	
There is consistency between data entry and data display	1	
Labels are close to the data entry fields (e.g. labels are right justified)	1	

Forms and data entry

Checkpoint		Comments
The content is up-to-date, authoritative and trustworthy	1	
The site contains third-party support (e.g. citations, testimonials) to verify the accuracy of information.	0	
It is clear that there is a real organisation behind the site (e.g. there is a physical address or a photo of the office)	1	
The company comprises acknowledged experts (look for credentials)		
The site avoids advertisements, especially pop-ups.	1	
Delivery costs are highlighted at the very beginning of checkout		
The site avoids marketing waffle		
Each page is clearly branded so that the user knows he is still in the same site	1	
It is easy to contact someone for assistance and a reply is received quickly	0	
The content is fresh: it is updated frequently and the site includes recent content		
The site is free of typographic errors and spelling mistakes	0	
The visual design complements the brand and any offline marketing messages	0	
There are real people behind the organisation and they are honest and trustworthy (look for bios)	1	

Trust and credibility

Checkpoint		Comments
The site has compelling and unique content		
Text is concise, with no needless instructions or welcome notes	0	
Each content page begins with conclsuions or implications and the text is written with an inverted pyramid style	0	
Pages use bulleted and numbered lists in preference to narrative text	1	
Lists are prefaced with a concise introduction (e.g. a word or phrase), helping users appreciate how the items are related to one another	0	
The most important items in a list are placed at the top	0	
Information is organised hierarchically, from the general to the specific, and the organisation is clear and logical	0	
Content has been specifically created for the web (web pages do not comprise repurposed material from print publications such as brochures)	0	
Product pages contain the detail necessary to make a purchase, and users can zoom in on product images		
Hypertext has been appropriately used to structure content	0	
Sentences are written in the active voice	0	
Pages are quick to scan, with ample headings and sub-headings and short paragraphs	1	
The site uses maps, diagrams, graphs, flow charts and other visuals in preference to wordy blocks of text	0	
Each page is clearly labelled with a descriptive and useful title that makes sense as a bookmark	0	
Links and link titles are descriptive and predictive, and there are no "Click here!" links	0	
The site avoids cute, clever, or cryptic headings		
Link names match the title of destination pages, so users will know when they have reached the intended page	0	
Button labels and link labels start with action words	1	
Headings and sub-headings are short, straightforward and descriptive	0	
The words, phrases and concepts used will be familiar to the typical user	1	The descriptions regarding "livello di sicurezza" aren't clear enough.
Numbered lists start at "1" not at "0"	0	
Acronyms and abbreviations are defined when first used	0	
Text links are long enough to be understood, but short enough to minimise wrapping (especially when used as a navigation list)		

Writing and content quality

Checkpoint		Comments
The screen density is appropriate for the target users and their tasks	0	
The layout helps focus attention on what to do next	0	
On all pages, the most important information (such as frequently used topics, features and functions) is presented on the first screenful of information ("above the fold")	0	
The site can be used without scrolling horizontally	-1	notifications page need scrolling horizontally
Things that are clickable (like buttons) are obviously pressable	0	
Items that aren't clickable do not have characteristics that suggest that they are	0	
The functionality of buttons and controls is obvious from their labels or from their design	0	
Clickable images include redundant text labels (i.e. there is no 'mystery meat' navigation)	0	
Hypertext links are easy to identify without needing to 'minesweep' (e.g. underlined)	0	
Fonts are used consistently	0	
The relationship between controls and their actions is obvious	-1	on the left side, there is "prosegui", and next to it on the right side, there is indietro which isn't logical
Icons and graphics are standard and/or intuitive (concrete and familiar)	0	
There is a clear visual "starting point" to every page	-1	For registration, the starting point is at the end of the page in small font.
Each page on the site shares a consistent layout	0	
Pages on the site are formatted for printing, or there is a printer-friendly version		
Buttons and links show that they have been clicked	0	
GUI components (like radio buttons and check boxes) are used appropriately	0	

Page layout and visual design pt. 1

Fonts are readable	-1	some instructions are written in small fonts.
The site avoids italicised text and uses underlining only for hypertext links	0	
There is a good balance between information density and use of white space	0	
The site is pleasant to look at	0	
Pages are free of "scroll stoppers" (headings or page elements that create the illusion that users have reached the top or bottom of a page when they have not)	0	
The site avoids extensive use of upper case text	0	
The site has a consistent, clearly recognisable look and feel that will engage users	1	
Saturated blue is avoided for fine detail (e.g. text, thin lines and symbols)	-1	Saturated blue is used a lot in this app.
Colour is used to structure and group items on the page	0	
Graphics will not be confused with banner ads	0	
Emboldening is used to emphasise important topic categories	0	
On content pages, line lengths are neither too short (<50 characters per line) nor too long (>100 characters per line) when viewed in a standard browser width window	1	
Pages have been designed to an underlying grid, with items and widgets aligned both horizontally and vertically	0	
Meaningful labels, effective background colours and appropriate use of borders and white space help users identify a set of items as a discrete functional block	-1	some fonts are small or in colors that aren't clear enough to be readable.
The colours work well together and complicated backgrounds are avoided	0	
Individual pages are free of clutter and irrelevant information	0	
Standard elements (such as page titles, site navigation, page navigation, privacy policy etc.) are easy to locate	0	
The organisation's logo is placed in the same location on every page, and clicking the logo returns the user to the most logical page (e.g. the home page)	0	
Attention-attracting features (such as animation, bold colours and size differentials) are used sparingly and only where relevant	0	
Icons are visually and conceptually distinct yet still harmonious (clearly part of the same family)	0	
Related information and functions are clustered together, and each group can be scanned in a single fixation (5-deg. about 4.4cm diam circle on screen)	0	

Page layout and visual design pt. 2

Checkpoint		Comments
The default search is intuitive to configure (no Boolean operators)	-1	There are no search bars.
The search results page shows the user what was searched for and it is easy to edit and resubmit the search		
Search results are clear, useful and ranked by relevance	-1	There are no search bars.
The search results page makes it clear how many results were retrieved, and the number of results per page can be configured by the user		
If no results are returned, the system offers ideas or options for improving the query based on identifiable problems with the user's input		
The search engine handles empty queries gracefully		
The most common queries (as reflected in the site log) produce useful results		
The search engine includes templates, examples or hints on how to use it effectively		
The site includes a more powerful search interface available to help users refine their searches (preferably named "revise search" or "refine search", not "advanced search)	-1	There are no search input in the app
The search results page does not show duplicate results (either perceived duplicates or actual duplicates)		
The search box is long enough to handle common query lengths		
Searches cover the entire web site, not a portion of it		
If the site allows users to set up a complex search, these searches can be saved and executed on a regular basis (so users can keep up-to-date with dynamic content)		
The search interface is located where users will expect to find it (top right of page)		
The search box and its controls are clearly labelled (multiple search boxes can be confusing)		
The site supports people who want to browse and people who want to search		
The scope of the search is made explicit on the search results page and users can restrict the scope (if relevant to the task)		
The search results page displays useful meta-information, such as the size of the document, the date that the document was created and the file type (Word, pdf etc.)		
The search engine provides automatic spell checking and looks for plurals and synonyms		
The search engine provides an option for similarity search ("more like this")		

Search

Checkpoint		Comments
The FAQ or on-line help provides step-by-step instructions to help users carry out the most important tasks	0	
It is easy to get help in the right form and at the right time	0	
Prompts are brief and unambiguous	-1	for the "VIA/ VIALE/ PIAZZA" part it's confusing if it is asking for the address or asking to enter one of these three words. it only says "inserisci" with no example.
The user does not need to consult user manuals or other external information to use the site	0	
The site uses a customised 404 page, which includes tips on how to find the missing page and links to "Home" and Search	0	
The site provides good feedback (e.g. progress indicators or messages) when needed (e.g. during checkout)	1	
Users are given help in choosing products		
User confirmation is required before carrying out potentially "dangerous" actions (e.g. deleting something)	0	
Confirmation pages are clear	0	
Error messages contain clear instructions on what to do next	-1	error message isn't helpful while for example completing "VIA/VIALE/PIAZZA" section.
Immediately prior to commiting to the purchase, the site shows the user a clear summary page and this will not be confused with a purchase confirmation page	0	
When the user needs to choose between different options (such as in a dialog box), the options are obvious	-1	for the "VIA/ VIALE/ PIAZZA" part it's confusing if it is asking for the address or asking to enter one of these three words. it only says "inserisci" with no example.
The site keeps users informed about unavoidable delays in the site's response time (e.g. when authorising a credit card transaction)	-1	it gets stucked on "caricando", and won't communicate any errors.
Error messages are written in a non-derisory tone and do not blame the user for the error	0	
Pages load quickly (5 seconds or less)	-1	Sometimes it gets stuck on "caricando" and won't communicate any error.
The site provides immediate feedback on user input or actions	-1	Sometimes it gets stuck on "caricando" and won't communicate any errors; the fields aren't checked until you press "prosegui".
The user is warned about large, slow-loading pages (e.g. "Please wait..."), and the most important information appears first	-1	Sometimes it gets stuck on "caricando" and won't communicate any error.
Where tooltips are used, they provide useful additional help and do not simply duplicate text in the icon, link or field label	-1	error message isn't helpful while for example completing "VIA/VIALE/PIAZZA" section.
When giving instructions, pages tell users what to do rather than what to avoid doing	0	
The site shows users how to do common tasks where appropriate (e.g. with demonstrations of the site's functionality)	0	
The site provides feedback (e.g. "Did you know?") that helps the user learn how to use the site	0	

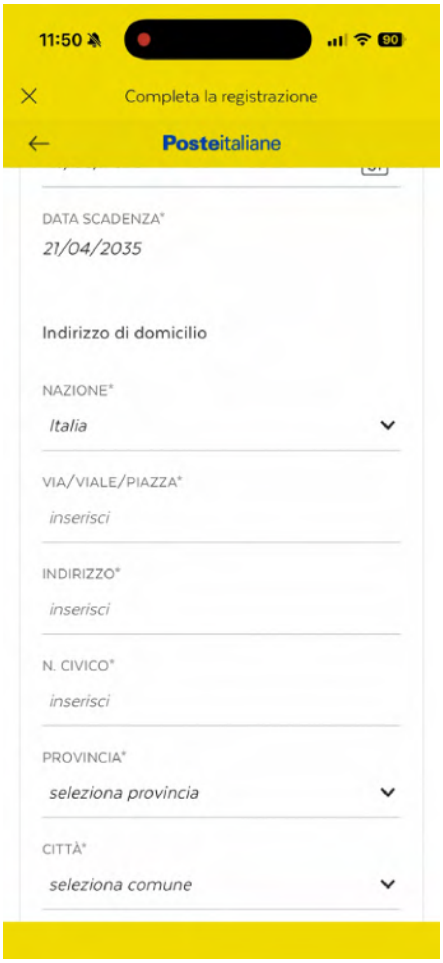
Help, feedback, error tolerance pt. 1

The site provides context sensitive help	0	
Help is clear and direct and simply expressed in plain English, free from jargon and buzzwords	0	
The site provides clear feedback when a task has been completed successfully	0	
Important instructions remain on the screen while needed, and there are no hasty time outs requiring the user to write down information	0	
Fitts' Law is followed (the distance between controls and the size of the controls is appropriate, with size proportional to distance)	0	
There is sufficient space between targets to prevent the user from hitting multiple or incorrect targets	-1	on the left side, there is "prosegui", and next to it on the right side, there is indietro which isn't logical
There is a line space of at least 2 pixels between clickable items	0	
The site makes it obvious when and where an error has occurred (e.g. when a form is incomplete, highlighting the missing fields)	0	
The site uses appropriate selection methods (e.g. pull-down menus) as an alternative to typing	-1	for the "VIA/ VIALE/ PIAZZA" part it's confusing if it is asking for the address or asking to enter one of these three words. it only says "inserisci" with no example.
The site does a good job of preventing the user from making errors	-1	for the "VIA/ VIALE/ PIAZZA" part it's confusing if it is asking for the address or asking to enter one of these three words. it only says "inserisci" with no example.
The site prompts the user before correcting erroneous input (e.g. Google's "Did you mean...")	-1	for the "VIA/ VIALE/ PIAZZA" part it's confusing if it is asking for the address or asking to enter one of these three words. it only says "inserisci" with no example.
The site ensures that work is not lost (either by the user or site error)	0	
Error messages are written in plain language with sufficient explanation of the problem	0	
When relevant, the user can defer fixing errors until later in the task	0	
The site can provide more detail about error messages if required	-1	error message isn't helpful while for example completing "VIA/VIALE/PIAZZA" section.
It is easy to "undo" (or "cancel") and "redo" actions	0	

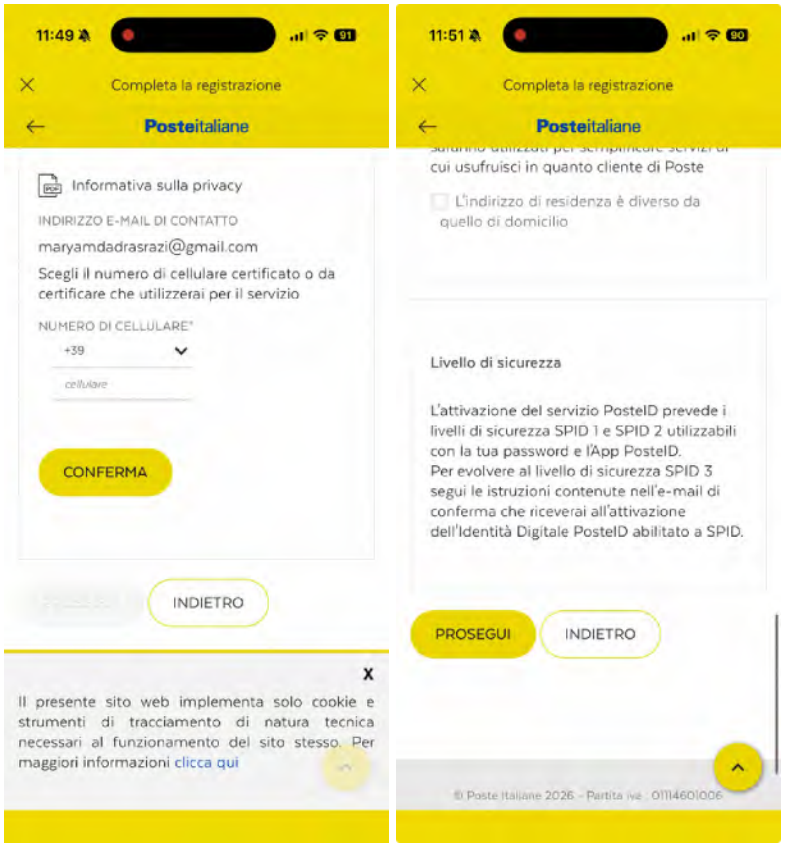
Help, feedback, error tolerance pt. 2



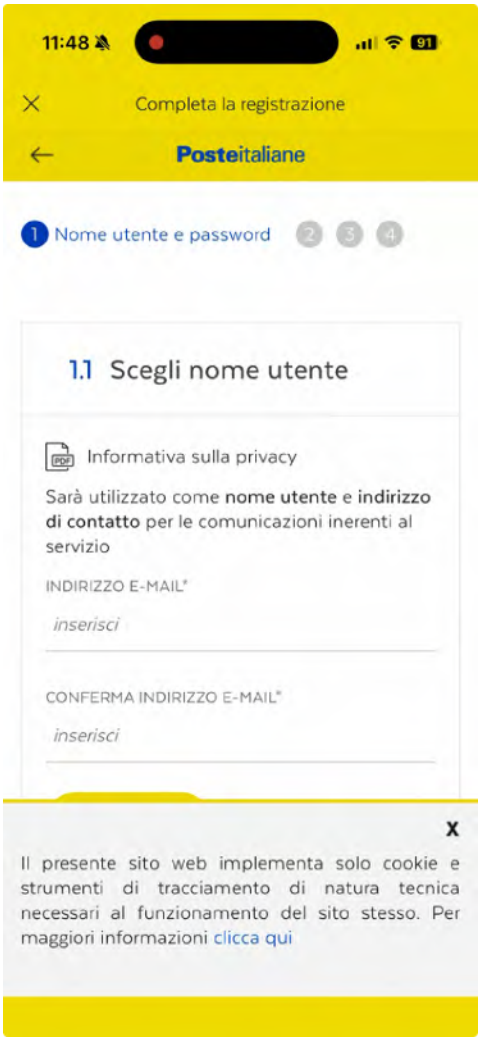
Home page PosteID - pop up for QR code scanning, notifications and ultimi accessi



The confusion of address field with no example



The counter-intuitive replacement of prosegui and indietro



Redundant text fields even for email

Preliminary User testing card

This test relates to the client's tool

Name of tool

CieID

Testing protocol

Testing session setup (devices, operating systems, online vs. in presence, your device or the user's, location of testing, etc.)

The test was conducted on the mobile application. The devices used were a tester's iPhone 11 lent to one of the intermediaries, a Samsung A155F belonging to one of the tested users, a Cubot smartphone with Android operating system (not designed for NFC technology) and an iPhone 17 belonging to one of the testers, lent to a tested user. The users were tested either in a home or in a university room.

Testing method

- Discount usability testing (2-5 people)

List of tasks to test

● Task 1: Configuration of CieID, level 1 and 2 of security ● Task 2 and 3: Recovery of PIN and registration of ID card on CieID, level 3 of security ● Task 4: Recovery of the PUK

Testing methodology

- Thinking aloud for segment #1 - **The Autonomy-Seeking Senior**
- Bottom line data for segment #2 - **The Trusted Digital Mentor**

Metrics of tests

- Success
- Efficiency

- Errors
- Satisfaction

Recruitment of subjects

- **Lucia**, 70 years old woman, scarce digital competences. She is a direct representative of the segment #1 - The Autonomy-Seeking Senior. As a newcomer to digital identity systems (SPID/CIE), she is navigating the digital transition after a lifetime of relying on traditional, analog interaction modalities. Family acquaintance recruited by a team member.
- **Camilla**, 27 years old woman, former operator for Bologna's digital social service. She used to provide digital assistance to elderly citizens living in the Municipality of Bologna. She pursued higher education and has strong IT skills. Camilla and the testers study in the same University.
- **Mauro**, 31 year old man: A UX designer working in the digital space who is enthusiastic about user experience and technology. Having transitioned from SPID to CieID, he is well-acquainted with the overall workflow.
- **Giovanni**, 60 years old man, average digital skills. He partially fits the segment #1 because of his age and poor eyesight. Despite that, he is quite used to navigating online and using mobile applications. He is a neighbor of Alice, one of the testers.

Modality of recruitment of subjects

The design team mainly relied on acquaintances and distant family relatives.

Motivation for choice of subjects

Intermediaries were chosen because of their digital skills and the experience acquired during work experiences assisting elderly citizens and providing them digital support at the office's front desks. We recruited one young woman we met studying in the same university course.

Users belonging to the segment #2 were chosen among the little acquaintances we have of people above 60 years old, which happened to be either distant relatives or neighbors.

Test #1: Lucia

Name (fake one)

Lucia

Date of test

01/10/2026

Assistant of test (team member)

Alice Spadavecchia

Context of test (location, duration, notable events etc.)

Home

Qualitative data of tests

- **Perceived satisfaction and emotion:** the user exhibited persistent anxiety throughout task execution, demonstrating a strong dependency on external assistance provided by the tester. Overall, the interaction evoked feelings of insecurity and general discomfort.
- **Frustration and cognitive pressure:** frustration arose primarily when interacting with thin typography and dense, complex instructions. Encountering unfamiliar tasks generated high cognitive pressure, directly contributing to user errors.
- **Mental model fit and clarity:** the interface failed to accommodate the participant's limited digital literacy, as the system did not align with familiar real-world or digital mental models.

Quantitative data of tests

Success:
Task 1 - 50% (completed with help)
Task 2 and 3 - 50% (completed with help)
Task 4 - 50% (completed with help)

Efficiency:
Task 1 - 03:43
Task 2 and 3 - 04:58
Task 4 - *day 1*: 01:23, *day 2*: 02:54

Errors (mistake and slips):
Task 1 - *mistakes*: 2, *slips:* 0
Task 2 and 3 - *mistakes*: 4, *slips*: 2
Task 4 - *mistakes*: 3, *slips*: 0

Requests for guidance:
Task 1 - 3
Task 2 and 3 - 6
Task 4 - 3

SUS score

30

Outcomes and reflections useful for design

- Need to adapt texts and instruction to the **user's competency level.**
- Enhance **guidance and clear indications** throughout task sequences, incorporating interactive tutorials to lead the user step-by-step through task completion.
- Improve the **learnability of the home page**, as the initial screen currently lacks sufficient information to guide users through the menu and other available options.

Test #2: Camilla

Name (fake one)

Camilla

Date of test

02/10/2026

Assistant of test (team member)

Alice Spadavecchia

Context of test (location, duration, notable events etc.)

Spazio Studentesco di Via Zamboni 34

Qualitative data of tests

- **Perceived satisfaction and emotion:** user feedback highlighted overall satisfaction, while noting room for improvement across specific interactions.
- **Frustration and cognitive pressure:** frustration stemmed primarily from system bugs encountered during the task execution rather than the UX design flaws. Overall, the user demonstrated confidence throughout the tests.
- **Mental model fit and clarity:** the user's mental model aligned well with the application's interaction design and user experience.

Quantitative data of tests

Success:
Task 1 - 100%
Task 2 and 3 - 90% (three scanning requests/NFC bug)
Task 4 - 100%

Efficiency:
Task 1 - 01:57

Task 2 and 3 - 02:25
Task 4 - *day 1*: 00:52, *day 2*: 01:98

Errors (mistake and slips):
Task 1 - *mistakes*: 0, *slips:* 1
Task 2 and 3 - *mistakes*: 1, *slips*: 2
Task 4 - *mistakes*: 0, *slips*: 0

Requests for guidance:
Task 1 - 0
Task 2 and 3 - 0
Task 4 - 0

SUS score

62.5

Useful suggestions from subjects

- Camilla highlighted the need for clearer instructions about where to position the ID card during NFC scanning, as one of the slips was caused by a misinterpretation of the provided guidance.
- During the creation of the “Codice App CieID” the interface misleads users into assuming the choice must the code must consist of digits only, whereas the explanatory text below states otherwise. The visual design of the interface should better match the requested input type.
- If possible, reduce the 48 hours waiting period for the recovery of the PUK by maintaining a high level of security.

Outcomes and reflections useful for design

- Provide more visual feedback during the NFC scanning, such as a timer indicating the time remaining to complete card reading.
- Match visual clues with written text instruction so users can immediately understand the request without committing a slip.

- Provide guidance, improve partial or complex explanations using jargon-free vocabulary, and make accessing assistance quick and easy so users do not need to rely on external help.

Test #3: Mauro

Name (fake one)

Mauro

Date of test

04/10/2026

Assistant of test (team member)

Maryam Dadrasrazi

Context of test (location, duration, notable events etc.)

Home

Qualitative data of tests

- **Perceived satisfaction and emotion:** The participant was generally content, though minor interaction refinements were noted.
- **Frustration and cognitive pressure:** Technical bugs caused occasional friction, but the user remained confident and unaffected by design flaws.
- **Mental model fit and clarity:** The app's navigation and flow matched the user's expectations.

(attach any accompanying material, including all quantitative tables)

Quantitative data of tests

Success:
Task 1 - 100%
Task 2 and 3 - 100%

Task 4 - 100%

Efficiency:
Task 1 - 02:55
Task 2 and 3 - 01:34
Task 4 - *day 1*: 01:23, *day 2*: 01:09

Errors (mistake and slips):
Task 1 - *mistakes*: 1, *slips:* 0
Task 2 and 3 - *mistakes*: 0, *slips*: 0
Task 4 - *mistakes*: 0, *slips*: 1

Requests for guidance:
Task 1 - 0
Task 2 and 3 - 0
Task 4 - 0

SUS score

*Use the provided Google form. Do **not** change words or order of the first 10 questions. Customize and expand as desired the other questions.*

62.5

***Useful suggestions from subjects** (only if provided and interesting)*

- While entering the series ID (containing both letters and numbers), the input field was confusing as it didn't specify whether to enter letters or numbers; this can be improved by clearly indicating the expected input type.
- The NFC error was another slip which could be enhanced with an animated GIF demonstrating the process and showing that the user should pause briefly during NFC scanning.

Outcomes and reflections useful for design

- Clarify expected input types for fields requiring both letters and numbers. Provide visual animated guidance for NFC scanning to help users pause correctly.

Test #4: Giovanni

Name (fake one)

Giovanni

Date of test

03/10/2026

Assistant of test (team member)

Alice Spadavecchia

Context of test (location, duration, notable events etc.)

Home

Qualitative data of tests

- **Perceived satisfaction and emotion:** The user demonstrated a moderate level of confidence while executing the tasks, although occasional reassurance was required for clarification. Overall, the user expressed general satisfaction, even though device limitations prevented the completion of the second task.
- **Frustration and cognitive pressure:** Generally, no signs of cognitive overload were observed. However, frustration arose when deciphering CAPTCHA codes during the web activation of CieID, where poor eyesight further increased task complexity.
- **Mental model fit and clarity:** The user demonstrated a clear understanding of the system's underlying mental model, although noting that certain domain-specific terms were overly technical.

Quantitative data of tests

Success:
Task 1 - 100%
Task 2 and 3 - 0% (Device lacking NFC hardware)

Task 4 - 100%

Efficiency:
Task 1 - 02:21
Task 2 and 3 - 01:07
Task 4 - *day 1*: 00:58, *day 2*: 02:03

Errors (mistakes and slips):
Task 1 - *mistakes*: 1, *slips:* 1
Task 2 and 3 - *mistakes*: 0, *slips*: 2
Task 4 - *mistakes*: 0, *slips*: 1

Requests for guidance:
Task 1 - 1
Task 2 and 3 - 1
Task 4 - 0

SUS score

70

Useful suggestions from subjects *(only if provided and interesting)*

- The user's device lacked NFC technology which prevented him from completing task 3. The app only signaled it while the task was already being executed. The user suggests to provide information at the outset of the task, to avoid wasting the user's time

Outcomes and reflections useful for design

- Automating the detection of NFC technology to adapt the home page to the capabilities of the user's device. The app could provide a prompt suggesting the most suitable authentication method based on the device.
- Avoid CAPTCHAs in favor of OTPs to facilitate code interpretation and reduce stress and frustration during authentication.

- Provide clearer explanations regarding the roles of PIN and PUK during registration, as well as guidance on where to find this information.

Metrics of success - quantitative table

A	B	C	D	E	F	G
Test	Success	Efficiency (measured in time)	Mistakes	Slips	Requests for guidance	SUS score
Task 1 - Camilla	100%	01:57	0	1	0	62.5
Task 2 e 3 - Camilla	90% (three scanning requests - BUG)	02:25	1	2	0	
Task 4 - Camilla	100%	day 1: 00:52 day 2: 01:98	0	0	0	
Task 1 - Lucia	50% (completed with help)	03:43	2	0	3	30
Task 2 and 3 - Lucia	50% (completed with help)	04:58	4	2	6	
Task 4 - Lucia	50% (completed with help)	day 1: 01:23 day 2: 02:54	3	0	3	
Task 1 - Mauro	100%	02:55	1	0	0	62.5
Task 2 e 3 - Mauro	100%	01:34	0	0	0	
Task 4 - Mauro	100%	day 1: 01:23 day 2: 01:09	0	1	0	
Task 1 - Giovanni	100%	02:21	1	1	1	70
Task 2 and 3 - Giovanni	0% (Device lacking NFC hardware)	01:07	0	2	1	
Task 4 - Giovanni	100%	day 1: 00:58 day 2: 02:03	0	1	0	

Preliminary User testing card

This test relates to the competitor's tool

Name of tool

PosteID

Testing protocol

Testing session setup (devices, operating systems, online vs. in presence, your device or the user's, location of testing, etc.)

The test was conducted on the mobile application. The devices used were an iPhone 11 belonging to one of the testers and lent to one of the intermediaries, a Samsung A155F belonging to one of the tested users, a Cubot smartphone with Android operating system (not designed for NFC technology) and an iPhone 17 belonging to one of the testers, lent to a tested user. The users were tested either in a home or in a university room.

Testing method

- Discount usability testing (2-5 people)

List of tasks to test

- Task 1: Initial on-boarding + use of NFC technology
- Task 2: completion of the task on posteID website and finalizing the account

Testing methodology

- Bottom line data

Metrics of tests

- Success
- Efficiency

- Errors
- Satisfaction

Recruitment of subjects

- **Lucia**, 70 years old woman, scarce digital competences. She is a direct representative of the segment #1 - The Autonomy-Seeking Senior. As a newcomer to digital identity systems (SPID/CIE), she is navigating the digital transition after a lifetime of relying on traditional, analog interaction modalities. Family acquaintance recruited by a team member to perform the preliminary test.
- **Camilla**, 27 years old woman, former operator for Bologna's digital social service. She used to provide digital assistance to elderly citizens living in the Municipality of Bologna. She pursued higher education and has strong IT skills. Camilla and the testers study in the same University.
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Modality of recruitment of subjects

We mainly relied on acquaintances and distant family relatives.

Motivation for choice of subjects

Intermediaries were chosen because of their digital skills and the experience acquired during work experiences assisting elderly citizens and providing them digital support at the office's front desks. We recruited one young woman we met studying in the same university course.

Users belonging to the segment #2 were chosen among the little acquaintances we have of people above 60 years old, which happened to be either distant relatives or neighbors.

Test #1: Lucia

Name (fake one)

Lucia

Date of test

01/10/2026

Assistant of test (team member)

Alice Spadavecchia

Context of test (location, duration, notable events etc.)

Home

Qualitative data of tests

- **Perceived satisfaction and emotion:** Users expressed general satisfaction upon task completion, but exhibited underlying insecurity and a frequent need for emotional reassurance. This was particularly evident during the web finalization flow, where uncertainty led users to rely on external prompts from the tester rather than proceeding independently.
- **Frustration and cognitive pressure:** Frustration and cognitive load spiked during NFC positioning in Task 1 and complex multi-step web forms (Task 2). The prolonged execution time on the web portal (over 4 minutes) and repeated requests for guidance reflect cognitive fatigue and fear of committing irreversible errors during registration.

- **Mental model fit and clarity:** The overall goal of setting up a digital identity aligned with users' mental models, but the cross-channel transition from the mobile app to the web browser created friction. Users expected a continuous onboarding journey rather than a fragmented flow requiring external browser interactions.

Quantitative data of tests

Success:
Task 1 - 90% (with external help)
Task 2 - 100%
Efficiency:
Task 1 - 02:47
Task 2 - 04:05

Errors (Mistake and slips):
Task 1 - *mistakes:* 1, *slips:* 1
Task 2 - *mistakes:*1, *slips:* 2

Request for guidance:
Task 1 - 1
Task 2 - 3

SUS score

62.5

Outcomes and reflections useful for design

- Although Task 2 achieved a 100% success rate and Task 1 reached 90%, the frequent requests for guidance (3 in Task 2, 1 in Task 1) reveal that task completion relies heavily on external assistance. Design must shift from "assisted success" to "unassisted autonomy" by embedding contextual UI cues.
- The need for external help during NFC scanning highlights a lack of physical affordance and feedback. Users hesitate when positioning the card against the device. The Introduction of

animated positioning guides and immediate, actionable error recovery prompts could help the user

Test #2: Camilla

Name (fake one)

Camilla

Date of test

02/10/2026

Assistant of test (team member)

Alice Spadavecchia

Context of test (location, duration, notable events etc.)

Spazio Studentesco di Via Zamboni 34

Qualitative data of tests

- **Perceived Satisfaction and Emotion:** As a digitally proficient intermediary, Camilla demonstrated complete autonomy, high confidence, and full task mastery throughout the session, executing all tasks smoothly without requiring any assistance or reassurance.
- **Frustration and Cognitive Pressure:** Cognitive pressure and frustration were minimal. However, the occurrence of 1 slip during Task 2 indicates that misleading UI signifiers (such as conflicting input requirements) can trigger accidental execution errors even for fully autonomous, expert users.
- **Mental Model Fit and Clarity:** Camilla’s mental model aligned seamlessly with the application's system architecture. Her ability to complete both tasks completely unassisted reflects strong digital literacy, confirming that current usability flaws cause minor

operational slips rather than conceptual confusion for intermediaries.

Quantitative data of tests

Success:
Task 1 - 100%
Task 2 - 100%

Efficiency:
Task 1 - 01:54
Task 2 - 02:57

Errors (Mistake and slips):
Task 1 - *mistakes:* 0, *slips:* 0
Task 2 - *mistakes:* 0, *slips:* 1

Request for guidance:
Task 1 - 0
Task 2 - 0

SUS score

87.5

Useful suggestions from subjects

Eliminate the forced redirection to the external Poste.it web portal to finalize registration. Users would greatly benefit from completing the entire SPID setup within the mobile app, avoiding cross-channel navigation friction.

Outcomes and reflections useful for design

- Redirecting users from the PosteID mobile application to the web portal to complete Task 2 significantly increases execution time (averaging over 4 minutes) and cognitive load. The redesign should consolidate the full registration journey into a single, self-contained mobile interface.

- Provide real-time visual feedback and an animated diagram showing where to position the ID card relative to the smartphone's accompanied by a more evident scanning progress indicator to prevent users from moving the card prematurely.

Test #3: Mauro

Name (fake one)

Mauro

Date of test

04/10/2026

Assistant of test (team member)

Maryam Dadrasrazi

Context of test (location, duration, notable events etc.)

Home

Qualitative data of tests

- **Perceived satisfaction and emotion:** As a UX designer and expert user, Mauro began the test with high confidence and autonomy. However, his initial positive emotion quickly degraded into visible annoyance and severe dissatisfaction as he encountered structural UI inconsistencies. Because of his professional background, his frustration was directed entirely at the application's poor design rather than his own abilities, resulting in a low SUS score (45).
- **Frustration and cognitive pressure:** Cognitive pressure spiked significantly during Task 2 due to poorly designed data entry forms. The redundant verification inputs forced an unnecessary cognitive load, forcing him to second-guess basic actions. Frustration peaked when the application became unresponsive ("caricando")

and failed to communicate a clear error state regarding the ambiguous "VIA/VIALE/PIAZZA" field, ultimately forcing him to abandon the task midway (50% completion) out of exasperation.
- **Mental model fit and clarity:** Mauro's established mental model for standard onboarding flows clashed immediately with the app's Information Architecture. The home page was heavily biased toward returning users (login), hiding the registration entry point at the bottom in a small font. Furthermore, the inverted placement of primary and secondary actions (e.g., 'Indietro' and 'Prosegui') violated his internalized left-to-right UX conventions, causing operational slips and breaking the expected interaction flow.

Quantitative data of tests

Success:
Task 1 - 100%
Task 2 - 50%

Efficiency:
Task 1 - 00:30
Task 2 - 06:10

Errors (mistake and slips):
Task 1 - *mistakes*: 0, *slips*: 0
Task 2 - *mistakes*: 1, *slips*: 2

Requests for guidance:
Task 1 - 0
Task 2 - 1

SUS score

45

Useful suggestions from subjects

- Unclear distinction between login and registration steps.

- Street type specification (VIA/VIALE/PIAZZA) is perplexing, preventing task advancement.
- Low-contrast gray text for confirmation instructions reduces legibility, leading users to mistake repeat fields for password inputs; redundant verification fields add unnecessary friction.
- Inverted placement of navigation controls violates standard left-to-right reading order, triggering accidental inputs.
- Vague validation and alert messaging during form submission.
- Intermittent application freezes encountered throughout data upload sequence.

Outcomes and reflections useful for design

- Clarify the entry point and visually separate the login and registration flows to prevent user confusion.
- Simplify the address input form by using standardized dropdowns or auto-complete for street types (e.g., VIA, VIALE, PIAZZA).
- Improve visual contrast for instructional and hint text, and streamline form inputs by removing redundant verification fields.
- Align navigation controls with standard UX patterns (e.g., standard left-to-right flow) to minimize accidental clicks.
- Provide clear, explicit inline validation and user-friendly error messages during form completion.
- Optimize performance and asynchronous data upload handling to eliminate app freezing during registration.

Test #4: Giovanni

Name (fake one)

Giovanni

Date of test

3/10

Assistant of test (team member)

Alice Spadavecchia

Context of test (location, duration, notable events etc.)

Home

Qualitative data of tests

- **Perceived satisfaction and emotion:** Despite failing Task 1 due to a hardware limitation (lack of NFC technology on his device), Giovanni maintained a very positive attitude toward the application, as reflected by his high SUS score (87.5). He demonstrated good digital confidence overall, though he required occasional reassurance when his device failed to scan the card and during the final web authentication steps.
- **Frustration and cognitive pressure:** In Task 1, cognitive friction was caused by the system's failure to inform him upfront that his device lacked NFC support.
- **Mental model fit and clarity:** Giovanni's mental model aligned well with standard mobile and web navigation patterns. However, the system failed to match his expectations regarding hardware compatibility (lack of initial NFC availability check).

Quantitative data of tests

Success:
Task 1 - 0% (Lack of NFC technology)
Task 2 - 100%

Efficiency:
Task 1 - 00:56
Task 2 - 03:27

Errors (mistakes and slips):

Task 1 - *mistakes:* 0, *slips:* 0
Task 2 - *mistakes:* 0, *slips:* 2

Request for guidance:
Task 1 - 1
Task 2 - 1

SUS score

87.5

Useful suggestions from subjects

- Provide an automatic device check upon app launch to detect NFC availability, instantly warning the user if NFC is unsupported and suggesting alternative identity verification paths without forcing unnecessary scanning attempts.

Outcomes and reflections useful for design

- Designing for inclusive onboarding requires the system to gracefully manage hardware limitations. When an app relies on NFC, it must proactively detect device capabilities and guide non-NFC users toward alternative registration methods, preventing confusion and dead-end task attempts.
- Giovanni's high SUS score (87.5) despite a 0% completion rate on Task 1 shows that users distinguish interface usability from hardware constraints. Redesign should leverage this positive attitude by ensuring that alternative web-based or manual onboarding routes are just as intuitive and unassisted.

Metrics of success - quantitative table

A	B	C	D	E	F	G
Test	Completion of task	Efficiency (measured in time)	Errors	Slips	Asking for guidance	SUS score
Task 1 - Lucia	90% (with external help)	02:47	1	1	1	62.5
Task 2 - Lucia	100%	04:05	1	2	3	
Task 1 - Camilla	100%	01:54	0	0	0	87.5
Task 2 - Camilla	100%	02:57	0	1	0	
Task 1- Mauro	100%	00:30	0	0	0	45
Task 2- Mauro	50%	06:10	1	2	1	
Task 1 - Giovanni	0% (Lack of NFC technology)	00:56	0	0	1	87.5
Taak 2 - Giovanni	100%	03:27	0	2	1	

Conclusions and design recommendation card

List of issues by importance

Issues based on impact and persistence. Scores go from 1 to 5 (1 indicating a minor issue, 5 indication a critical issue)

#1 - Inaccessible Domain-Specific Terminology: recurrent domain-specific terms for the identification of different levels of authentication.
Impact: 5
Persistence: 5

#2 - Sub-Standard Visual Accessibility and Inadequate Color Contrast Ratio: non compliance with visual accessibility standards. Inadequate contrast between thin white typography and background elements severely impairs legibility for elderly citizens.
Impact: 5
Persistence: 5

#3 - Absence of Dynamic Feedback and System Status Visibility during NFC Interaction: missing visual cues about the required action to properly position the card during NFC scanning and/or lack of a timer to visualize how much time the reading task requires.
Impact: 5
Persistence: 4

#4 - Concealed Visual Signifiers and Low Discoverability for Novice Users: the menu icon is thin and hidden to citizens suffering from poor eyesight causing problems in the navigation from the home page to secondary section of the application.
Impact: 4
Persistence: 4

#5 - Non-Persistent Onboarding and Flawed On-Demand Learnability: tutorials appear once, during the first use of the application. They are never repeated during the execution of complex tasks (such as the NFC scanning), causing the user to rely on external help or navigate the application until the tutorial section is found.
Impact: 4

Persistence: 4
#6 - Asynchronous PUK Recovery Journey: the recovery of the PUK to reset the PIN is based on an asynchronous process requiring 48 hours wait and another 48 hours period to use it, forcing the user to follow strict time constraints.
Impact: 4
Persistence: 2
#7 - Misleading Visual Affordances and Conceptual Model Mismatch: during level 3 of authentication and the entering the PIN page, instruction to recover it are provided but redirections are not provided with links but only through natural language instruction in bold, resembling links, leading the user to try to click on them.
Impact: 3
Persistence: 2
#8 - Isolated Support Channels and Absence of Contextual Help: assistance contact such as e-mail and telephone number are only provided in a section hidden inside the menu. There is no contextual help during critical tasks.
Impact: 2
Persistence: 3
#9 - Externalized Instructional Media and Fragmented Context Switching: while searching for a tutorial the user is redirected to a YouTube video instead of providing them with a simple step by step tutorial directly inside the application.
Impact: 1
Persistence: 3
#10 - Non-Standardized Data Entry: “Inserisci il PIN” and “Crea Codice CieID” data entry forms do not have a consistent design.
Impact: 1
Persistence: 1

Summary table of SUS scores
(replace with actual names and scores, add columns as needed)

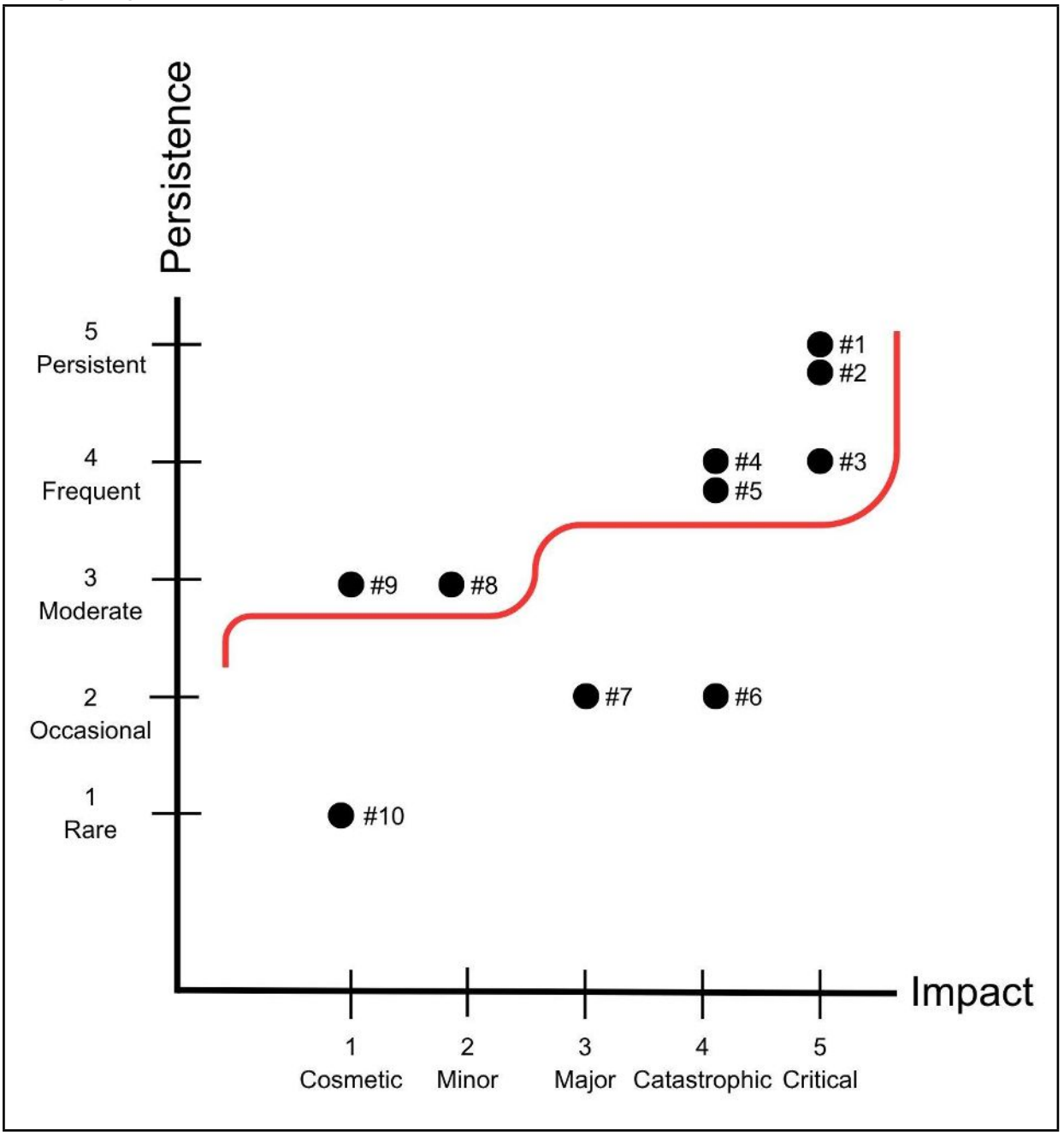
Question	Lucia	Camilla	Mauro	Giovanni
Question #1	2	2	2	4
Question #2	4	3	3	2

Question #3	2	3	4	4
Question #4	4	2	1	2
Question #5	2	2	4	3
Question #6	3	3	2	2
Question #7	2	2	2	4
Question #8	4	3	3	2
Question #9	2	2	4	4
Question #10	3	3	2	3
Total Score	30	62.5	62.5	70
Average	56,25			
Variance	318.75			

If variance is significant, suggest reasons and propose activities to reach a reliable final score.

The high variance (318.75) shows how much the application is polarized based on users' prior experience with similar applications, relying heavily on individual digital skills rather than adapting the user to varying levels of digital literacy. For instance, Lucia's SUS score (30) highlights how age-related inexperience and poor eyesight are not properly accommodated by the current UX of CieID, forcing users belonging to **Segment #1** to rely on external help by intermediaries.
The redesign should focus on balancing the level of accessibility, aiming for an inclusive digital tool that adapts to users' needs and skills.

Urgency curve



Design recommendations

Based on the assessment and testing we decided to focus on the improvement of the interface of CieID, especially focusing on the homepage, which seems to be a substantial obstacle in the initial execution of tasks.

Redesign of issue #1: Adoption of a non-technical vocabulary

Given the insistent use of specific terminology in the first tutorials and on the homepage we believe that a simplification of the concepts is needed

since the language obstacle leads elderly users to rely on external help before the actual execution of the task.
The adoption of a user-friendly vocabulary will help the user in the first step, reducing the initial requests for guidance highlighted both in the interviews and in the user testing (see Giovanni). Replacing the first two text chunks in the homepage with a reformulation of the concepts in order to facilitate the interpretation of the tasks can be particularly beneficial for older users or those unfamiliar with the concept of multi-level authentication.

Redesign of issue #2: Adjusting text sizes to reduce visual effort
Since texts, including the chunks belonging to tutorials, are extremely thin and small in size (12-14px), contributing to the low contrast problem between the blue and white (palette chosen for PA application), users with poor eyesight tend to ignore initial explanations and important information because they are often illegible. Focusing on improving the legibility of texts can help the user focusing on tutorials and instruction, relying on their own interpretation.
The requests for guidance during the test of older users (Lucia and Giovanni), were often related to the inability of reading small texts without proper glasses or due to small texts that were ignored even if very significant for the completion of the task i.e. while the user was on the “Inserisci il PIN” page and needed to recover it, Giovanni and Lucia asked for help instead of reading the verbose chunk of text that explained how to recover it because it was small and looked unnecessary.

Redesign of issue #3, #5, #8 and #9 - Including dynamic feedback and visual cues during NFC scanning alongside with contextual assistance
To overcome the wariness regarding NFC technology, which many older users have never used, clear visual cues and dynamic feedback must be provided. Firstly, it is necessary to provide a clear tutorial every time the user tries the action. When accessed once, do not appear again the second time. If the user did not manage to conclude the task the first time, no other guidance is provided.
Another frequently encountered error is the user detaching the card from the phone while the reading was not completed yet. No dynamic feedback

is provided. The redesign must include a timer that communicates exactly the time until the end of the task to avoid errors and repetitions.
The extraneity to NFC technology should be also solved with an integration of clear, legible explanation on where to position the card during the scanning, since many errors occur because the card has been positioned incorrectly.
Finally, the UX can avoid users' anxiety by clearly inserting a question mark icon providing assistance contact that the user can immediately use and the possibility to replay the initial tutorial.

Redesign of issue #4 - Replacing hidden navigation menu with high-discoverability tab bars

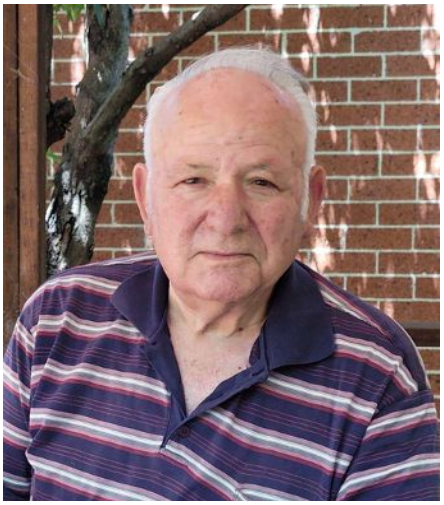
The application navigation is not obvious to novice users and non-experts. It is characterized by fine lines and could be mistaken for an icon of another nature. The necessity to make navigation clear from the very first visit to the app leads to a redesign focus on the adoption of tab bars, in order for the user to immediately know which sections are available, what are the functions of the app and where to find contacts and tutorials.

PHASE B

Persona and use case card

This persona is the protagonist

Photo of persona



Name of persona

Nicola Fontana

Meaningful quote

“Voglio continuare ad essere indipendente il più a lungo possibile”

Corresponding segment

The Autonomy-Seeking Senior - Direct user (1C + 2B + 3C +4B + 5A)

Description of persona

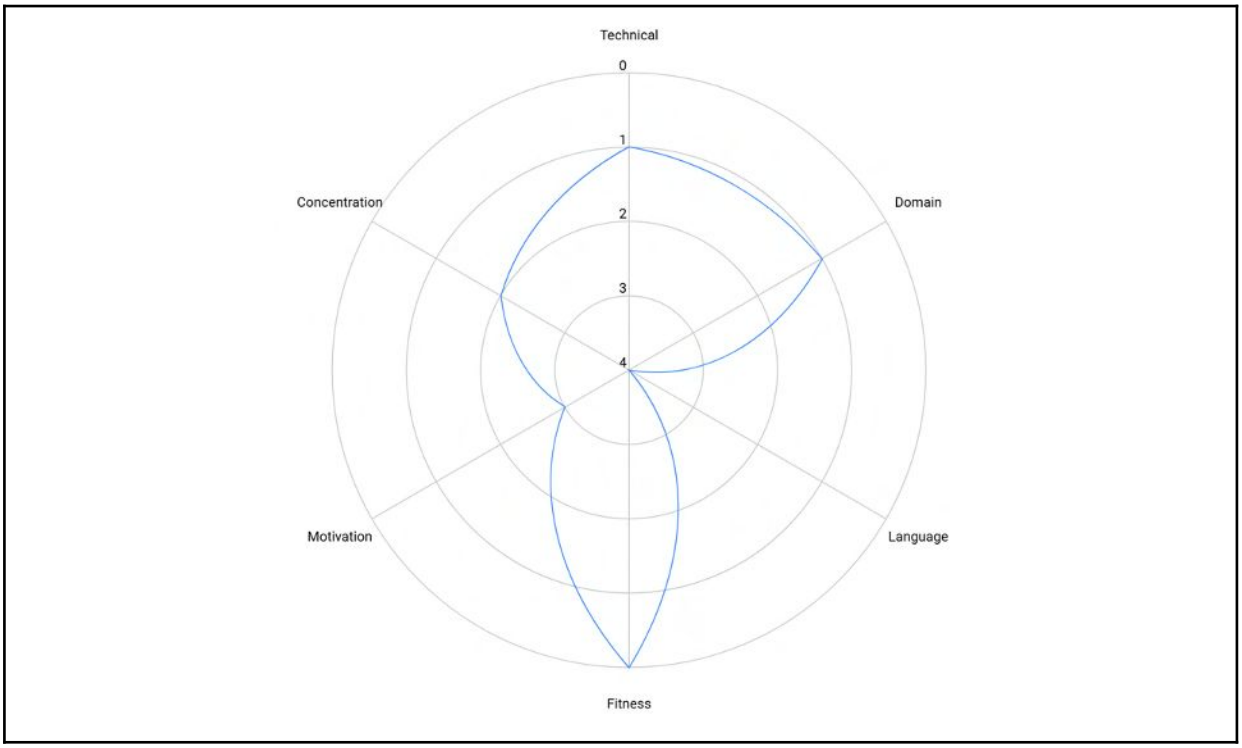
Basic facts

- 78 years old man
- Male
- Lower secondary education
- Retired mechanic
- Widow with three sons
- Suffers from slight cataract that prevents him to properly see and age-related recall limitations

Background and psychological characteristics

Nicola Fontana is an elderly man, retired for about ten years, living alone in his home in Teramo, Abruzzo. He was born and raised there, working as a mechanic since he was 15 years old and dedicated the majority of his life to his work, which he was fond of. Nicola became independent before reaching the age of majority. He used to help his family economically and has felt the responsibility of its well-being since he was very young. He married young and had three kids with his wife, Carola. They decided to live close to both their families by buying a house in Teramo, in the place they loved. Since his wife passed away, two years ago, he's been living alone. His sons call and visit him often since they are concerned about their father's mental and physical health but he demonstrates to still be able to take care of himself. He is still very autonomous in everyday tasks and doesn't rely on anyone. Nicola grew up that way and it is very difficult for him to ask for help. One of the things that puts him in a difficult place is technology. He knows how to call and message his loved ones but doesn't access the internet often since he is not very interested and finds some of the operations difficult to carry out. When obliged to ask for help, he feels like a burden and he really wishes to be independent with technology too.

Chart of personality traits



Additional interesting details

- Very skeptical about technology, Nicola has always preferred a face to face interaction and paper over a screen but, as a matter of pride, he really wished to be autonomous.
- He feels insecure while reading instructions that use very specific terminology and wishes to be able to understand even complex concepts.
- He gets discouraged by long instructions, and the need for glasses often deters him from starting a task when he doesn't have them at hand, leading him to give up if the matter isn't urgent.

Description of use case #1

Narrative of use case

Nicola had to renew his ID card. The municipality operator handed to him the new version the CIE (Carta d'Identità Elettronica) accompanied by a sheet of paper with two codes composed of 4 digits each. The operator suggests that he uses them to register his CIE on the CieID application to get access to all public administration online services including online health services such as FSE (Fascicolo Sanitario Elettronico) or the INPS portal to consult the pension slip.
Nicola is interested in the service since the idea of performing tasks online avoiding relying on strangers is appealing so, once at home, he downloads the app and is immediately introduced to a tutorial. In order to read it he has to wear glasses since he hardly distinguishes the written texts from the background, due to the low contrast.
While scrolling the tutorial he finds himself wondering what are CIE credentials (level 1 and level 2) and what it means to access via card reading (level 3). He starts to be annoyed since it is not clear to him what he must do with the CIE.
Once the tutorial is finished he finds on the screen two options recalling the concepts he did not understand before. He clicks on the "Accesso semplificato" options. This redirects him to "Inserisci credenziali CIE". He fills in the first field but does not have a password. He clicks on the option to activate credentials and is redirected to the webpage of the

Ministry of the Interior. He is slightly disoriented but still manages to fill the fields with Codice Fiscale and serial number of the CIE. The harsh part consists in filling the gap under the “Risolvi in captcha”, a hard to decipher code which he finds very confusing.
After the activation of the credential he is finally able to activate the card but the time required to understand the task and effort did put him in a difficult position, reinforcing his idea that technology is a complex matter.

Description of use case #2

Narrative of use case

Nicola is curious about the level 3 of security. Still unfamiliar with the concept he returns to the app after the level 1 and 2 registration to try to reach the maximum level of security, since skeptical about privacy and security guaranteed by technological devices.
Clicking on the right option he is not shown the tutorial again on how to perform the task he starts feeling uncertainty and anxiety.
He is asked to insert the PIN, composed by two 4 digit sequences divided in two different sheets of paper. Scarce and partial instruction lead him to search for the code on the CIE. After several attempts he manages to insert the right code and he is required to place the card against the back of the phone. Without clear guidance he tries to carry out the task but the absence of visual cues lead Nicola to improperly position the card in the middle of the phone instead of the upper part. After many trials, he manages to find the right position autonomously, wishing the task would be less time consuming. During the execution he separates the CIE from the device too soon. A feedback signals an error regarding the incompleteness of the action and the necessity to carry out the same task once more. Nicola tries again and he finally manages, after several trials, to keep the card close for as long as necessary. The lack of dynamic feedback confuses Nicola about the time required.

Concluding remarks

Impact of personas on design

Encourage reliance on the application instead of the intermediary:
The redesign must provide tools to allow the user to rely on the application, which should guide him and assist the user through the use of tutorials and visual cues. The system must focus on making the elder user independent and lighten the burden of the intermediary's role.

Accommodate users' level of digital skills and domain related knowledge:
Users' skills and literacy level must be accommodated. The redesign must aim at the ability of the user to comprehend the scope of the application, the features available and the tasks execution. It must minimize doubts, anxiety and the need to request external guidance.

Reduce possibilities to make errors:
By insisting on the insertion of tutorials and facilitating the retrieval of them and by adjusting size of text chunk, focusing on just a few, straightforward instructions, the redesign must minimize the number of errors, leading the user to rely on the app and not fear the system.

Impact of use case #1 on design

Simplify complex and domain-related concepts:
Leverage on a simplified vocabulary, introducing authentication concepts in the homepage in a straightforward language.

Centralize verification workflow:
Centralizing the entire verification workflow natively within the mobile application eliminates external site redirections while replacing visual CAPTCHAs with seamless OTP verification, ensuring a continuous, accessible and frictionless user journey

Impact of use case #2 on design

Include dynamic feedbacks and easy assistance retrieval:
During NFC card reading, the user should be guided by a timer that clearly signals the time necessary for the complete and correct execution of the task. Question mark icons to retrieve assistance contacts and also

including a FAQ section with answers to frequently asked questions should also be available in every page of complex tasks in order to immediately provide a way for the user to get assistance.

Provide frequent tutorials and clear explanation:
The app should always provide a (skippable) tutorial at the beginning of complex tasks in order for the user to be guided each time and avoid the stress of recalling forgotten operations. The tutorial of the NFC card reading must show unequivocally how to position the card on the back of the phone, highlighting the upper part of the dispositive to avoid misunderstandment.

This persona is NOT the protagonist

Photo of persona



Aisha Benali

Meaningful quote

“Sono sempre pronta a dare una mano, soprattutto alle persone che ne hanno più bisogno”

Corresponding segment

The Trusted Digital Mentor - Intermediary (1D + 2C + 3A + 4B + 5C)

Name of persona

Description of persona

Basic facts

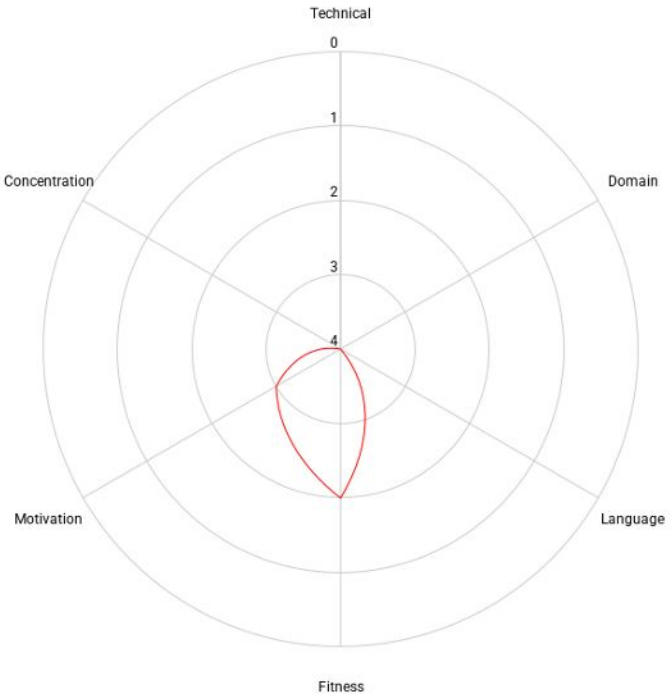
- 28 years old
- Female
- Italian
- Master’s degree holder
- Digital civil service operator

Background and psychological characteristics

Aisha is a young girl, recently graduated in a Master’s degree program and currently employed in the digital civil service program. She works at a front desk inside a library in Reggio Emilia, providing assistance to citizens, especially the elderly, who need help with tasks that require the use of an electronic device. She is a kind and helpful young woman, focused on carrying out her work properly. She is frequently in contact with people over the age of 70 who ask her for help, especially with public administration applications. She is always happy to help, but she often fails in her goal of enabling these citizens to use these technologies independently. It is frustrating for her to frequently see the

same people asking for the same kind of assistance, as she would like to help them become autonomous so that they no longer have to rely on her. She knows that for these elderly citizens, coming to the front desk is often a burden, an effort that could be avoided if only there were a way to teach them how to use these systems once and for all. She strongly desires that applications be accessible to everyone, and not just to those with a high level of education or to digital natives.

Chart of personality traits



Additional interesting details

- Chronically online, she knows the web and how it works inside out, yet struggles to find a simple way to explain concepts that, to her, are already fundamental.
- She prefers to get things done quickly, wasting as little time as possible

Description of use case #3

Narrative of use case

Aisha is working her shift at the library when a woman over 70 approaches her to ask for help with the cieID application. The lady wishes to proceed with the card's third-level authentication but says she is stuck in the process. When she clicks on "Accesso con lettura carta" she is asked for her PIN, which she cannot recall. She mentions, with some difficulty, that she read the instructions directing her to the “Gestione carta” section. She says she tried clicking on the text in bold, following the mental model she had developed from her limited experience with applications; however, what she thought was a link didn’t work, and the lady claims to be stuck. She cannot find this function on the homepage either and is turning to Aisha precisely because she does not know where to start.
Aisha picks up her mobile phone and points out the small, thin "hamburger" icon that hides the menu; she shows her that, tucked away inside, there is the "Gestione carta" section, along with other potentially useful features. The woman understands, and, seizing the opportunity, asks Aisha to recover the PIN for her. Aisha tries to explain the steps, but the woman has already lost interest; she believes the application is beyond her grasp and does not trust her own abilities. After a few attempts and observing the woman’s "learned helplessness," Aisha decides to take over the task entirely. In this instance, too, Aisha knows that the woman has learned almost nothing and will, in all likelihood, turn to her again, because the application has failed to instill any sense of self-efficacy in the woman.

Concluding remarks

Impact of personas on design

Integration of the Help-People-Help-People (HPHP) paradigm into system design:
Implement an explicit highly visible navigation to enable elderly direct users to independently discover support channels, contextual tutorials

and secondary services without relying on an intermediary to navigate the system.

Include scaffolded guidance to bridge the digital literacy gap between intermediaries and elderly users:
intermediaries frequently possess high domain and technical competence, which creates a cognitive bias known or an empathy gap toward senior citizens. When faced with complex administrative jargon, tech-savvy intermediaries often find it difficult or time-consuming to rephrase concepts into plain language. As a result, out of frustration or a desire for fast execution, intermediaries tend to perform tasks on behalf of the senior rather than instructing them, leaving the elderly citizen in a state of learned helplessness. The redesign should directly equip intermediaries with contextual microcopy, plain-language scripts, and step-by-step coaching prompts embedded within the workflow

Impact of use case #3 on design

Bottom Navigation Bar (Tab Bar):
The redesign must replace the indirect navigation drawer to a persistent and visible bottom navigation bar that exposes primary functional pathways directly on the primary viewport with high contrast icons and explicit text labels, so that the interface offloads memory requirements onto visual signifiers.

Aligning UI signifiers with user mental models - converting static microcopy into direct in-context recovery shortcuts:
On the "Inserisci PIN" screen, the instructional microcopy informs users who have forgotten their PIN to navigate to the "Gestione Carta" section. However, the phrase "Gestione Carta" is formatted strictly as static bold text rather than an actionable UI element.
To respect established interaction mental models and facilitate error recovery, the static text reference is converted into a contextual interactive hyperlink (deep-link shortcut). Tapping "Gestione Carta" should trigger a direct navigation path to the PUK/PIN recovery section, providing an immediate, frictionless recovery pathway right at the point of failure.

This persona is NOT the protagonist

Photo of persona



Name of persona

Tina Zandi

Meaningful quote

“Aspetti per favore che traduco quello che mi ha detto!”

Corresponding segment

The International IT Student - Secondary User: (1D + 2C + 3D + 4B + 5C)

Description of persona

Basic facts

- 26 years old
- Iranian
- Enrolled in an English-taught Master's degree related to Information Technology
- Italian language proficiency is at a basic A2 level
- Urgently needs to set up her digital identity credentials (like CieID) to access public administration services

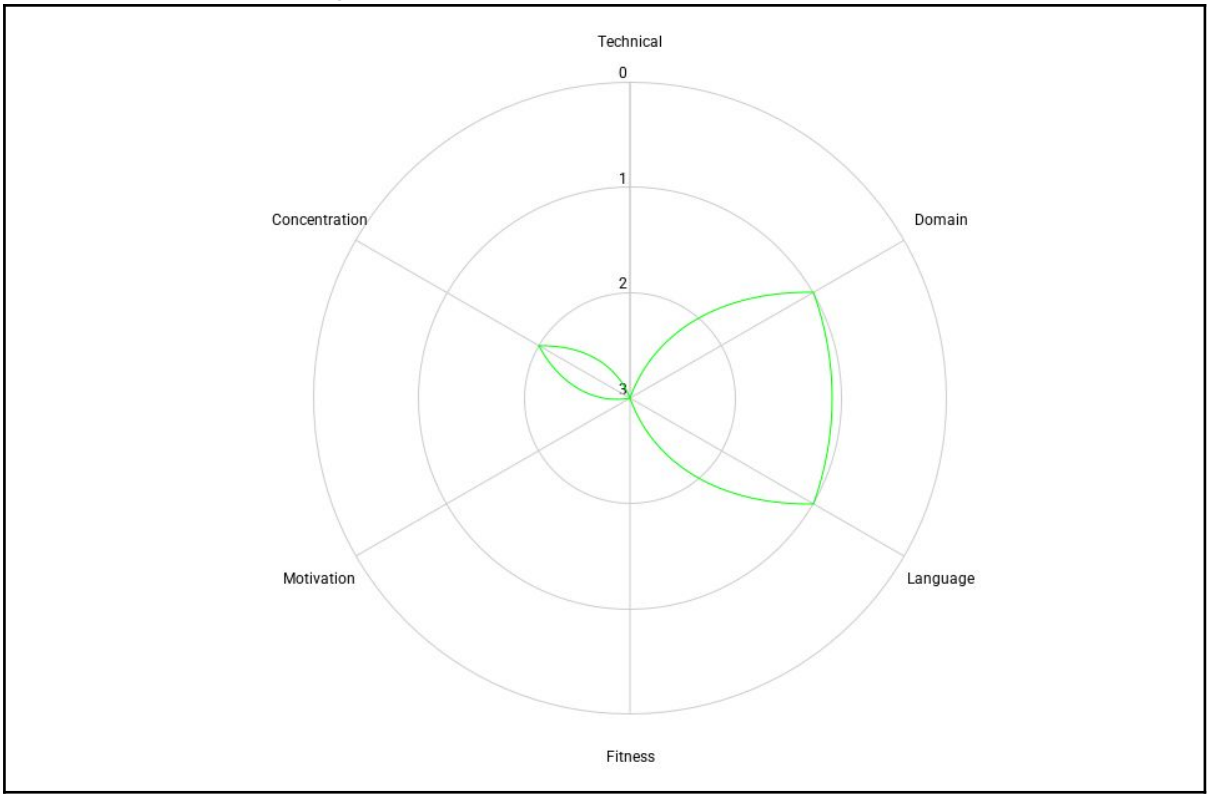
Background and psychological characteristics

Having navigated complex bureaucratic systems in Iran, Tina is not intimidated or surprised by the convoluted nature of Italian bureaucracy. Instead, she approaches it methodically, actively constructing a clear conceptual model of the process flow to understand exactly what action is required at each step.

Professionally, her background in the IT sector and her experience teaching English have trained her to be highly adaptable; she is excellent at learning languages and her brain is well-adapted to translating concepts on the fly. Furthermore, because Farsi is rarely supported in global software systems, she is completely accustomed to using applications in non-native languages.

To compensate for her A2-level Italian, Tina heavily leverages her technical expertise. She relies on "knowledge in the world" (such as familiar UI layouts, visual signifiers, and standard mapping) rather than "knowledge in the head" (memorizing Italian bureaucratic vocabulary). She instinctively follows standard UX patterns to deduce how an application works, allowing her to get her job done efficiently even when she cannot fully read the text.

Chart of personality traits



Description of use case #4

Narrative of use case

Tina is at the university library and needs to activate her CieID (“Livello di sicurezza 3”) to access the Comune di Bologna portal and finalize her residenza request. She downloads the app and immediately looks for an English language option, but finding none, she is forced to proceed in Italian. Thanks to her IT background and reliance on “knowledge in the world”, she navigates the initial screens using standard UX mental models without reading every word.

However, the flow abruptly halts on the “Inserisci il PIN” screen prior to the NFC card scan. A dense wall of bureaucratic Italian text asks for her 8-digit CIE PIN, while a static note at the bottom mentions using the PUK code in “Gestione carta” if the PIN is forgotten. Tina holds her municipal paper receipt containing the PUK, assuming this is the only code she needs to complete the process. She had actually set up her PIN previously, but the confusing terminology makes her think the PUK is the final master code.

Stuck on the text-heavy screen and afraid of making a mistake, she asks her Italian classmate, Marco, to translate. Marco explains that the app is asking for the PIN first, then the PUK. Since Tina doesn't realize she already has the PIN, Marco has to read through a long, complex paragraph detailing the PIN recovery process. Exhausted by trying to translate the bureaucratic jargon into English, Marco sighs, takes the phone, and says, “It’s too complicated to explain, I'll just request the recovery for you”. He taps through the forms, initiating a recovery via mail that forces a frustrating 48-hour waiting period to complete the task.

The ultimate irony occurs two days later: Tina realizes she actually had her PIN saved in her notes all along. The app’s confusing information architecture, the intimidating “livello 3” jargon, and the lack of clear visual distinctions between the codes misled her. Because the interface relied on dense text instead of visual guidance, Marco was forced to act as a quick “executor” rather than a mentor, resulting in an unnecessary 48-hour delay and leaving Tina feeling dependent rather than empowered.

Concluding remarks

Impact of personas on design

Prioritize “Knowledge in the World” over “Knowledge in the Head”: Because Tina has low linguistic competency but high technical competency, the interface must rely on universal iconography, and clear visual hierarchies rather than text-heavy explanations.
Implement clear Conceptual Models: Provide a visual roadmap or progress tracker for complex bureaucratic processes. If Tina can visually see the overall structure of the task, her IT background will allow her to deduce the required steps intuitively, bypassing the need to read dense Italian paragraphs.
Design for Intermediation (HPHP): To prevent the intermediary from taking the phone and becoming an “executor”, minimize walls of text. Text should be broken down into scannable, bite-sized bullet points so an intermediary can glance at the screen and instantly explain the concept to Tina.

Impact of use case #4 on design

Multilingual Support (Italian/English toggle): Implement a direct and easily accessible option to switch the interface language from Italian to English. For an international user with high technical competency but low linguistic competency, translating technical bureaucratic jargon (like “livello di sicurezza 3” or PIN/PUK recovery procedures) is the primary friction point. A simple language toggle eliminates the need for an intermediary to translate walls of text, restoring the user's full autonomy.
Visual Distinction of Credentials: The confusion between PIN and PUK must be resolved visually. Design distinct graphical representations (e.g., different icons, color coding, or animations showing where the code comes from) for the PIN and the PUK so users do not rely solely on text labels to tell them apart.
Contextual Visual Guidance for Complex Tasks: The “Livello di sicurezza 3” activation, NFC card positioning, and credential recovery flows must be redesigned with graphical, step-by-step animations and timers so users do not have to rely on dense text instructions.
De-jargonize the Interface: Replace intimidating bureaucratic terms like “Livello di sicurezza 3” with plain, accessible language and pair them

with visual indicators of security levels. This reduces cognitive load for both the direct user trying to guess the meaning and the intermediary trying to translate it on the fly.

This persona is NOT the protagonist

Photo of persona



Name of persona

Matteo Rinaldi

Meaningful quote

“Prima o poi prenderò l'appuntamento dal medico, da solo!”

Corresponding segment

The First-Time Bureaucracy Experiencer - Secondary User: (1D + 2B + 3B + 4B + 5C)

Description of persona

Basic facts

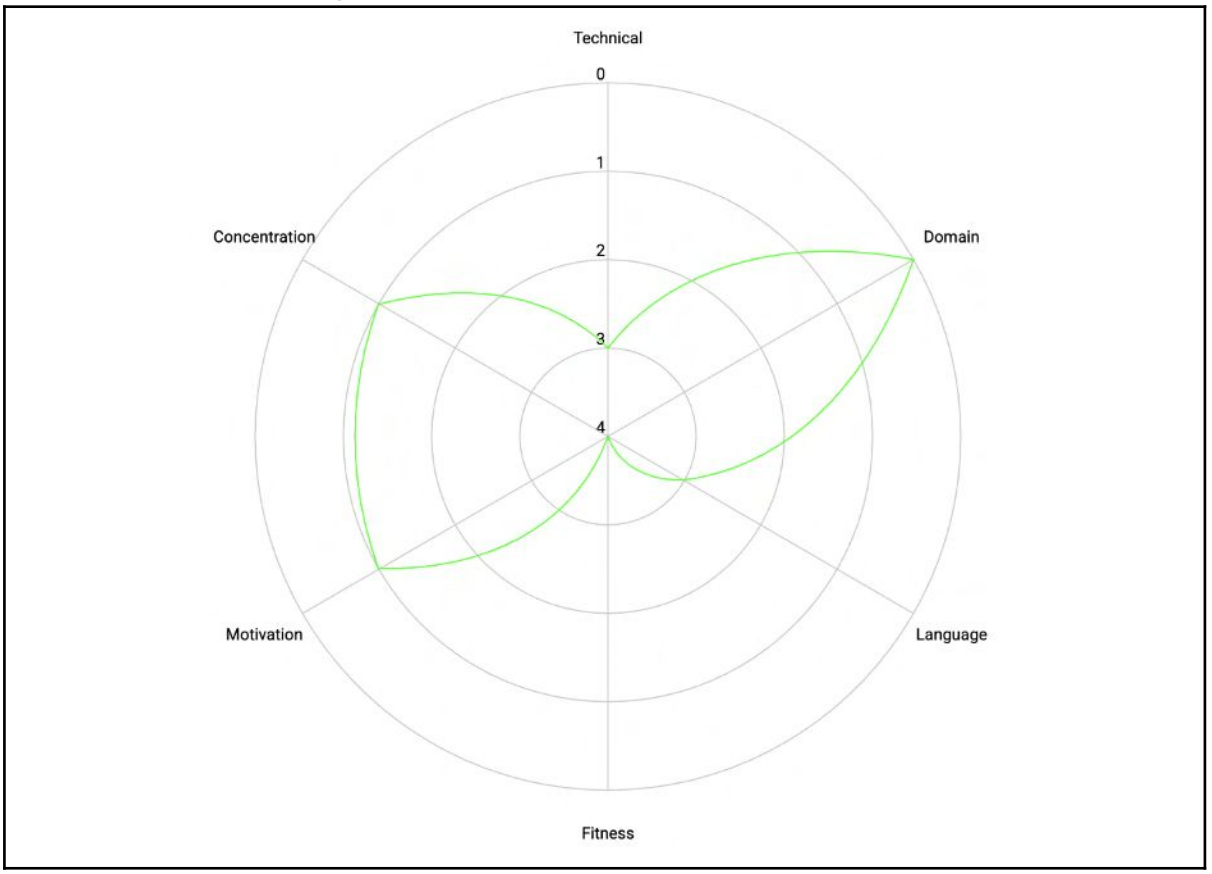
- 18 years old
- Italian
- High school senior preparing for university admission
- Mobile-native user
- Needs to set up digital identity (CieID / SPID) for the first time independently

Background and psychological characteristics

Matteo is a classic "smartphone-only" young user: he is hyper-fluent in social media, gaming, and entertainment apps that feature frictionless onboarding (such as one-tap "Sign in with Google/Apple" or FaceID), but has zero experience with public administration or adult bureaucracy. As observed in front-desk interviews, 18-year-olds receiving their CIE for the first time frequently require assistance because they have never encountered concepts like multi-factor security tiers, municipal envelopes, or split PIN/PUK codes.

Psychologically, Matteo has a very short attention span and low patience for institutional walls of text. He skips introductory tutorials automatically, assuming he can figure out any app through trial and error. When confronted with dry bureaucratic jargon ("Livello 1, 2, 3") or asked to retrieve a physical paper receipt given to him weeks earlier at the municipality, he immediately loses focus and delegates the problem to an adult or front-desk operator rather than reading instructions.

Chart of personality traits



Description of use case #5

Narrative of use case

Matteo needs to register for his university entrance exam before the deadline, which requires logging in via CieID. He downloads the CieID app and immediately skips the 5-step welcome tutorial without reading a single word. Landing on the Home Page, he is presented with two options, "Accesso semplificato (Livello 1 e 2)" and "Registra la carta (Livello 3)", without any plain-language explanation of what they mean. Assuming he can just type his email like on Instagram, he taps around randomly until the app demands an 8-digit PIN and NFC card registration. Matteo has his new plastic CIE card in his wallet, but he has no idea where the 8-digit PIN is. When the app mentions that the first half of the code was given at the municipality and the second half arrived with the card, Matteo panics and calls his older sister, Chiara (28, an experienced digital user acting as his intermediary), into his room. Chiara asks him for the paper envelopes from the *Comune*. Matteo rummages through his desk, finds only the second envelope, and complains that the app is broken. Chiara tries to show him where to check his PUK or recover his credentials inside the app, but because those options are hidden inside a thin hamburger menu and separated from the active task screen, Matteo gets impatient and shoves his phone into her hands, saying, "Fai tu, io non ci capisco niente di queste cose del Comune". Instead of Chiara guiding Matteo to become an autonomous adult citizen, the lack of bite-sized, engaging visual prompts and the hidden recovery tools force Chiara to act as an executor while Matteo goes back to scrolling on his tablet.

Concluding remarks

Impact of personas on design

Capture Attention with Visceral, Bite-Sized Cues ("Capture me!" and "Explain!"): Because Matteo has low domain competence and low concentration, the interface must avoid long introductory tutorials that users skip. Instead, use micro-interactions, bold visual highlights, and progressive disclosure to keep his attention anchored to one simple action at a time.

Bridge the Domain Gap for Young Novices: Never assume that high smartphone familiarity equals bureaucratic literacy. Translate institutional concepts ("Livello 3", "PIN/PUK") into everyday mental models familiar to young users (e.g., comparing CIE NFC scanning to Apple Pay/Google Wallet contactless payments).

Impact of use case #5 on design

Pre-Task "Document Checklist" with Visual Previews: Before starting card registration, display a 5-second visual checklist showing exact illustrations of the physical items needed (the plastic CIE card + the two municipal paper sheets) so impatient users do not start the flow unprepared and get stuck mid-way.

Interactive Co-Use Mode for Intermediaries: Provide clear, step-by-step visual cards that allow an intermediary (like an older sibling or front-desk operator) to point at the screen and guide the young user through finding their codes without taking the phone out of their hands

This persona is NOT the protagonist

Photo of persona



Name of persona

Elena Ferri

Meaningful quote

“Al lavoro uso sempre gli stessi tre pulsanti sul computer dell'ospedale. Se sbaglio a cliccare qualcosa qui sul telefono, rischio che i miei figli restino senza mensa scolastica!”

Corresponding segment

The Anxious Routine-Bound Parent - Secondary User: (1A + 2A + 3B + 4B + 5B)

Description of persona

Basic facts

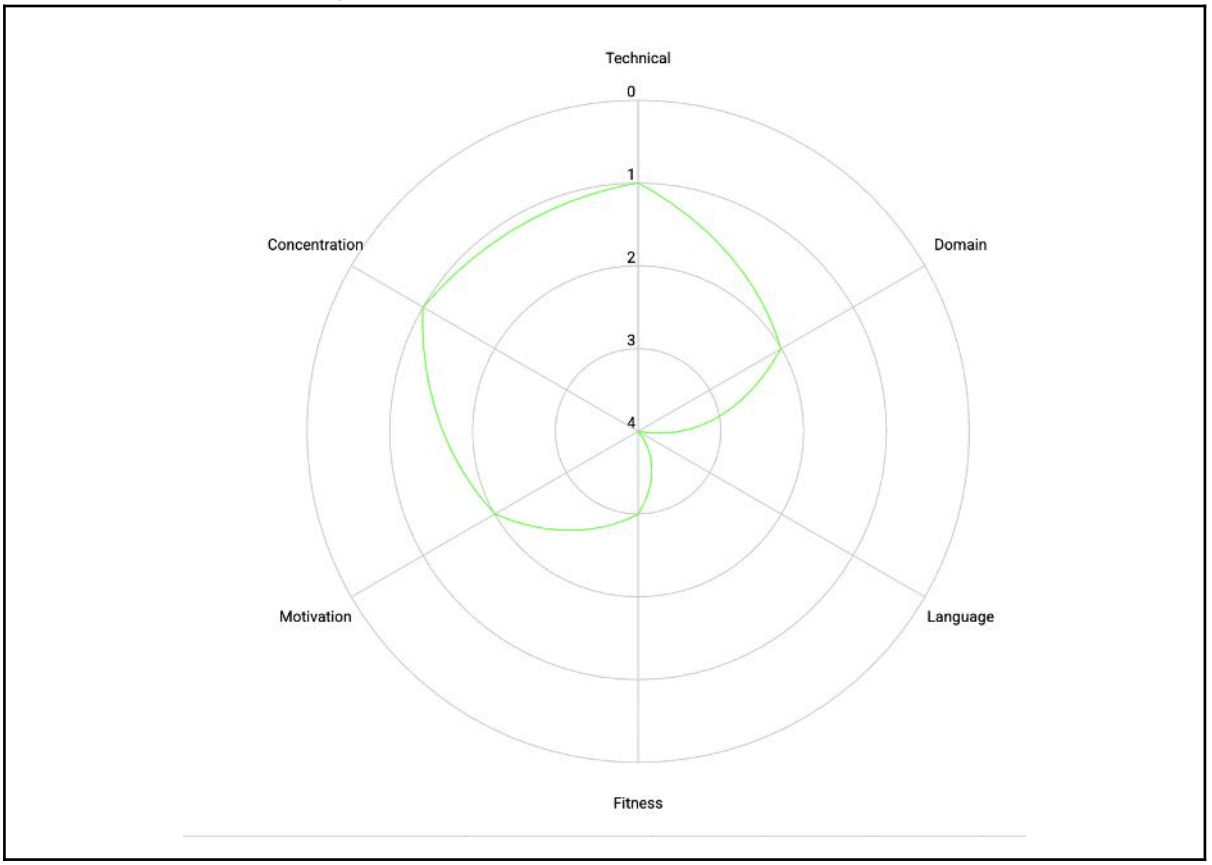
- 39 years old, Italian, mother of two children aged 7 and 10.
- Works full-time in the administration office of a public hospital in Bologna.
- Uses a modern smartphone (bought for her by her husband), which technically supports NFC, though she only uses it for WhatsApp, photos, and phone calls.
- Needs to use CieID to access the municipal school portal to enroll her kids in the school canteen (mensa scolastica) and pay the bollettini.

Background and psychological characteristics

Elena represents a crucial paradox in technical competence: although she works in a hospital administration office in front of a computer every day, she only knows the 15% of the legacy hospital software strictly required for her daily routine. Outside of those memorized steps, she is deeply uncomfortable with technology and experiences high digital anxiety whenever an interface changes or asks her to configure something new.

Even when online services work well, parents like Elena do not trust their own ability to use them and strongly prefer calling the school office or relying on an operator out of fear of making an irreversible bureaucratic mistake that could affect their children. Furthermore, balancing a full-time hospital job and two young children leaves her perpetually rushed and mentally fatigued, meaning she has zero time or calm to decipher cryptic menus or recover lost codes.

Chart of personality traits



Description of use case #6

Narrative of use case

During her 30-minute lunch break at the hospital administration office, Elena tries to log into the municipal portal to submit her children's school canteen application before the deadline. When the portal asks her to authenticate via CieID, she opens the app on her smartphone. Because she hasn't used the app since her husband set it up for her months ago, she cannot remember her "Codice App ID" or her CIE PIN, and the screen gives her no clear clue on which code is being requested.
Afraid of locking her card after reading a warning about a 5-trial limit, Elena looks at the "Inserisci il PIN" screen, where the text reads "Vai in Gestione carta" in bold. Thinking "Gestione carta" is a clickable link, she taps it repeatedly, growing frustrated when nothing happens because it is just static text. Because the real "Gestione carta" and "Recupero PUK" options are hidden inside a thin hamburger menu on the Home Page, she has no idea how to proceed without abandoning her task.
Panicking that her lunch break is ending, she calls over her younger colleague from the IT desk, Lorenzo (the intermediary), begging him to fix it. Lorenzo wants to teach her how to reset the code so she can manage school payments from home in the future. However, when he sees that recovering the PIN requires navigating out of the flow, finding the hidden menu, and performing three separate NFC card scans without any on-screen progress bar, he realizes there isn't enough time before her shift resumes. Lorenzo takes her phone and card, performs the three NFC scans himself in two minutes, and logs her in. Elena completes the school form, but she learns nothing from the interaction,confirming her belief that she is "bad with technology" and will need Lorenzo to do it for her again next time.

Concluding remarks

Impact of personas on design

Instill Trust Through Error Prevention and Reversibility: Because anxious users like Elena fear making irreversible mistakes affecting their family, the UI must provide clear, reassuring feedback at every step, make actions easily reversible, and avoid anxiety-inducing warnings without constructive solutions.
Support Interrupted Workflows and Low Concentration: Busy parents and office workers need a clear visual progress indicator showing where they are in a task and the ability to pause and resume without losing their place.

Impact of use case #6 on design

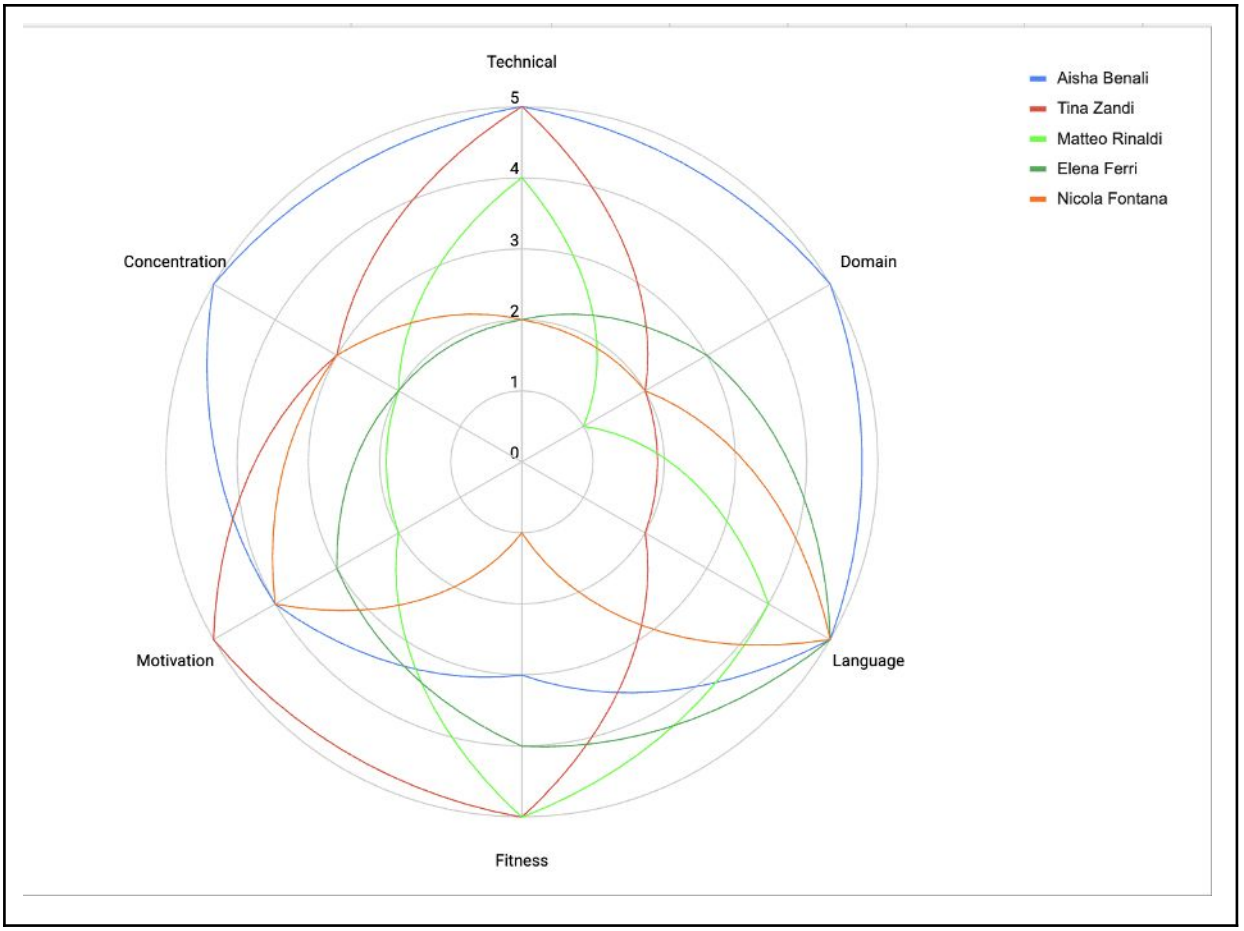
Eliminate False Affordances and Provide Direct Contextual Links: Make inline instructions like "Vai in Gestione carta" actual clickable buttons that take the user directly to the recovery flow without forcing them to exit to the Home Page and hunt for a hidden hamburger menu.
Streamline Credential Recovery for Rapid Mentoring: Reduce the friction of the PIN/PUK recovery flow (eliminating redundant triple NFC scans) and surface a visible "Serve aiuto?" contextual overlay right on the PIN screen. This allows an intermediary in a time-crunched setting (like a colleague on a lunch break) to quickly coach the user through the recovery in under a minute rather than taking over the device.

Cast card

Cast with roles and segments:

- ***Protagonist***: Nicola Fontana - The autonomy-seeking senior (1C + 2B + 3C + 4B + 5A)
- ***Secondary persona:*** Aisha Benali - The trusted digital mentor (1D + 2C + 3A + 4B + 5C)
- ***Additional persona:*** Tina Zandi - The international IT student (1D + 2C + 3D + 4B + 5C)
- ***Additional persona:*** Matteo Rinaldi - The First-Time Bureaucracy Experiencer (18 year old) (1D + 2B + 3B + 4B + 5C)
- ***Additional persona:*** Elena Ferri - The Anxious Routine-Bound Parent (1A + 2A + 3B + 4B + 5B)

Cumulative chart of personality traits



Impact on design

Impact of cast on design

Req-1. **Develop a trustable system serving as a point of reference for the user (driven by Nicola and Elena):** This design dimension focuses on establishing a transparent, reliable, and reassuring interface capable of guiding the user through sensitive digital identity management tasks (such as Level 3 security activation). The application must never leave the user in a state of ambiguity: every critical action, waiting period, or potential error state must be preceded by explicit confirmations and clear reassurances.

Req-2. **Accommodate users' level of digital and linguistic skills (driven by Nicola and Tina):** focus on ensuring the interface remains accessible and intuitive regardless of a user's technical background or native language. The system must eliminate linguistic and cognitive barriers. The application achieves this by adopting plain, action-oriented microcopy over administrative jargon and providing accessible multi-language support.

Req-3. **Design to minimize errors (driven by Nicola, Aisha, Matteo and Elena):** proactively prevent interaction slips and operational dead-ends during digital identity management. By offering real-time feedback and forgiving interaction pathways, the design minimizes user errors, reduces anxiety and prevents intermediary task takeover.

Req-4. **Focused on HPHP paradigm and system mediation (driven by Aisha):** Transforms the intermediary's role from a passive executor into an empowered digital mentor. By actively mediating assistance, the system ensures the device remains in the primary user's hands, fostering long-term digital autonomy.

Req-5. **Bridging the gap between the elder user's technical knowledge and the technical language (driven by Nicola and Tina):** This design dimension focuses on translating complex administrative and technical jargon into accessible, everyday language. By aligning system vocabulary with users' mental

models, the design eliminates cognitive translation friction and fosters self-reliance.

Req-6. **Prioritize “knowledge in the world” over “knowledge in the head” (driven by Tina and Matteo):** minimizes memory load by embedding clear visual cues, persistent signifiers. By relying on visible recognition rather than recall, the system enables users to navigate and complete tasks intuitively without memorizing complex procedures.

Req-7. **Foster user self-efficacy and prevent learned helplessness (driven by Nicola):** empowers users to complete tasks independently by rebuilding digital confidence through forgiving, error-tolerant interaction paths. The interface uses step-by-step progress tracking, positive reinforcement, and safe exploration options to eliminate anxiety and prevent users from surrendering control to intermediaries.

Req-8. **Design for language accessibility (driven by Tina and Aisha):** the interface features an on-demand multi-language selector and bilingual scaffolding prompts for an inclusive, cross-language system.

Impact of use cases on design

Req-9. **Reformulation of complex and technical jargon (use case #1, #3, and #5):** Replaces intimidating administrative terminology with plain, action-oriented language grounded in physical artifacts.

Req-10. **Centralize verification workflow and replace CAPTCHA with OTP verification (use case #1):** unify fragmented security checks into a single guided workflow while replacing visually barrier-heavy CAPTCHAs with simple One-Time Passwords (OTP).

Req-11. **Include bottom navigation bar (use case #3):** replace the indirect navigation drawer to a persistent and visible bottom navigation bar that exposes primary functional pathways directly on the primary viewport with high contrast icons and explicit text labels, so that the interface offloads memory requirements onto visual signifiers.

Req-12. **Align UI signifiers with user mental models (use case #3 and #6):** To respect established interaction mental models and facilitate error recovery, the static text reference is converted into a contextual interactive hyperlink (deep-link shortcut).

Req-13. **Include multilingual support (use case #4):** Implement a direct and easily accessible option to switch the interface language from Italian to English.

Req-14. **Provide frequent tutorial and clear explanation (use case #2):** The app should always provide a (skippable) tutorial at the beginning of complex tasks in order for the user to be guided each time and avoid the stress of recalling forgotten operations

Req-15. **Include dynamic feedbacks and easy assistance retrieval (use case #2):** During NFC card reading, the user should be guided by a timer that clearly signals the time necessary for the complete and correct execution of the task. Adopts question mark icons to retrieve assistance contacts and also including a FAQ section with answers to frequently asked questions.

Req-16. **Include pre-task document checklist and co-use coaching cards (use case #5):** Display a visual preview of required physical documents (plastic CIE card and municipal paper receipts) prior to starting registration, paired with step-by-step visual cards so intermediaries can guide young novices without taking their phone.

Req-17. **Streamline credential recovery with contextual help overlays (use case #6):** Reduce redundant NFC scans during PIN recovery and add a visible "Serve aiuto?" contextual overlay on the PIN screen to support rapid mentoring during interrupted or time-pressured workflows.

PHASE C

Information Architecture card

Was a specific design phase for the Information Architecture of the tool appropriate and necessary? If yes, why? If no, why?

A specific **Information Architecture redesign** phase was essential because the original tool's usability barriers were deeply rooted in its structural hierarchy, navigation pathways and labeling system. Resolving these friction points required fundamentally rethinking how features, vocabulary, and recovery flows are organized and exposed to users. Specifically, the IA redesign addresses **core navigation friction**, **false affordances**, and **cognitive disorientation** through four primary interventions:

1. replacing the thin hamburger menu icon with an explicit **bottom navigation bar** clearly showing the sections of the application.
2. transforming the static textual reference to "Gestione carta" section on the "Inserisci il PIN" screen, into an active link, providing a **direct pathway** to the credential recovery section.
3. Integrating a persistent **contextual help icon** on screens related to a complex task (question mark) offering repeatable step-by-step tutorials and task specific FAQs with common questions and answers.
4. reframing complex technical vocabulary by replacing abstract administrative jargon on the home page with plain, **action-oriented labels**, aligning the labeling system with users' mental models.

These changes aim to improve the user experience and facilitate discoverability and learnability.

Was a top down or bottom up approach selected (or both) for the Information Architecture of the application?

- Mixed approach

If your choice is different from the existing tool, then justify your choice:

The redesign mainly focuses on discoverability to enhance the novice user experience by explicitly displaying every section and feature offered by the application. In this way the system does not rely entirely on users'

digital skills or “knowledge in the head”, which elderly users may lack. A reframing of technical vocabulary is necessary to introduce the user to the main features of the app with plain language, reducing uncertainty and anxiety. This top-down approach is complemented by a bottom-up strategy that rethinks fine navigation details to save time and effort, providing immediate, task-level support through targeted contextual assistance and FAQs sections.

What are the main decisions about the organization and labeling system? Are they different from the existing tool?

The redesign is based on the premise that, in an app designed to guide the user through the completion of tasks, all sections of the site should be prominent and clearly visible. Consequently, the redesign of the main navigation shifts elements previously tucked away inside a "hamburger" menu to a **clearly visible bottom navigation bar**. This exposes core application sections directly from the homepage, reducing tap-depth and eliminating reliance on memory ("Knowledge in the Head"). About the labeling system, we retained the standard section labeling across internal pages but reframed the initial homepage access cards to better **align with users’ digital literacy**. Cryptic domain terminology ("Level 1 and 2" vs. "Level 3") was replaced with concrete, action-oriented labels:

- “Accesso semplificato - configura il tuo dispositivo per accedere senza lettura carta (livello 1 e 2)” is now replaced by “*Accesso semplificato - accedi con credenziali CIE*”
- “Accesso con lettura carta - registra la carta per accedere con il massimo livello di sicurezza (livello 3)” is now replaced by “*Accesso completo con Carta d’Identità Elettronica (CIE)*”.

What are the most relevant metadata fields? Are they different from the existing tool?

Unlike the existing tool, which relied on minimal, domain-heavy metadata our redesign introduces task-oriented metadata fields directly mapped to our primary IA interventions:

- **Task Category** (drives the bottom bar): classifies screens into core macro-destinations (“Home”, “Gestisci Carta”, “Tutorial”, “Supporto”), enabling the persistent Bottom Navigation Bar to highlight active sections and manage global routing.
- **In-Context Recovery Target** (enables the PIN inline link): associates instructional text on the "Inserisci PIN" screen with an active target location ID (“CredentialRecovery”), transforming static text into a direct functional deep-link at the point of failure.
- **Help Context ID** (powers the ? assistance icon): links each complex task screen to its specific tutorial steps and FAQ set, ensuring that tapping the header ? icon dynamically pulls up relevant, task-specific scaffolding.
- **Access Method Type** (reframes homepage vocabulary): replaces cryptic administrative classification metadata ("Level 1/2 vs 3") with user-centered action tags (“Credential login" vs "CIE card scan"), aligning homepage microcopy with citizens’ mental model.

How do you plan to design the navigation system? **Is it different from the existing tool?**

Yes, the proposed navigation system fundamentally differs from the existing tool by transitioning from a concealed, high-friction model to a persistent, multimodal navigation architecture structured around three complementary layers.
First of all, regarding the global navigation, the redesign replaces the drawer with a persistent bottom navigation bar, exposing permanently the main application section (“Home”, “Gestione carta”, “Recupero PUK”, “Tutorial”, “Crediti”, “Privacy”, “Assistenza”). This provides **continuous visual orientation**, reduces tap-depth and lowers cognitive load through immediate recognition.
The redesign also implements **in-context deep-linking** at failure points. On the “Inserisci PIN” screen, the reference to “Gestione carta” becomes an active shortcut link, allowing direct modal navigation to credential recovery without breaking their workflow.
Finally, the new version embeds a **standardized help icon** (question mark) in the top right header of every screen involving complex tasks. Tapping it opens a task level drawer with step-by-step tutorials and

contextual FAQs providing immediate, non intrusive guidance directly where the task takes place.
These three navigation layers replace a rigid, drawer based pathway with an open, forgiving and self correcting navigation ecosystem designed specifically for low digital literacy users.

Browsing aids

The new CieID redesign addresses the visibility of browsing aids and replaces misleading UI cues with more appropriate tools that redirect the user. Specifically it introduces a persistent bottom navigation bar, in-context line deep-links, persistent contextual help (?) icon, visual progressor indicator during complex tasks to reduce disorientation and cognitive load.

Search aids

The application lacks search aids. The redesign does not address this aspect since the browsing aids allow the user to reach every section of the application and carry out any task.

Contents and Tasks

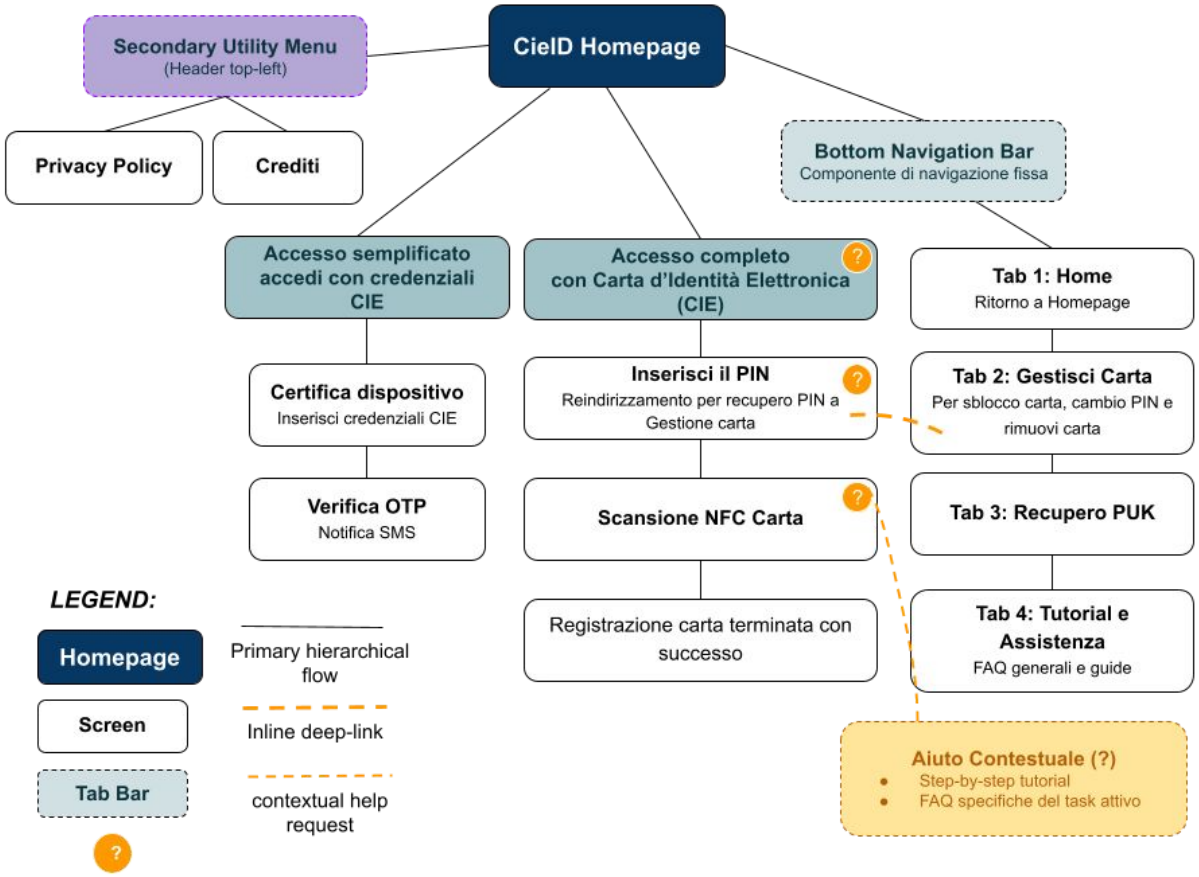
The redesign restructures the relationship between application contents and user tasks by adopting a task-centric, contextual architecture. Labels are organized around explicit user goals rather than abstract administrative categories ("Level 1, 2, 3"), aligning interface vocabulary directly with citizens' mental models. Instructional content (micro-tutorials, task guidance, and FAQs) is directly embedded within active task workflows via the persistent “?” help signifier and inline deep-links, providing immediate assistance right at the point of execution or failure. Complex multi-step tasks (e.g. NFC card reading) are decomposed into linear, manageable sub-tasks supported by clear visual progress indicators (“Step X of Y”), reducing cognitive load and preventing user disorientation.

Invisible components

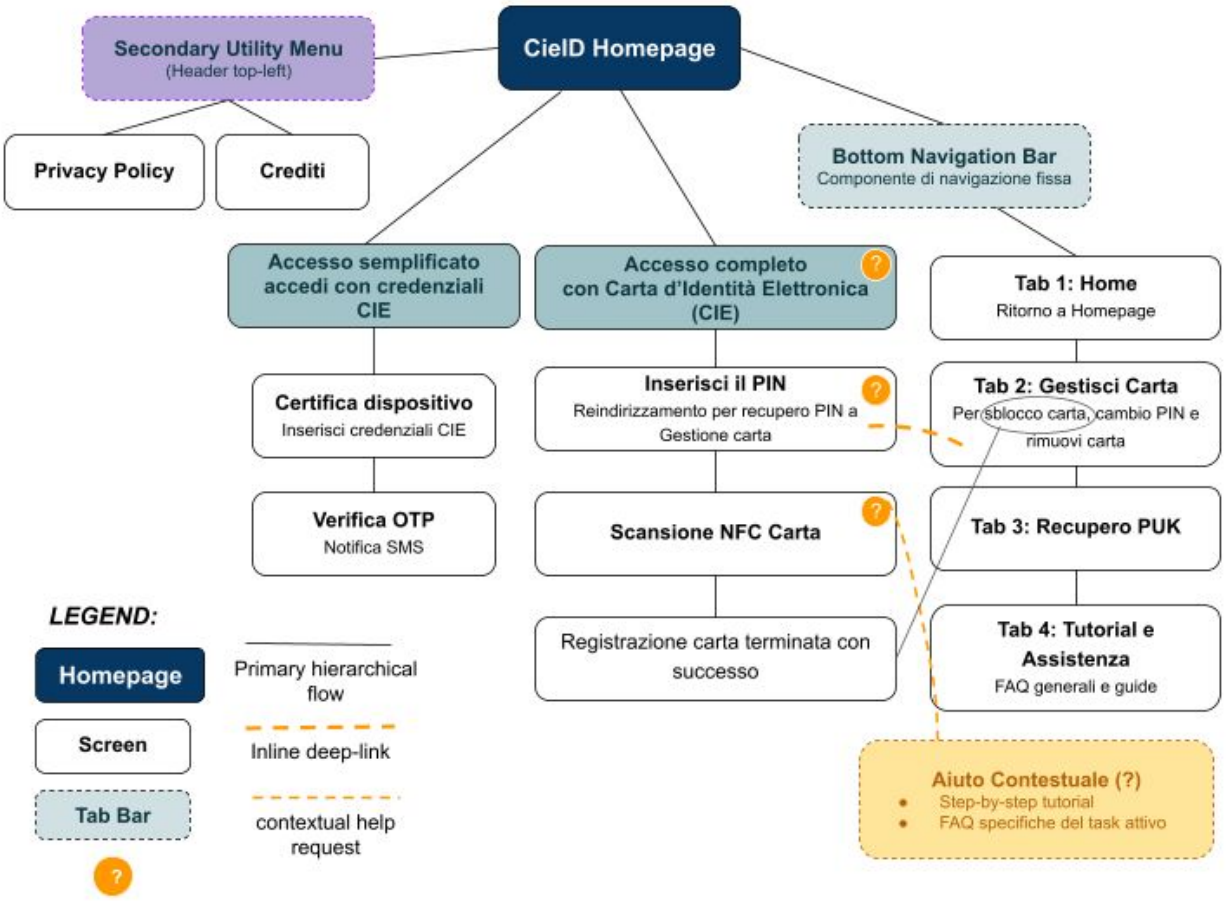
The invisible components of the Information Architecture encompass the underlying metadata, routing logic and system state tracking that power the user experience behind the scenes. They consist of background

attributes (“Task Category”, “Help Context ID”, “In-Context Recovery Target”) that dynamically map each screen to its specific tutorial package, error recovery targets and background monitoring of workflow progress (e.g. Step 2 of 4).

Blueprint for CieID redesign v.0/1

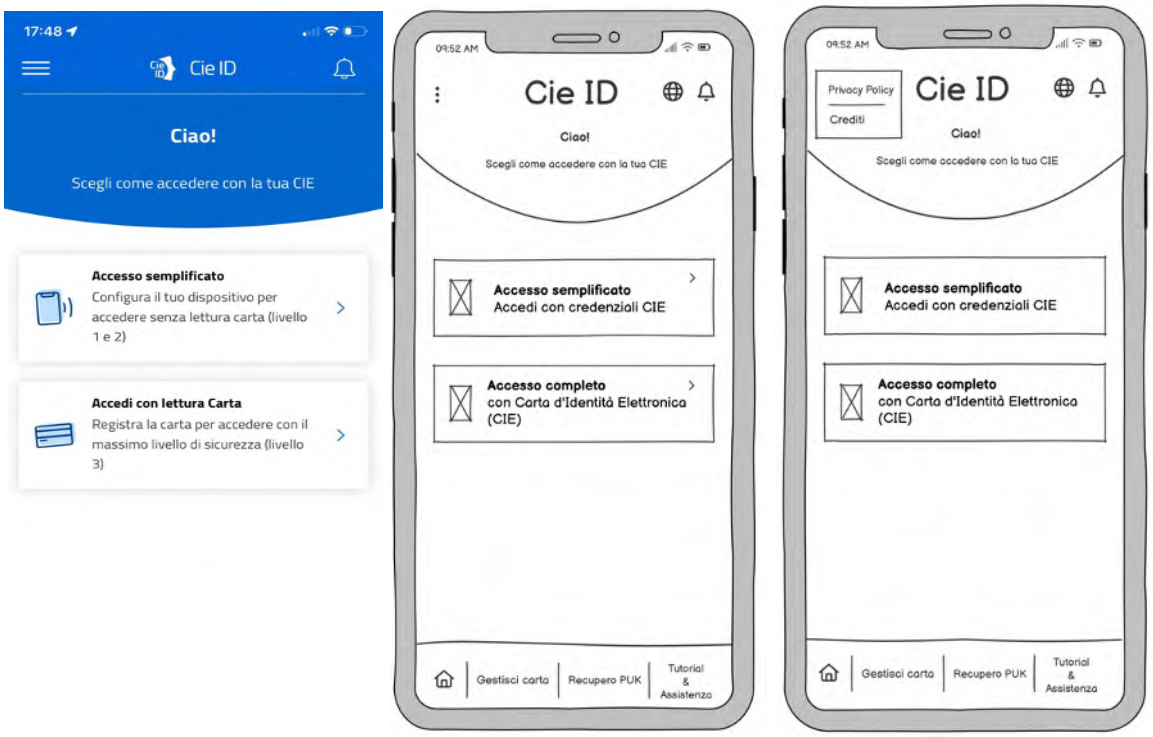


Blueprint for CieID redesign v.2



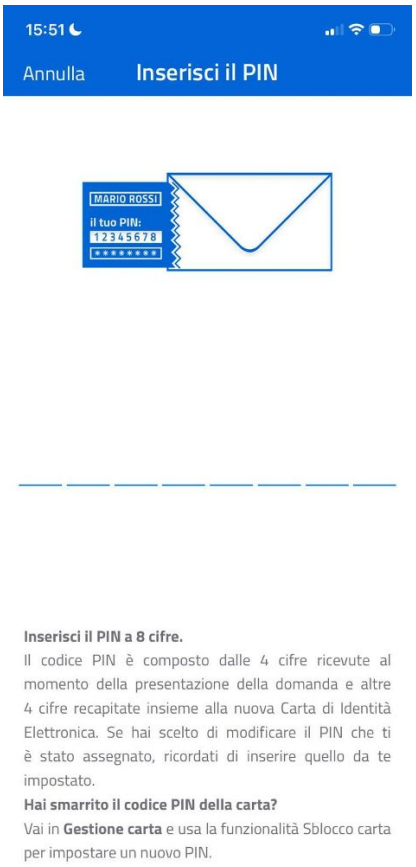
Before & After CieID
Redesign v.0

Homepage



The version 0 of the new redesign of CieID addresses the initial ambiguity of the homepage by **reformulating complex technical jargon** such as “Configura il tuo dispositivo per accedere senza lettura carta (livello 1 e 2) and “Registra la carta per accedere con il massimo livello di sicurezza (livello 3)” replacing them with straightforward and clear explanation.
The second intervention is focused on the **modification of the menu**, hidden under the hamburger icon in the original design. In version 0 the main sections contained in the site are explicit inside the bottom navigation bar while secondary information such as “Privacy Policy” and “Credits”, which do not address any task are placed inside the three dots icon top-right corner of the screen.
Finally, the **globe icon** next to the notification icon (top-left) represents the possibility to switch language from Italian to English, specifically addressing immigrant needs.

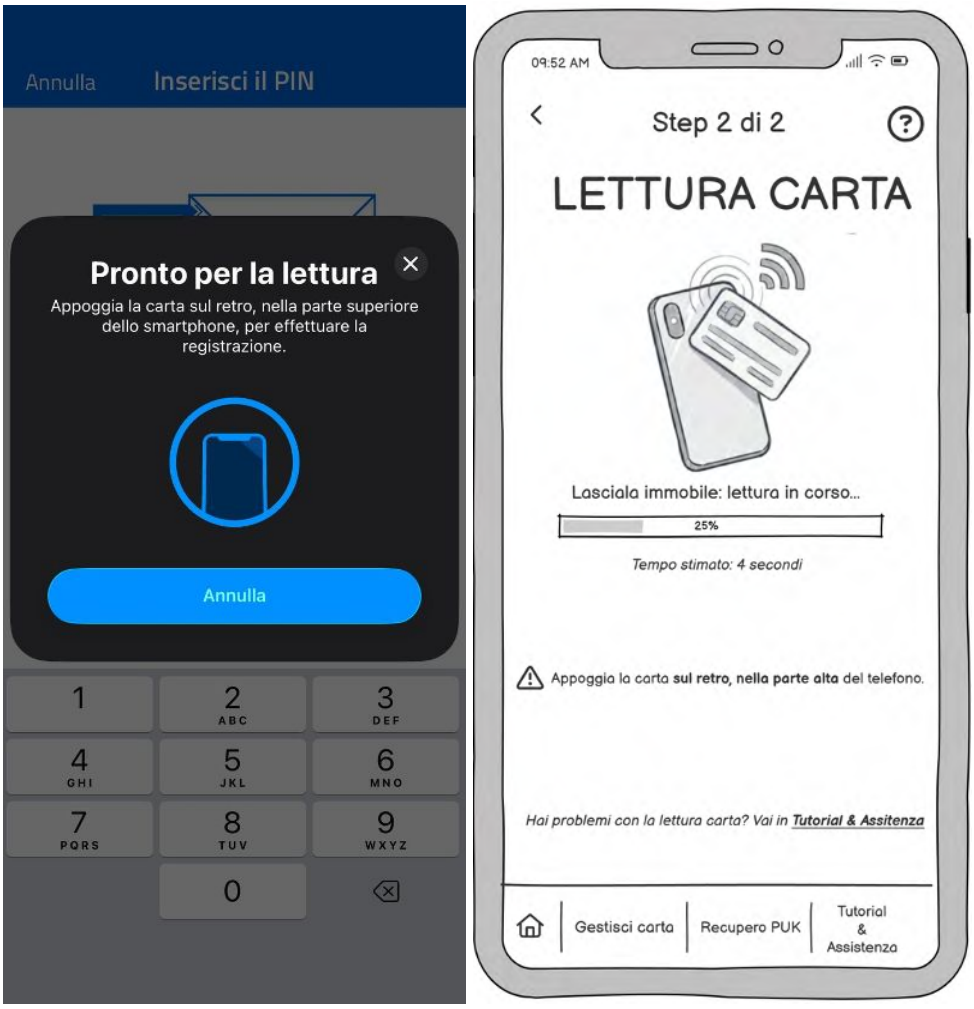
Inserisci PIN screen





The “Inserisci PIN” screen has been redesigned to **prioritize clear explanation**, previously condensed in a verbose and low contrast text chunk that doesn't capture the attention of users. In the version 0 the information about where to find the PIN are prioritized and placed in the middle of the screen.
As this is the first stage of the third-level authentication process, the screen indicates the user's position within the sequence of operations by displaying “Step 1 of 2”. In the top-right corner of the screen the question mark icon is the **shortcut to directly access FAQs, assistance and tutorials** which are selected according to the current task.
Under the entry field of the code the redesign transforms the static redirection “Vai in Gestione carta” of the original version into **an inline deep link**, reducing the taps required to reach the destination screen.

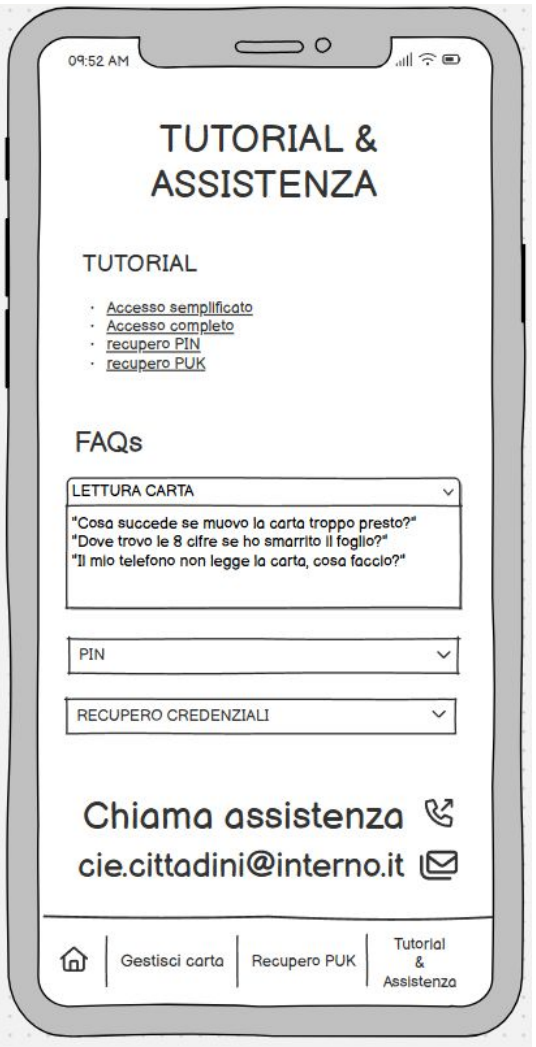
Lettura carta screen



The second step of the third level authentication is focused on the card reading through NFC scanning. To help the user orientate the redesign creates another screen displaying “Step 2 di 2” instead of placing a mini-screen alongside the "Inserisci PIN" screen.
There are **clear visual cues** that suggest to the user how to perform the task. First of all an illustration shows clearly where the card must be placed according to the device. Then a **timer** is displayed to avoid errors due uncertainty about timing.
At the bottom of the screen a warning icon is followed by a written suggestion on where to place the card and for how long.
Finally, two **contextual help** are provided, one inline deep-link redirecting the user to “Tutorial & Assistenza” and the question mark icon on the top right of the screen.

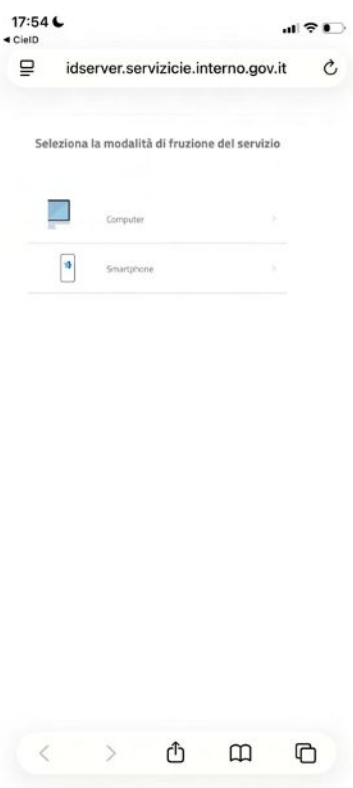
Tutorial & Assistenza screen

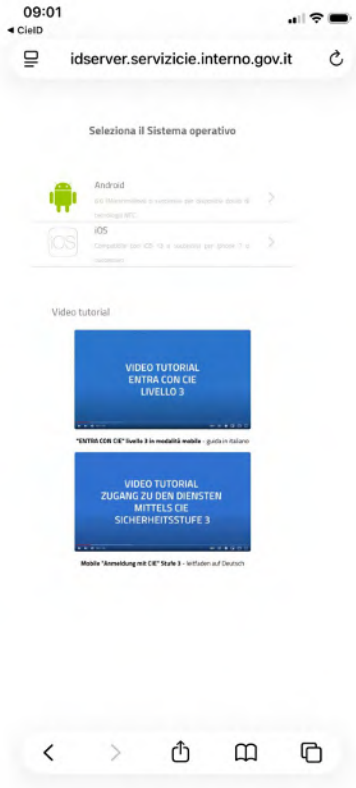




Assistance and tutorials in the version 0 of the redesign are contained in the same screen instead of being divided in two different sections. All the tutorials available are linked at the top of the screen followed by a FAQs section comprising answers on doubts about the most frequent task. In the bottom of the page the assistance number is provided by clicking on “Chiama assistenza” and by clicking on the email the user is automatically redirected to their email provider.

Tutorial









The redesign addresses also the issue of the tutorial being external to the application. In order to simplify the access to explanation and added information we chose to **integrate each tutorial directly into the application** without external links to YouTube.

CieID Redesign v.1

Result of the first inspection (heuristic analysis)

Homepage



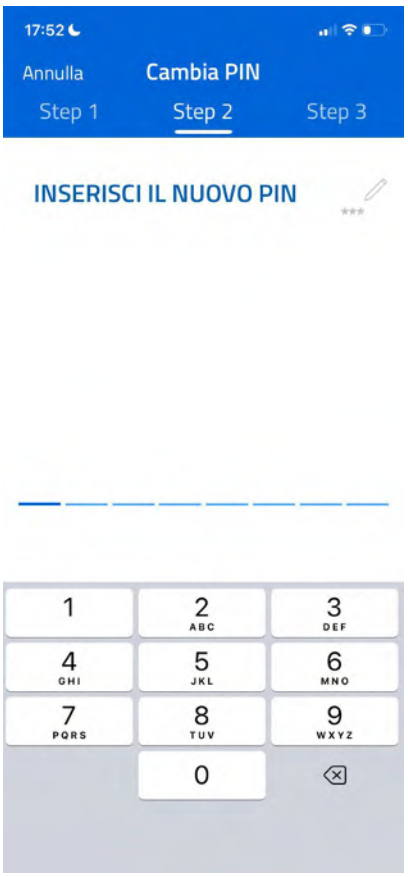
The new version provides the user with the same jargon free labeling of tasks in the home page but adds the possibility to consult the **checklist of the requirements** for each task, in order to guide the user's choice and help them

prepare to carry out the access.
The bottom navigation bar has been improved by the removal of an item (“Recupero PUK”), a step the user would likely use less frequently, in order to ensure the best possible readability. Secondary items remain accessible by expanding the navigation bar with one click. The two hidden items are under “Aiuto e recupero", a more generic labeling which contains “Recupero PUK” and “Assistenza e Tutorial”.

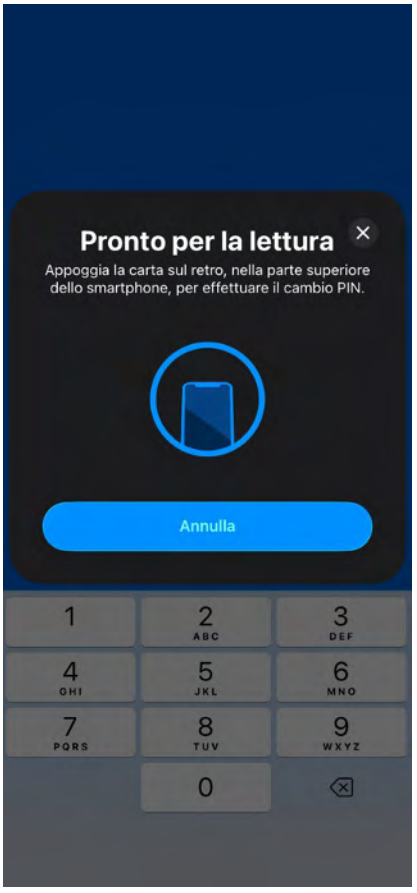
PIN recovery flow

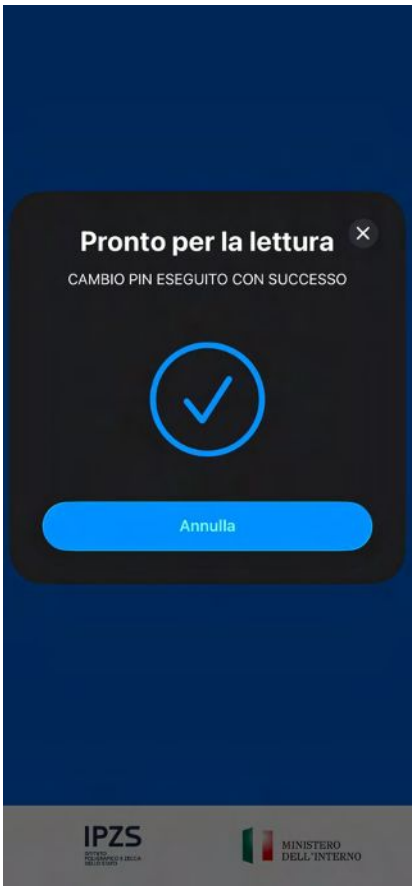


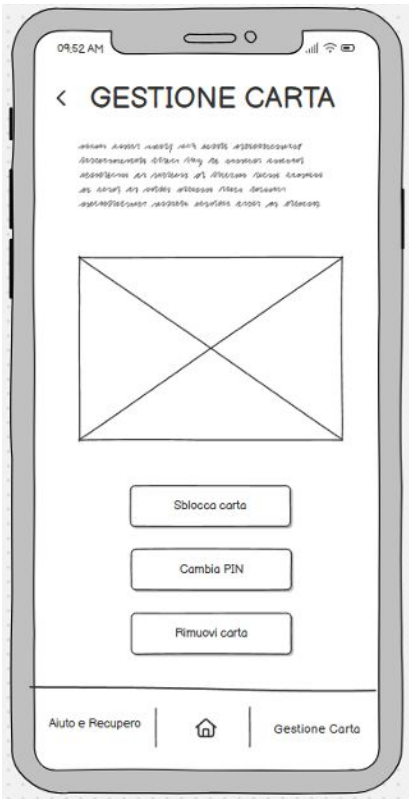


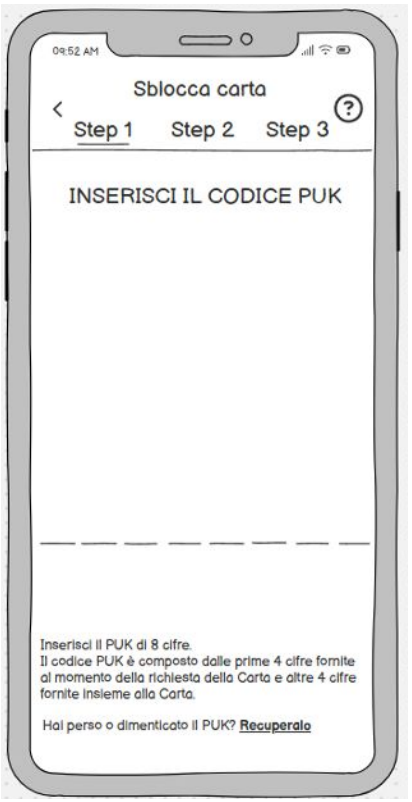

17:52
Annulla
Cambia PIN
Step 1
Step 2
Step 3
INSERISCI IL NUOVO PIN
1
2 ABC
3 DEF
4 GHI
5 JKL
6 MNO
7 PQRS
8 TUV
9 WXYZ
0


17:52
Annulla
Cambia PIN
Step 1
Step 2
Step 3
RIPETI IL NUOVO PIN
1
2 ABC
3 DEF
4 GHI
5 JKL
6 MNO
7 PQRS
8 TUV
9 WXYZ
0


Pronto per la lettura
Appoggia la carta sul retro, nella parte superiore dello smartphone, per effettuare il cambio PIN.
Annulla
1
2 ABC
3 DEF
4 GHI
5 JKL
6 MNO
7 PQRS
8 TUV
9 WXYZ
0


Pronto per la lettura
CAMBIO PIN ESEGUITO CON SUCCESSO
Annulla
IPZS
MINISTERO DELL'INTERNO


09:52 AM
GESTIONE CARTA
Sblocca carta
Cambia PIN
Rimuovi carta
Aiuto e Recupero
Gestione Carta


09:52 AM
Sblocca carta
Step 1
Step 2
Step 3
INSERISCI IL CODICE PUK
Inserisci il PUK di 8 cifre.
Il codice PUK è composto dalle prime 4 cifre fornite al momento della richiesta della Carta e altre 4 cifre fornite insieme alla Carta.
Hai perso o dimenticato il PUK? Recuperalo

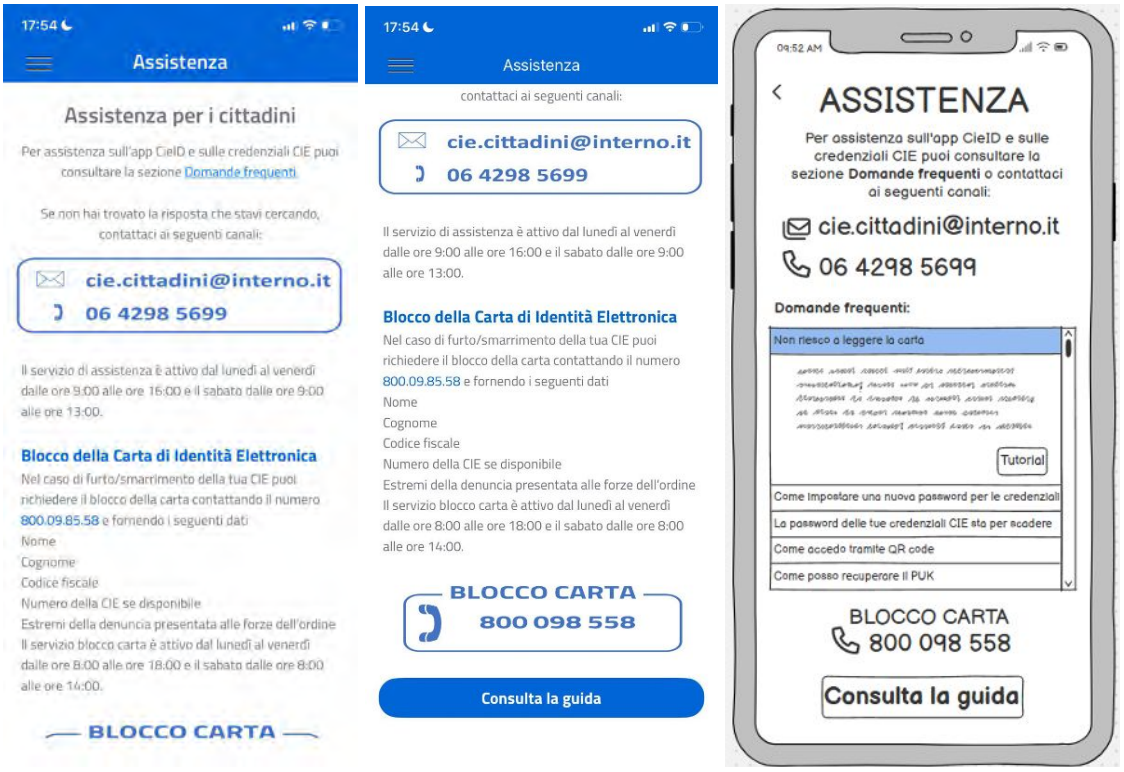

09:52 AM
Cambia PIN
Step 1
Step 2
Step 3
INSERISCI IL NUOVO PIN
space
return


09:52 AM
Cambia PIN
Step 1
Step 2
Step 3
RIPETI IL NUOVO PIN
Prendi nota e tieni al sicuro il nuovo codice PIN, sarà necessario per le prossime richieste di autenticazione
space
return



The PIN recovery flow has undergone some modifications. First of all in the “Sblocco carta” screen we tried to conform the descriptive text of the PUK as much as possible to the PIN’s description in the “Inserisci il PIN” screen to leverage the internal consistency of the application.
The version 1 also adds a **question mark on every screen** during the task to provide a direct link to the support page.
During the card scanning a **visible illustration**, providing a much clearer demonstration of how to hold the card up to the phone is provided, accompanied by a warning message that explains, in plain and natural language, how to carry out the scanning.
Finally and most importantly, the screen incorporates a **progress bar** that provides **dynamic feedback to track task completion**.

Assistenza & Tutorial screen

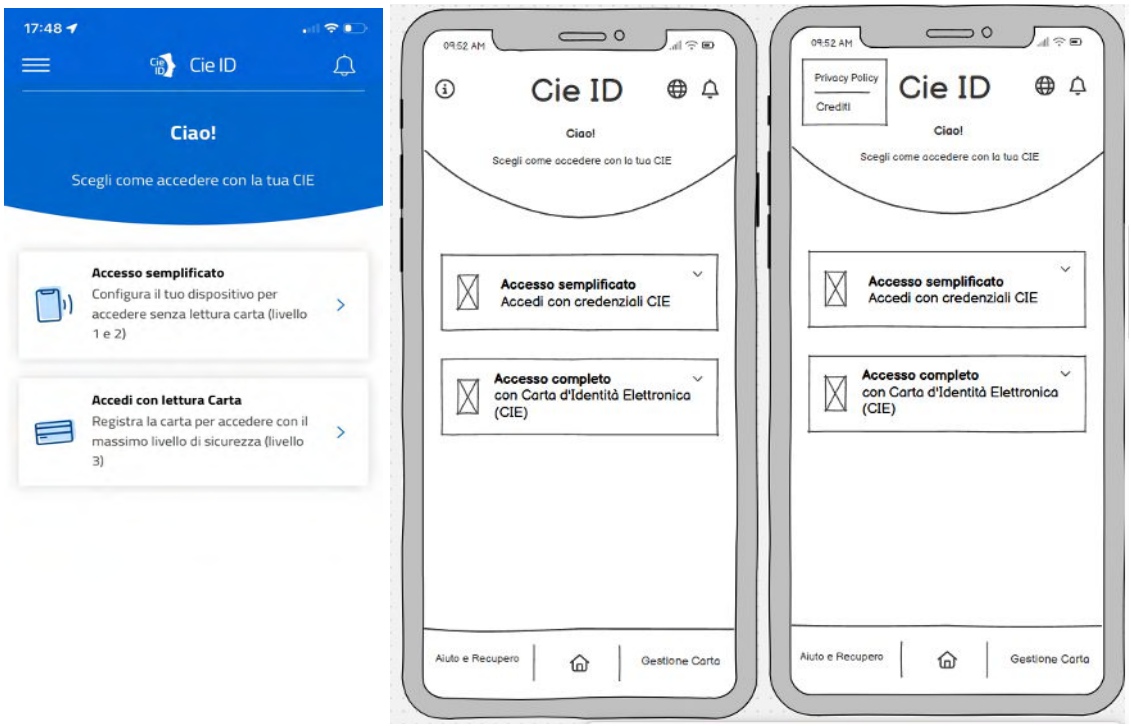


The “Assistenza & Tutorial” screen has been reorganized in a clearer layout and structured in order to insert all the essential information the user might need. The contact information is highlighted in the same way as in the original version. Additionally, a section for frequently asked questions (FAQs) has been added; in the original version, this could only be accessed by being redirected to an external webpage.
For each FAQ linked to a specific task, a **contextual link to the relevant tutorial** has been integrated. This ensures that all essential information is contained within the application, avoiding redirects that might disorient the user.

CieID Redesign Final Version 2

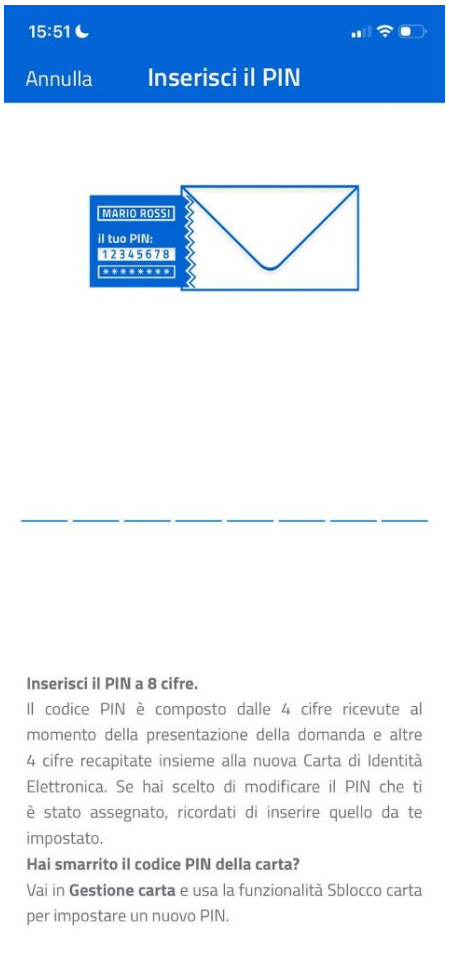
Result of the Inspection and the Formative Testing

Homepage



The final version of the homepage retains the features of the previous version, incorporating only one minor change: the replacement of the overflow menu icon with the **info icon**.

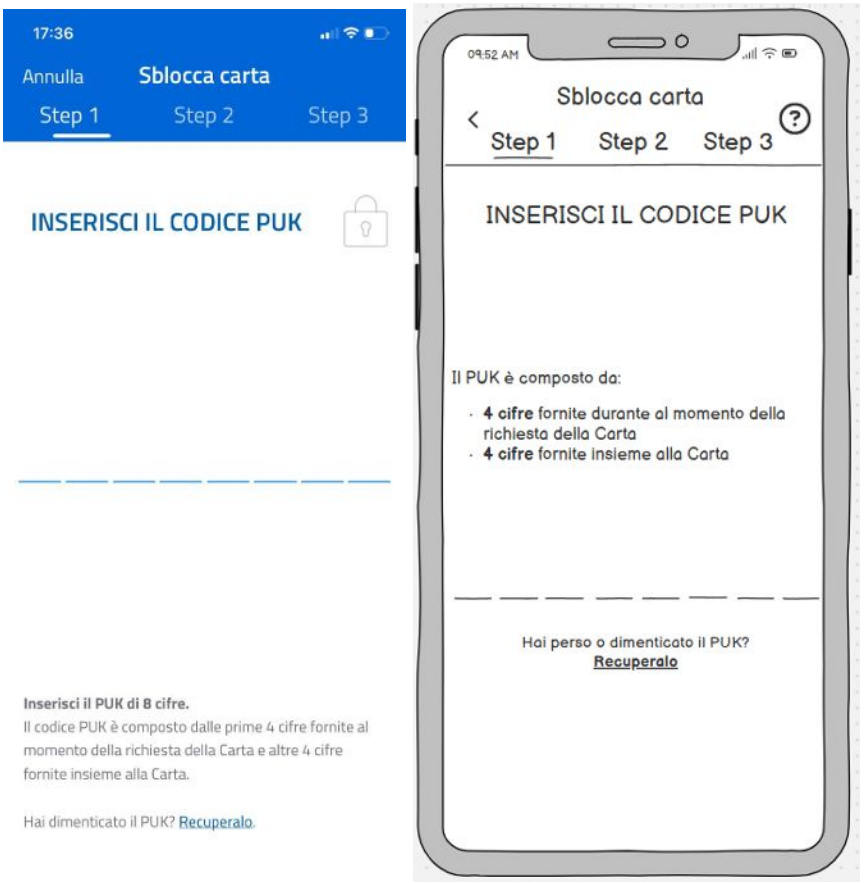
Inserisci PIN screen





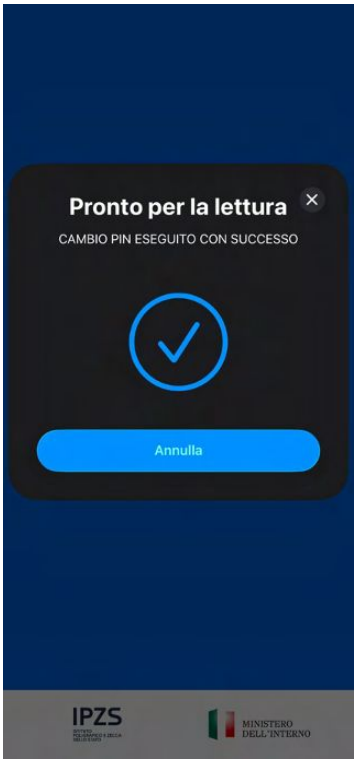
The “Inserisci PIN” screen page incorporates a task-step tracking system consistent with the one used on other task-related screens, ensuring aesthetic coherence throughout the application.
The deep inline link below no longer redirects the user to the "Gestione carta" page; instead, it leads directly to "Sblocco carta", a **streamlined path toward PIN recovery** that simplifies the process by minimizing user uncertainty and the number of choices required.

Sblocco carta screen



The “Sblocco carta” final redesign version highlights the **PUK description above the entry field** such as the description of the PIN in the “Inserisci PIN” screen to maintain visual coherency.

Cambia PIN and Registra carta









Previously, the process for changing a card's PIN required the user, who had already scanned the card, to return to the home page and click on "Accesso completo" again, in order to enter the PIN and scan the card a second time.
In this latest redesign, given that the two tasks are closely linked and

interdependent, we have decided to make the **card-unblocking and registration steps consecutive**. This approach was chosen because all security standards are maintained, while the sequential nature of the tasks saves the user time and effort, allowing them to complete both tasks more easily and efficiently.

Design summary card

Design versions as submitted

- **Version 0**

Folder name:	**CieID v.0**
Date / time of completion:	20/10/2026 17:00

- **Version 1**

Folder name:	**CieID v.1**
Date / time of completion:	22/10/2026 10:30
Types of evaluation	Inspection, Formative Testing

- **Final Version 2**

Folder name:	**CieID v.2**
Date / time of completion:	25/10/2026
Types of evaluation	Inspection, Summative Testing

List of issues handled during redesign, ordered by importance

- **Accessibility and vocabulary** (Issue #1 and #2): the redesign specifically addresses the technical jargon in the homepage, omitting misleading and confusing terminology such as “livello 1 e 2”, “livello 3”, “Configura”, favoring simple and concise explanations. To ensure all users can grasp the most important information, we decided to use black for all text, replacing the light gray used in the original application. Key information is positioned in the center of the screen, with essential terms highlighted to aid understanding. This approach enhances accessibility, including for users with visual impairments (see CieID Redesign Final Version 2

wireframes - Homepage, Inserisci PIN screen, Sblocca carta screen).

- **Visual signifiers and discoverability (issue #4):** the redesigned version of CieID prioritizes discoverability especially by providing a visible bottom navigation bar. All essential features are now visible to the user, replacing the hidden menu ith an explicit one and including secondary information inside the info icon in the spot previously reserved to the hamburger menu. Another important implementation is the insertion of a clear illustration on how to place the CIE during the scanning. It is more explanatory than the original, which was rather abstract, since it relied on the text to provide the explanation. Finally a progress bar is integrated in the same screen in order for the user to exactly know the status of the process. This helps to disambiguate the already unknown NFC scanning process installing trust in the user (see CieID Redesign Final Version 2 wireframes - Homepage, Cambia PIN screen).
- **PIN recovery and "Accesso completo" flow (Issue #3 and #7):** the cumbersome and repetitive process for changing a PIN and registering a card, which previously required users to independently locate the relevant pages (without any direct links) and scan the card twice in a row, has been significantly streamlined. Now, the PIN change and card unlocking triggered by the initial scan allow the user to complete the card registration process directly, without needing to perform any further scans. (see CieID Redesign Final Version 2 wireframes - Cambia PIN, Registra carta screen).
- **Assistance and contextual help (Issue #5 and #8):** each screen involved in the execution of a task provides a direct pathway to access assistance contacts and contextual tutorials. It eliminates all redirects to external sites and resources - such as those in the original version that sent users to YouTube videos or web-based FAQs - by integrating all information directly into the application. This ensures that users remain oriented and can access all the instructions needed to complete their tasks without ever leaving the application environment (see CieID Redesign Final Version 2 wireframes - Assistenza & Tutorial screen).

PHASE D

Inspection card

Name of version

CieID v.0

Inspection context data

20/10/2026 18:00

Inspection method

- Heuristic Analysis

Output of inspection

- **Enhancement recognition rather than recall:** the hidden menu has been made visible and each function of the app is explicit in a bottom navigation bar.
- **Clear and simple presentation of the features:** elimination of technical jargon, reducing text length, and using clear, direct terms to increase user autonomy in selecting tasks.
- **Inclusion of a multilingual toggle:** provide a way for the user to select the English version of the app. Specifically implemented to address the needs of people belonging to The international IT student (1D + 2C + 3D + 4B + 5C) Segment, or more generally to immigrant citizens.
- **Visibility of system status:** the redesign helps the user navigate the system and perform tasks through visual cues.
- **Redundancy in the “Lettura carta” screen:** during this step the user can already access the “Tutorial & Assistance” screen by clicking on the question mark. The link in the bottom of the page is redundant and unnecessary.
- **Lack of requirement list:** in the homepage the two tasks could be integrated with a list of requirements that the user could consult in order to help them choose between the tasks.
- **Disorganized structure in the “Tutorial & Assistenza” screen:** the structure does not link the tutorial to the contextual task. The

links to the video and the FAQ section are disconnected and are not related to each other.
- **Excessive number of labels in the navigation bar:** several labelling do not allow an optimal visibility and the required dimension of the font-size.

Design recommendations

- **Add checklist under tasks in the homepage:** implement a toggle to expand the sections inside the homepage to visualize what the user needs to carry each access.
- **Reduce elements in the navigation bar:** consolidate information to reduce the number of labels in the navbar and improve readability.
- **Avoid redundancy:** remove inline deep-link inside the “Lettura carta” screen.
- **Reorganize and rename “Tutorial & Assistenza” respective screen and label:** rename the section “Assistenza & Tutorial” to mirror the order of the screen organization and reorganize it in a functional way.

Name of redesigned version

CieID v.1

Inspection card

Name of version

CieID v.1

Inspection context data

21/10/2026 10:00

Inspection method

- Informal Action Analysis

1) *Complete list of action necessary to complete the task (Happy path)*

- Task 2 and 3 (see 06a Preliminary user testing)

Action 1 - Visualize the requirements to complete “Accesso completo”
Action 2 - Click on “Accesso completo” and land on “Inserisci il PIN” screen
Action 3 - Click on “Gestione Carta” deep inline-link to reset lost PIN.
Action 4 - Click on “Sblocca carta” button
Action 5 - Insert PUK code
Action 6 - Insert new PIN
Action 7 - Repeat new PIN
Action 8 - Click on proceed on the screen warning you to remind your PIN.
Action 9 - Scan the CIE card to reset the PIN.
Action 10 - Land on the “Gestione carta”
Action 11 - Go back to the home page
Action 12 - Click on “Accesso completo” and land on “Inserisci il PIN” screen
Action 13 - Insert the new PIN
Action 14 - Scan the CIE card

Output of inspection

- **Generic inline deep-link:** the user is redirected to “Gestione carta” while trying to reset the PIN but lands on the screen and still has to choose between three options: “Sblocca carta”, “Cambia PIN” and “Registra carta”. Several options can mislead the user and lead to failing trials.

- **Redundancy of actions:** in order to reset the PIN the user has to scan the CIE. After the scanning the user is redirected to the initial “Gestione carta” screen requiring them to go back to the “Accesso completo” task autonomously without a guide.

Design recommendations

- **Standardize PUK description:** conform the description and information about the PUK with the one describing how to find the PIN, to leverage on aesthetic and visual coherency.
- **Standardize steps signifiers:** adopt a standard way to visualize the system status during the execution of a task.
- **Make connected tasks consecutive, avoiding interruptions:** when resetting the PIN the user is asked to scan the card. Then the same task is required again to finish the “Accesso completo”. Connect the two actions by allowing the user to complete the “Accesso completo” with just scanning, reducing redundancy and optimizing the process. This will save the user additional time transforming the task into a quick and efficient series of action.
- **Change the label order in the navigation bar:** put “Gestione carta” on the right part of the navigation bar since it is likely most needed than “Aiuto & recupero”.

Name of redesigned version *(as outcome of inspection, only if existing)*

CieID v.2

Formative test card

Name of version

CieID v.1

Testing session setup (devices, operating systems, online vs. in presence, your device or the user's, location of testing, etc.)

The test was conducted in presence, in the user's house, using a low-fi prototype on the tester's mobile device.

Testing method

- Discount usability testing (2-5 people)

List of tasks to test

- Recovery of PIN and registration of ID card on CieID, level 3 of security (Task 2 and 3 in 06a Preliminary User Testing).

Testing methodology

- Bottom line data (the subject is left alone and performs the tasks on his/her own - no interaction with the assistant)

Metrics of tests

- Success
- Efficiency
- Errors
- Satisfaction

Recruitment of subjects

- **Giovanni**, 60 years old man, average digital skills. He partially fits the segment #1 because of his age and poor eyesight. Despite that, he is quite used to navigating online and using mobile applications. He is a neighbor of Alice, one of the testers.

Motivation for choice of subjects

The design team mainly relied on acquaintances and distant family relatives.

Test #1: Giovanni

Name (fake one)

Giovanni

Date and time of test

23/10/2026

Assistant of test (team member)

Alice Spadavecchia

Context of test (location, duration, notable events etc.)

House

Qualitative data of tests

- **Perceived satisfaction and emotion:** the user demonstrated a general increase in confidence while performing the task; the time saved and the suggestions provided by the application facilitated task completion, thereby improving the user experience.
- **Frustration and cognitive pressure:** A persistent point of friction, compared to the original version of CieID, occurs immediately after the card is unlocked. At this stage, users continue to feel disoriented and unsure of what the next step should be. Users have also expressed annoyance at having to scan the card twice within a span of just a few seconds. Overall, however, the level of frustration and the frequency of errors have decreased significantly since the original version of the app.

- **Mental model fit and clarity:** The user demonstrated an understanding of how the system works, except for a couple of instances where they required guidance because they got stuck. Compared to the original version, they appreciated its clarity and the fact that it met their expectations.

Quantitative data of tests

Success:
Task 2 and 3 - 100%

Efficiency:
Task 2 and 3 - 02:15

Errors (mistake and slips):
Task 2 and 3 - mistakes: 1, slips: 1

Requests for guidance:
Task 2 and 3 - 1

SUS score

87.5

Outcomes and reflections useful for design

- **Make the task 2 and 3 consecutive:** the gap between the unblock of the card and PIN recovery and the access process must become consecutive to ensure that the user has to perform the card scanning just once, saving time and mental effort.
- **Visually align the system status indicator on each page:** provide a unified way to visualise system status in each page involved in the execution of a task.
- **Replace overflow menu with info icon:** a minor modification to conform the homepage with web standards.

Name of redesigned version

CieID v.2

Inspection card

Name of version

CieID v.2

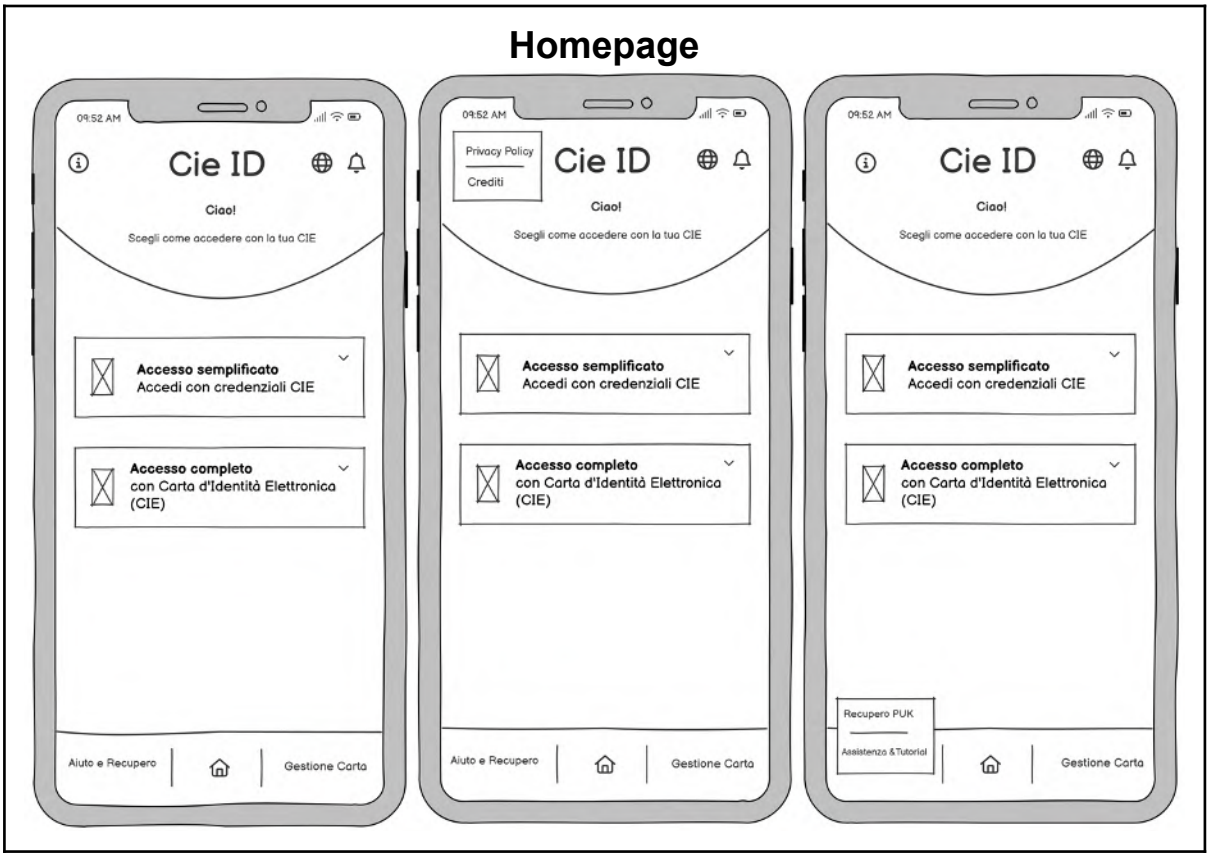
Inspection context data

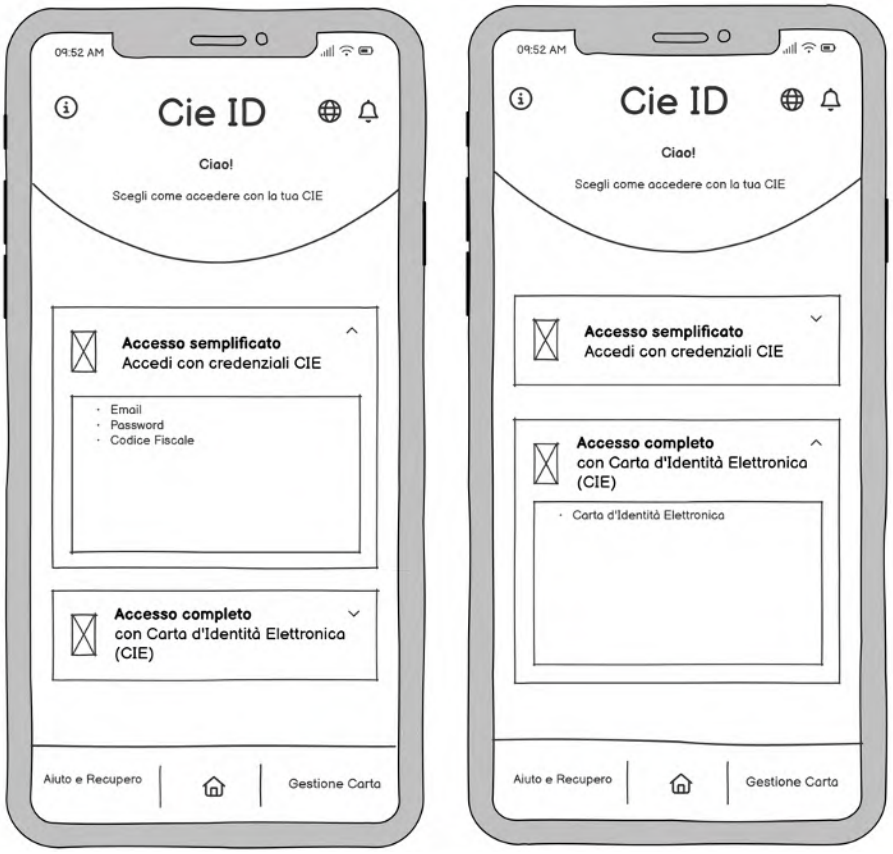
24/10/2026 14:00

Inspection method

- Cognitive Walkthrough

Low-fi wireframes - CieID v. 2





Inserisci PIN



Sblocco carta



Lettura carta e registrazione



Homepage after “Accesso completo”



Persona (for a complete overview see 02a Persona card)

Nicola Fontana, **The Autonomy-Seeking Senior** - Direct user (1C + 2B + 3C +4B + 5A).

Basic facts:

- 78 years old man
- Male
- Lower secondary education
- Retired mechanic
- Widow with three sons
- Suffers from slight cataract that prevents him to properly see and age-related recall limitations

Scenario

Nicola opens the CieID app on his smartphone to log in. Looking at the checklist, he learns that "Accesso completo" requires both his physical CIE and his PIN, which he believes he remembers. He tries entering the PIN but realizes it is incorrect. He decides to recover it and is immediately redirected to the "Sblocco carta" page by tapping the visible deep-inline link on the "Inserisci PIN" screen.
Having already retrieved his PUK a few months earlier, he enters it and creates a new PIN. When prompted to scan his card, he knows exactly what to do thanks to an instructional illustration and dynamic visual feedback that guide the action and prevent anxiety. Upon scanning, he successfully unblocks his card and completes its registration at the same time. At the end of this straightforward sequence, he has successfully completed two tasks in a single flow.

Happy path

- Task 2 and 3 (see 06a Preliminary user testing)

Action 1 - Visualize the requirements to complete “Accesso completo”
Action 2 - Click on “Accesso completo” and land on “Inserisci il PIN” screen
Action 3 - Click on “Sblocco carta” deep inline-link to reset lost PIN.
Action 4 - Insert PUK code
Action 5 - Insert new PIN
Action 6 - Repeat new PIN
Action 7 - Scan the CIE card to reset the PIN and register the card.
Action 8 - Go back to the home page

Output of inspection

- **Flow simplification:** the sequence dropped from 14 to 8 actions eliminating confusing navigation loops. The added inline link to “Sblocco carta” redirects the user exactly at the beginning of the PIN reset task.
- **Single-scan Flow:** Merged PIN reset and Accesso completo card registration into a single, cohesive process avoiding repetitions of the same actions when unnecessary. Reduced NFC scanning actions from 2 to 1 single scan, removing the most error-prone gesture.
- **Dynamic Scanning Feedback:** Real-time visual confirmation during NFC reading to prevent premature card removal and minimize physical errors.

Design recommendations

- **Persistent and On-Demand Contextual Guidance:** retain the initial onboarding guide/visual tutorial while ensuring it remains permanently accessible during subsequent app sessions. This offloads memory requirements (Knowledge in the World) and provides continuous support for digitally anxious or senior users whenever they need a quick refresher.

Summative test card

Name of final version

CieID v.2

Testing session setup (devices, operating systems, online vs. in presence, your device or the user's, location of testing, etc.)

The test was conducted on the final prototype on Figma on a PC, and was tested at home.

Testing method

- Discount usability testing (2-5 people)

List of tasks to test

- Recovery of PIN with PUK and registration of ID card on CieID

Testing methodology

- Bottom line data

Metrics of tests

- Success
- Efficiency
- Errors
- Satisfaction

Recruitment of subjects

- **Mauro**, 31 year old man: A UX designer working in the digital space who is enthusiastic about user experience and technology. Having transitioned from SPID to CieID, he is well-acquainted with the overall workflow.
- **Lucia,** a 70 years old woman, scarce digital competences. She is a direct representative of the segment #1 - The Autonomy-Seeking Senior. She is a family acquaintance recruited by a team member.

Motivation for choice of subjects

- **Mauro**: he is well-acquainted with the overall workflow. He had tested the first version of the app before, allowing him to compare this new version and provide his feedback.
- **Lucia**: she is selected among the tester's acquaintance to represent Segment #1. She is a senior Italian citizen which has been tested both with the original CieID app and the redesigned version to provide feedback about the usability of the app and the satisfaction while using it.

Test #1: Mauro

Name (fake one)

Mauro

Date and time of test

31/10/2026

Assistant of test (team member)

Maryam Dadrasrazi

Context of test (location, duration, notable events etc.)

Home

Qualitative data of tests

- **Perceived satisfaction and emotion:** Having previously tested the original version of the application, Mauro approached the CieID V.2 prototype with high curiosity and expressed strong overall satisfaction (SUS 77.5). He specifically appreciated the simplified explanations and the transformation of previously static text into clickable hyperlinks, noting only minor visual UI refinements related to the Figma prototype environment.
- **Frustration and cognitive pressure:** Cognitive load and frustration were virtually eliminated throughout the combined PIN recovery (via PUK) and CIE card registration workflow. Because complex operations were broken down into clear, linear steps (Step 1, Step 2, Step 3) with explicit explanations of where to find the 4+4 digit codes, Mauro completed the entire end-to-end flow in 01:10 without a single mistake, slip, or request for external guidance.
- **Mental model fit and clarity:** The redesigned Information Architecture aligned seamlessly with Mauro's expectations and UX mental models. Exposing primary sections in the bottom navigation bar and placing an active recovery link directly on the PIN entry screen created an intuitive, uninterrupted pathway from error recovery to final card registration.

Quantitative data of tests

Success:
Task 2 and 3 - 100%

Efficiency:
Task 2 and 3 - 01:10

Errors (mistake and slips):
Task 2 and 3 - *mistakes*: 0, *slips:* 0

Requests for guidance:
Task 2 and 3 - 0

SUS score

77.5

Outcomes and reflections

The user expressed high satisfaction with the overall flow, simplified explanations, easy-to-understand steps, and clickable hyperlinks provided where needed, noting that the navigation matched their expectations well. Overall improvements were considered satisfactory, with only minor UI element inconsistencies highlighted for further refinement.

Test #2: Lucia

Name (fake one)

Lucia

Date and time of test

03/11/2026

Assistant of test (team member)

Alice Spadavecchia

Context of test (location, duration, notable events etc.)

Home

Qualitative data of tests

- **Perceived satisfaction and emotion:** the user demonstrated autonomy during the execution of the task and the reduced execution times and redundancy avoidance significantly improved user’s experience and reduced anxiety.

- **Frustration and cognitive pressure:** bottom line data gathered during the execution of the tasks highlight the limited cognitive loading required to manage the new task flow. The frustration and guidance requests have been reduced with CieID v.2. The user is always guided through the task flow and when necessary, assistance is easily findable in the information architecture. This results in an experience free from complications.
- **Mental model fit and clarity:** The new version of the app aligns as closely as possible with users' mental models, adhering to UI conventions and helping users navigate and perform tasks on the site without the need for complex reasoning. All important and necessary elements are easily accessible, and during testing, users demonstrated both familiarity and confidence.

Quantitative data of tests

Success:
Task 2 and 3 - 100%

Efficiency:
Task 2 and 3 - 01:53

Errors (mistake and slips):
Task 2 and 3 - *mistakes*: 0, *slips:* 0

Requests for guidance:
Task 2 and 3 - 1

SUS score

85

Outcomes and reflections

The user demonstrated greater confidence in their ability to use the system compared to the initial test on the original app. Execution times were significantly reduced, and decreasing the number of required scans from two to one allowed the user to consolidate two particularly challenging actions into a single step, while still maintaining safety

standards. The user nonetheless partially relied on the tester's help, seeking confirmation and approval while carrying out the task.
The user still relied on the tester’s help, seeking confirmation and approval while performing the task; however, such requests decreased significantly compared to the first attempt with the original application (6 vs 1). This demonstrates how the new experience design enhances user autonomy, reducing the burden on intermediaries strictly adhering to the Help-People-Help-People initiative.

Summary table of SUS scores

Question	**Lucia**	**Mauro**
Question #1	3	4
Question #2	5	2
Question #3	5	4
Question #4	5	2
Question #5	5	4
Question #6	5	2
Question #7	5	4
Question #8	5	2
Question #9	3	5
Question #10	5	2
Total Score	85	77.5
Average	81,25	
Variance	28.13	

Conclusions

SUS average and variance

SUS average: 81,25 | variance: 28.13

Other quantitative metrics

Task completion rate (success): 100% success rate across all participants. Both the digital mentor (Mauro) and the senior user (Lucia) successfully completed the core workflow of PIN recovery via PUK and CIE card registration without task abandonment or critical failure.

Time-on-task (efficiency): High operational efficiency was observed for both user profiles. Mauro completed the entire workflow in 01:10, while Lucia required only 01:53. This represents a drastic reduction in task duration compared to the baseline test (respectively 01:34 and 04:58), where task completion was severely hindered by navigational friction and disorientation.

Error Rate: 0 interaction errors (0 mistakes and 0 slips) recorded for both participants throughout the session. This confirms the total elimination of data-entry missteps and premature card removal during NFC scanning.

Requests for Guidance (Autonomy): Mauro required 0 requests for guidance, demonstrating complete operational independence. Lucia made only 1 minor request for guidance (a brief verbal confirmation during step 1), marking a significant improvement in autonomy compared to baseline testing, where continuous facilitator intervention was necessary.

Conclusions

SUS Scores, threshold milestone and variance context

The average SUS score of 81.25 represents a significant milestone for the CieID redesign, placing the prototype comfortably above the

standard usability benchmark of 68. To properly understand the score variance (28.13), it is valuable to evaluate individual participant feedback relative to their baseline testing of the original application:

• Mauro (Segment #2: Digital Mentor / tech-savvy user) assigned the redesign a score of 77.5 - sitting 9.5 points above the usability threshold and marking a 32.5-point increase from the score of 45 he gave to the original CieID app during preliminary testing.

• Lucia (Segment #1: Senior / digitally anxious user) awarded the new version an 85 - exceeding the benchmark by 17 points and achieving a remarkable 55-point leap from her initial score of 30 on the original app.

The variance of 28.13 is a direct mathematical result of conducting discount usability testing with a two-subject sample. Rather than indicating inconsistent usability, this spread simply reflects the natural difference in perception between a tech-savvy mentor and a senior novice. Most importantly, both individual scores (77.5 and 85) independently exceed the 68 threshold, confirming substantial qualitative and quantitative improvements across both user segments.

Objective behavioral and quantitative metrics

The high SUS scores are strongly validated by the objective behavioral metrics recorded during the core test task (**recovering the PIN via PUK and registering the ID card on CieID)**, providing that the redesign is a functional success. When analyzing actual task performance, **Mauro** achieved a **100% task completion rate** in just 01:10, with zero interaction errors (**0 mistakes and 0 slips**) and **0 requests for guidance**. Similarly, Lucia achieved a 100% completion rate in 01:53 minutes, completely eliminating the errors and frustration experienced during the baseline test.

The redesign successfully bridged the gap between failure and autonomy: both the recovery of the forgotten PIN via PUK and the subsequent CIE card registration were completed smoothly without any intervention or takeover by the assistant. Ultimately, while digital identity workflows remain conceptually demanding for vulnerable citizens, the

quantitative data proves our redesign provides the structural safety net required to navigate them independently.

Qualitative shift and validation of the core design interventions

Summative testing demonstrated a clear qualitative shift that directly validates our four core Information Architecture and UI design interventions, eliminating the navigational dead-ends and cognitive friction observed in the baseline. By replacing the hidden hamburger menu with an explicit, high-contrast **Bottom Navigation Bar**, the redesign aligned immediately with the user's navigational mental model and eliminated exploratory guesswork. This structural clarity was reinforced at the primary point of failure by converting the static reference to *"Gestione carta"* on the *"Inserisci il PIN"* screen into an **interactive inline hyperlink** (*“Sblocca carta”*)**,** an intervention Mauro explicitly praised for providing clickable hyperlinks and easy-to-understand steps right where needed. Furthermore, decomposing complex workflows into **linear, numbered steps** - supported by an NFC progress indicator, upper-phone positioning graphics, and a persistent “?” contextual help icon - eradicated the anxiety of premature card removal and disorientation during card scanning. Redundancy in tasks 2 and 3 has been replaced by single-scan flow, halving the necessity to perform a complex action as the scanning of a card. It has also been facilitated with clear illustrations, leaving no doubts to the user. Finally, reframing abstract bureaucratic jargon (*"Livello 1, 2, 3"*) into **plain, action-oriented homepage cards (*"Accesso semplificato"* and *"Accesso completo"*)** empowered both the direct user and the intermediary. This transformed a frustrating bureaucratic hurdle into a predictable, self-correcting user experience, reducing remaining participant feedback to minor visual polish on the Figma prototype.